$$\sigma = PR/t \qquad (25\text{-}4)$$

where σ is the skin stress, P is the internal pressure, R is the radius of curvature, and t is the skin thickness. Thus:

$$P = t\sigma/R \qquad (25\text{-}5)$$

(In the case of a closed cylinder the hoop stress is still given by PR/t but the stress in the axial direction is $PR/2t$).



Fig. 25-8. The IAE Deployed Off the SPARTAN Spacecraft of NASA.

The structural films used in the consideration of large apertures are in the vicinity of 0.0005 inch thick, the radii of curvatures are in the low to high tens of meters and the skin stresses required to remove stowage wrinkles can vary from about 4 MPa to 6 MPa.

Thus, the inflation pressure to turn a large, purely inflatable body of revolution into a stiff structure in space is so small it may be considered to be a decent vacuum on Earth. Concomitantly, the mean free path of intermolecular collisions in the inflation gas becomes larger, thus retarding its escape from the inflated structure (there is no explosion of the inflatable, not that anyone could hear, anyway). This is a significant result when one considers that inflatable structures in space must peacefully coexist with micrometeoroids and debris.

The consequences of these findings (at least on a theoretical level), along with the continued development of structural films such as Mylar and Kapton (DuPont is the largest manufacturer, but smaller companies have made further improvements) have played an important role in the interest that these structures have generated the last twenty or more years in the field of Space Mission Design. But should the mission designer be too enthusiastic to use an inflatable payload in space, as I have been most of my career, they're kindly reminded that this area is much more complex and difficult than it seems. It is an admixture of physics, art, very careful engineering, much daring and a good dose of humility, preferably learned before large amounts of capital are expended. I have encountered many situations where exorbitant claims are made for inflatable membrane missions by well meaning, but inexperienced mission planners and engineers.

Table 25-0, Fig. 25-8, Eq. 25-5

25.2.3 The Design Process

As noted above, this is similar to the design process that any payload system must undergo. For inflatables there are some differences that need to be taken into account. The thin film materials used are non-linear in that (a) their modulus changes with applied stress; (b) as manufactured, their modulus is different in the longitudinal direction from that in the lateral direction (and this introduces yet another level of complexity); and (c) the inflation process can produce fairly large deformations of the material, making it difficult for the usual FEM codes (e.g., NASTRAN) to converge to a solution successfully and efficiently. Thus, the design of an inflatable structure can be, and in most cases is, a cyclical process, particularly when accurate surface figures are required in the deployed system (e.g., antennas, solar concentrators).

A. Mission Requirements.

Most important for the deployed structure are Mission Lifetime, Mission Orbit (micrometeoroid distributions, UV, atomic oxygen, particle radiation and thermal environment), deployment and mechanical/gravitational loads. For the stowed configuration, requirements would be the available stowage volume and allowable mass of the payload, launch loads (acceleration, vibration, and ascent profiles) which must be taken into account in the design of the *inflatable containment structure* (also referred to as the "*canister*") and the inflatable payload mode of retention prior to deployment. Once these are defined, the search for the proper inflatable materials and thickness that will withstand orbit environment can begin. A very sizable amount of work has been done and reported on a number of different membranes and their behavior in most of the orbit environments but we will mention a few words on micrometeoroids here. There have been terrestrial experiments to attempt defining the damage function of micrometeoroids on stretched membranes, as it was thought that plastic films under stress would be torn by the impingement of these ubiquitous particles, due to the low tear resistance of the films. These experiments, conducted with hypervelocity particles (5 km/sec) of an average diameter around 1 micron have shown that the films do not tear. Rather, the impinging particle creates a "cookie-cutter" hole, whose diameter asymptotes out to 3 times the particle diameter [Thomas and Griese, 1980]. When convoluted with the micrometeoroid size and intensity distribution, this gives a fair, though approximate determination of the damage magnitude (the accumulated hole area) on the membrane as a function of time. Knowing the operational pressure, an analysis can be performed to determine the inflatant loss as a function of time and the amount of "make-up" gas carried along to supplement the loss, when necessary. Analyses performed at L'Garde for large inflatable-only bodies of revolution suggest that a mission life beyond 5 years in the space environment is not practical for these structures, unless ways to ameliorate the gas leakage are found in the future. Some concepts have been considered on this topic, including self-annealing membranes or in-

flatable gases that carry a sealing agent, but unfortunately nothing definitive has been produced so far.

B. The "Inverse Problem"

Once a material meeting all the requirements delineated above is found, the design process can begin in earnest. The *Inverse Problem* is a somewhat arcane term describing the relatively easy to understand analytical geometry process of starting out with the shape of the conceptual structure and working it backwards (hence, inverse problem) to the flat segments (gores) that need to be cut from the roll of film, so that when seamed together and inflated to their operational pressure, they will produce the desired shape with the desired accuracy. L'Garde, Inc. developed finite element codes to deal with this problem. For instance, FAIM (Finite-element Analysis for Inflatable Membranes) [Palisoc and Huang, 1997] deals with the "forward" transformation, starting with the un-inflated (unstressed) shape and proceeding to predict the inflated shape. When the latter converges to the conceptual one, the flat gore shape resulting from the "inverse" transformation is deemed to be correct. Figure 25-9 shows the stresses calculated for a lenticular membrane structure when properly inflated. We note that a commercially available code, ABAQUS started making much headway into the non-linear and large deformation arena a few years ago, and is now used widely by several entities in the space community.

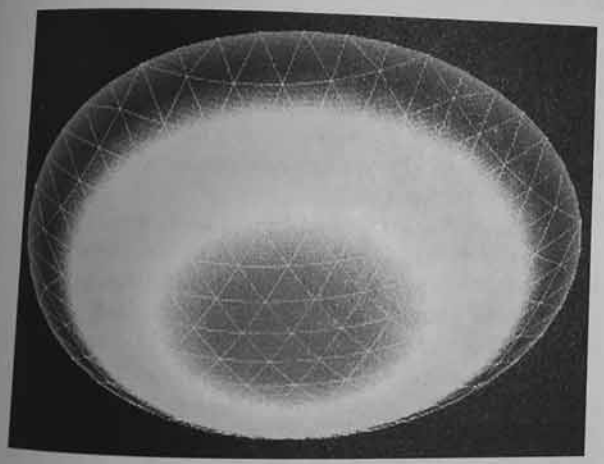


Fig. 25-9. A Reflector Stress Field Calculated by FAIM.

C. Design

Membrane Segments: After the non-linear analysis iterations (if required) are done, the flat segment shapes are designed and cut. It must be noted here that membranes have non-zero Coefficient of Thermal Expansion (CTEs) (the best ones are in the proximity of 10 ppm/°F) but in addition they possess a hygroscopic coefficient of expansion, rendering them susceptible to moisture. When high accuracy is required, the environmental conditions in cutting, inspection, and seaming of the segments (gores) to their final configuration, the temperature and humidity levels during these processes must be specified so that shape correctness is assured. Finite element thermal codes **must** be used to determine membrane temperatures and thermal gradients on orbit, as these will affect the shape accuracy of the membrane or inflatable structural component. It must be noted here that NeXolve has recently reported being able to manufacture polyimide membranes with an almost zero CTE [Poe and Patrick, 2008].

Stowed Configuration: In parallel with the segment design, the folded (stowed) configuration must be determined, so that deployment proceeds in as controlled a way as possible. No other process in payload design is as dependent on its initial conditions as this one and much care must be exercised in envisioning the folding pattern and ensuring that the folded package meets the allowable stowage volume requirement, and that deployment can proceed as designed. These objectives are usually met by tests involving simple material coupons, seamed coupons, and subscale tests of the final payload configuration.

Inflation System: This is comprised by an inflation gas accumulator and the appropriate valves to inflate the system either through a pre-programmed command, or an electronic timer that is initiated at the appropriate moment in the mission time table, or an uplink command. The inflation gas most often used is pure dry nitrogen, but CO_2 also can work under the proper thermal environments and inflation rates. The simplest inflation system is a tank containing a pre-determined amount of inflation gas under pressure, which opens upon command by activating an electro-explosive device (EED or "squib"), thus allowing the gas to enter the inflatable volume and bring it to its proper pressure (this is referred to as a "blow-down" inflation system). Naturally, the complexity of the inflation system increases considerably when more than one inflatable chamber is present, particularly if the operational pressures are different, and yet have to be inflated by the same system to improve the volume and mass of the entire payload. For instance, a very small pressure is needed to inflate a lenticular antenna reflector, but its deployment and support structural parts require a much higher pressure. Then, the gas accumulator must be followed by a pressure regulator that feeds the appropriate inflation ports through thoroughly tested valves. In addition, pressure and temperature sensors may be included to ascertain and control system status before and after deployment. It should be apparent that inflation system components should be stored and assembled in clean rooms. Otherwise, a minute dust particle trapped in a normally closed valve can quickly spell catastrophe. Finally, during launch ascent and at times beyond, it must allow for the escape of the entrapped atmospheric air that enters the system during packaging. This is accomplished via a vent valve that stays open until just before the inflation command is given, at which time it is programmed to close.

Stowage System (canister)

The canister contains all the subsytems making up the inflatable payload: the stowed membrane and its deployment/support structure, the inflation system, requisite

microprocessor, electronics and instrumentation including sensors. Its function is to provide the proper environment during launch and act as the "pedestal" from which deployment proceeds. It also provides the interface between the payload and the spacecraft, if the payload is to be connected to the latter during the mission; or it can act as the spacecraft itself if the payload is to be a free-flier upon separation from the main bus. It should be obvious that the configuration of the inflatable payload subsystems, launch, pre- and post-deployment environments, are all necessary inputs in the canister design.

Testing

The testing of all hardware components proceeds along the normal testing procedures as any other payload. It includes mechanical and environmental materials coupon testing, component and subsystem functioning of all mechanical and electronics components, and repeated functioning of the canister door opening to ascertain unencumbered payload exit from its stowed configuration. Pre- and post-vibration testing is necessary to uncover critical issues. The greatest testing difficulty arises when large membrane apertures are involved. For instance, the NASA/JPL/L'Garde Inflatable Antenna Experiment reflector was only 15 m in diameter, yet it required well over 300 m^2 of high bay area to determine its overall surface accuracy and inflation requirements (Fig. 25-10); and the NASA 400 m^2 solar sail membrane required the NASA Plumbrook Facility to test its deployment. Obviously, it would be very difficult if not impossible to find or build facilities that would accommodate even larger area apertures that appear in conceptual future mission designs. This does not mean however that these missions cannot proceed. Sub-scale tests can be conducted for a variety of cases, as long as the correct scaling laws can be applied [Greschik, Mikulas, and Freeland, 1999].

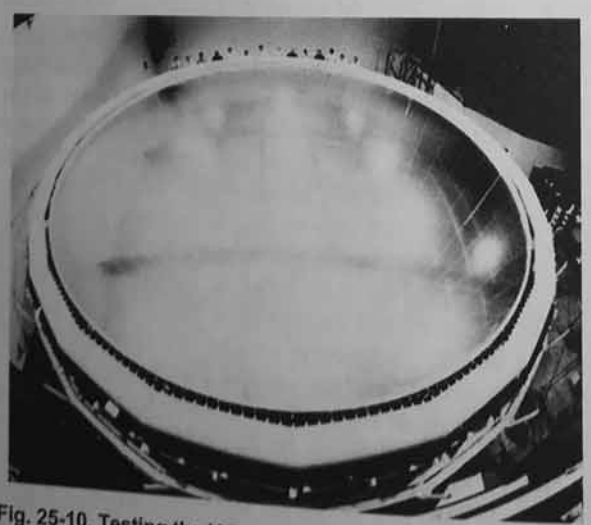


Fig. 25-10. Testing the IAE Reflector at L'Garde.

25.2.4 Engineering Inflatable-Rigidizable Structures

Whereas purely inflatable structures such as large solar concentrators, antennas, targets, large balloon-shaped objects, could theoretically operate for weeks, months, or even years in space due to the low inflation pressure required (inflation gas must be carried on board depending on mission life), the same cannot be said for inflatable structures used for deployment and support of functional payloads. These structures can be toroidal, such as supports of a lenticular reflector or flat membrane, or cylindrical in form, such as deployment and support struts. For the sake of structural efficiency and the stiffness required of support structures, their curvature radius is usually rather small by comparison to their other dimensions, and their skin thickness usually an order of magnitude or more larger than that of the pure inflatables. Thus much larger pressures (significant fractions of an atmosphere) are needed to achieve the required structural performance, in which case a micrometeoroid collision can deplete the inflatant quickly and cause structural failure. This is what gives rise to Inflatable-Rigidizable Structures. These are manufactured out of material laminates usually comprised of a structural fabric impregnated with a resin, which upon inflation becomes rigid using any of a number of physical properties encountered in space, or that are designed into the material itself.

The Echo Satellite series used a form of rigidization involving Mylar film metalized by 0.001 inch of vapor deposited aluminum (VDA). A slight overpressure applied upon reaching the operational pressure work-hardens the aluminum into a spherical shell which keeps its shape in the absence of gravitational loads, even upon losing all inflation pressure. The same rigidization technique has been used by L'Garde in the flight of an orbital Optical Calibration Sphere that flew in the late 1990s and a passively deployable solar array that was qualified for flight but never flew due to funding limitations (Fig. 25-11). ILC-Dover has also produced a number of terrestrial prototypes.

Rigidizable structures can be classed as irreversibly and reversibly rigidizable. The distinction becomes understood with reference to the pre-launch testing of these objects. The former are "one-shot" mechanisms, meaning that after one structural test, it cannot be used again. Thus, an identically designed and fabricated mechanism must be used for the next test in the series, be it an additional development test, qualification, or as a flight test item. In many cases, this procedure can be worrisome, particularly when an expensive mission is at hand (EEDs are also one-shot mechanisms, but the space community feels much more secure using these, since they have been used for decades now and each batch undergoes significant testing to attain a 0.999 reliability rating). To do this with inflatable structural elements would be a very expensive proposition, due to the large amount of "touch" labor involved in the construction of the test articles, thus impractical if there is no budget to support batch testing (I have never seen that kind of budget available in this field; thus if there is no other way to accomplish the mission, some engineers may well opt to take the chance and use a one-shot structure, and some have). On the other hand, reversibly rigidizable structures can undergo a fair



Fig. 25-11. A Solar Array with Inflatable-Rigidizable Deployment/Support Structure. Work-hardened aluminum is used for the struts. The carbon composite Canister (top) and its lid (bottom) complete the structure.

number of testing cycles by reversing the rigidized state back to the pliable and thus foldable and stowable state, and finally be flown. Table 25-1 shows a comparison between different rigidization techniques. Recently, great strides have been made in rigidization techniques, and in UV (ultra-violet) rigidization processes and resins.

Table 25-1. Rigidization Processes and Properties.

Metal Foil Work-Hardening by Overpressure	Repeatable, high stiffness, unsuitable for large loads
Water-based Resins	Reversible, low stiffness, high strength, poor for optics
Thermosetting	Irreversible, high stiffness, high strength
UV Radiation	Irreversible, high stiffness, high strength, non-uniform
Sub-Tg (glass transition temperature) Resins	Reversible, high stiffness, high strength, orbit tailorable
Particle Radiation	Irreversible, brittle, and fries the spacecraft

25.2.5 Reflector Mass and Stowage Volume vs. Diameter

Table 25-2 shows an approximate value for the mass and stowage volume of an inflatable reflector as a function of its aperture diameter. The deployment/support structural components are inflatable-rigidizable, and the assumed packaging factor is 3 (i.e., the volume allowed for stowage of the flexible payload is 3 times the volume of the flexible material). A reflector *f/D* of 0.4 is also assumed. All the major necessary components for the full functioning of the payload (inflatable mass, inflation system and gas, electronics, and canister) are included.

Table 25-2. Mass and Stowage Volume of an Inflatable Reflector as a Function of Aperture Diameter. [Palisoc et al., 1998]

Diameter	Mass	Stowed Vol.
3	5.39	0.0283
5	9.38	0.0422
10	24.37	0.0834
30	152.98	0.3324
50	391.42	0.7164
100	1,468.05	2.2663

25.2.6 Summary

Inflatable space structures have been considered, experimented with, fairly developed and flown over several decades (Fig. 25-13). Other than short duration missions, they have not reached the stage where they can be used by mission engineers, though they have been included in numerous future mission concepts. Quite possibly, these concepts will not be achieved without inflatable structural elements being employed, unless launch vehicle prices decrease considerably, or robotic assembly in space becomes fully developed and cost efficient. The reason for this avoidance is mainly that inflatable materials developed to date have several issues associated with them and more development and flight experience is necessary before this class of structures reaches full maturity. However, what exists today can be used in cases where masses and stowage volumes are critical and there is no other way for the mission to occur, even for small spacecraft and cubesats (see for instance Fig. 25-12). This will also help the present and future space scientists to further their dreams of what now is considered impossible.

25.3 SmallSats

Rick Fleeter, *Brown University*

The idea that there is something countercultural about an initiative to make products smaller, lower cost, more reliable and more useful is the mystery that is at the core of the most transformative development in space since Apollo. Our world has embraced innovations in efficiency. We accept some compromises of inflight services for the availability of low cost airline tickets for local, national and international travel. Small, economical automobiles have always coexisted with large luxury models

Table 25-2, Fig. 25-11, Eq. 25-5◀

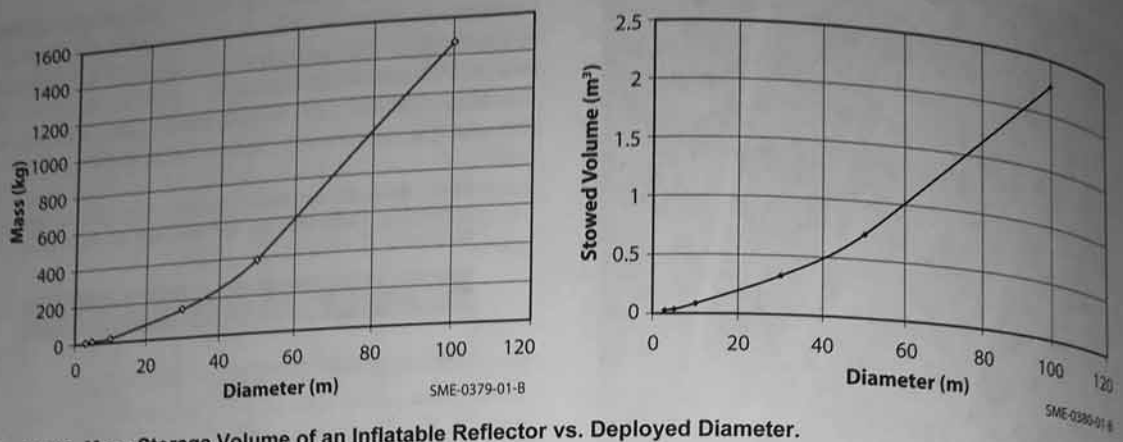


Fig. 25-12. Mass Storage Volume of an Inflatable Reflector vs. Deployed Diameter.



Fig. 25-13. A Deployed Membrane is Used as a Giant Star-occulter in this Concept of a Terrestrial Plane Finder. (Graphics courtesy of Princeton-Lockheed-Martin)

and SUVs without much apparent philosophical conflict. Nobody is disappointed that they can't sit in front of their iPod to have their long hair blown back as its powerful blast of sound washes over them. Big sound systems abound at concerts and in the home theaters of enthusiasts, though tiny personal MP3 players have dwarfed sales of those classical systems.

Since proposing in the mid 1980s that space reexamine the use of small, tightly focused, quickly developed, simple, low cost systems for those missions they can accomplish, there has emerged a myth that these are the indie, outlaw systems. That to accomplish better, cheaper, and faster is tantamount to magic. Impossible! Arthur Clarke wrote in 1962 "Any sufficiently advanced technology is indistinguishable from magic" [1962]. But isn't space where advanced technology is born?

25.3.1 We Were There First

While President Eisenhower's scientific advisors were assuring him of the physical impossibility of creating an artificial Earth orbiting satellite, the Soviet Union developed and launched the Sputnik spacecraft in October of 1957. Sputnik was a relatively heavy satellite at 84 kg, though 52 kg were batteries and most of the other 32 kg was a very heavy spherical shell structure. Besides its accomplishment of a single, simple, significant mission in space, transmission of radio signals at two frequencies to determine their detectability on Earth, Sputnik had several other features reinvented in subsequent small spacecraft. That heavy shell was a pressure vessel maintaining the mission electronics in compressed nitrogen environment fitted with fans to provide a more benign thermal environment for its radios, a method periodically employed since. Its 1 Watt radio power is the level still most often used for low Earth orbit small satellites, and its short mission lifetime was critical in keeping its design simple. The Sputnik Launch Vehicle has also become interesting again, a single stage rocket with strap on boosters, which was optimized for simplicity and low cost rather than high performance.

Released from the burden of its impossibility, American engineers succeeded, in January, 1958, in launching the first of the Explorer series of small satellites, with a mass of 14 kg about 6 kg of which were batteries. Pioneer also had a single mission, characterizing some basic elements of the space environment, and a short, 3 month operating life. With what little could be done with 6 kg of 1950s vintage electronics, it managed to detect the existence of regions of augmented radiation subsequently known as the Van Allen Radiation Belts.

Technological progress, the Space Race, defense applications of large rockets for weapons delivery and the desire to put humans in orbit and ultimately on the surface of the Moon, focused American and Soviet space activity on ever larger rockets and spacecraft. However during the Apollo period, roughly the 1960s, small satellites scored numerous space firsts including the first weather satellite, Tiros, weighing 122 kg and launched in 1962. The first geosynchronous communications satellite, Syncom, developed by Hughes (now Boeing) for $4M, weighed just 35 kg and began operations from a sub-GEO orbit on the day it was launched, February 14, 1963, though operations from GEO orbit awaited the launch later that year of Syncom 2.

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

25.3.2 Bigger is Better?

Weather, communications, and exploration, robotic and human, trace their roots to microspace, but all grew into much larger systems with greater capabilities including transporting humans into orbit and to the Moon, high resolution imagery across a range of wavelengths and atmospheric conditions, day and night, including the use of radar illumination, and probes capable of operating 10 to 20 years winding their way among the inner and outer planets sending back images and other physical data. The impressive results obtained in these first 3, classical, applications of space, for a while seemed the logical future for the field, their level of performance impossible from smaller platforms. Communications systems developed metrics such as the number of bits relayed to Earth per kilogram/cost of satellite which favored the development of ever larger platforms to realize the scale efficiencies both of larger space platforms and larger transportation systems.

25.3.3 Smaller is Better?

Space, compared with the pocket of your dress shirt, is quite large allowing more than ample room for satellites any size we care to make them. The Moon has roughly the volume of 250,000,000,000,000,000,000,000 (two hundred fifty billion trillion) iPhones, and still it is not crowding anybody particularly. Of what possible good is a small, less capable spacecraft compared with the miracles providing pinpoint navigation (GPS) constant imagery of weather and storm warnings (GOES and many others) and day/night all weather high resolution Earth imagery (Cosmo-Skymed) all of which accomplished via the major space systems which have now become considered our "classical" space technology?

The PC technology wars of the 1980s and early 1990s were never won, they just became irrelevant. The operating system running on your laptop doesn't really matter to the internet, and the world's major space agencies sponsor missions mostly according to their utility, not to support use of any one size class. Which carrier you choose for your cell phone no longer means much, and cars around the world offer similar safety, performance, and reliability. As technologies mature, they exit the product selection spotlight. Clients focus on utility. There is no "better", there is what takes you where you want to go today.

Applications

In the 1960s space had three applications: communications, remote sensing, and exploration. In the 1970s, the almost silent navigation revolution was affected by GPS. In the 1980s, first at the University of Surrey (Sec. 23.1), a 5th application emerged. The Cubesat (Sec. 25.4) pioneered by Bob Twiggs made widespread the use of satellites, exclusively small satellites, for education. More satellites are now developed annually for that application than the other 4 combined. Whereas space was once considered the field where new technology was born, it has now become their old age home. Large systems require such enormous investment, the fear of a failure has all but eliminated adoption of new technology. Small satellites have filled that niche and accommodate a 6th application, technology R&D.

Looking at another field once considered only the domain of techies, geeks, and nerds, but now in the hands of most of the planet's population, i.e., computing, our 7th application might be entertainment. And it is likely this latest of our applications spaces will happen in microspace, not because of any special technological advantage, but because entertainment is a speciality outside engineering. It happens when "the rest of us" start playing with spacecraft. Microspace being simpler, cheaper, and doable on a desktop adapts it better for experimentation by people who may not be experts in space engineering but who know how to engineer mystery, fantasy, thrills, to stimulate human emotion and to inspire us.

An example of the power of small satellites to put space in the hands of more people is the internationalization of spacecraft production. Though some countries may have aspirations to eventually develop major spacecraft, or may be customers and users of large satellites built abroad, the entry point for new national entrants in space is in small missions, often beginning in cooperation with a more experienced foreign partner:

- Malaysia developed the Tiungsat series initially with SSTL, progressing to the launch of Razaksat, a much more capable spacecraft
- Israel built not only series of small satellites as university projects starting with the Gurwin and Techsat series, and for communications (Amos) as well as small remote sensing satellites, but also the Shavit launch vehicle tailored for those missions
- Swedish Space Corporation (SSC) has been a leader in small satellite applications since the launch of the Viking in 1986 which had the dual missions of scientific research in upper atmospheric physics and the development of technologies and infrastructures for small space missions. SSC has successfully developed and launched 6 small satellites with 4 more in development with missions including geosynchronous servicing, GEO telecommunications, and formation flying
- Other countries now active in small space include Brazil, Korea, India, Kenya, South Africa, Singapore, Taiwan, and almost all the countries of Europe, many including Germany, Denmark, France, and Italy with strong academic programs
- The US played a leading role in small satellite applications in industrial realization of defense and science applications and in academics. Stanford and Cal Poly initiated the CubeSat architecture now adopted worldwide, first for academic experiments but now being exploited for science and military missions, often launched by the Peapod dispenser devel-

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

oped at the Naval Post Graduate School. American entrepreneurs developed numerous startup companies based on small and innovative space including Orbital, Defense Systems (now part of Orbital), AeroAstro (now part of Comtech), Andrews Space, SpaceDev and MicroSat Systems (both now part of Sierra Nevada), SpaceQuest, and Pumpkin

- Outside the US, Clydespace (Glasgow, Scotland) exemplifies the opportunities small space offers for entrepreneurs. There are many more entrepreneurial companies supporting the systems developers with services and subsystems tailored for smaller low cost missions

Budgets

It seems that it must have been easier to balance a household budget before the explosion in uses for money, including cable TV, digital cable, satellite, high speed internet, myriad cell phones, smart phones, computing platforms, plethora of cooking technologies (e.g., microwave, convection, sun tea) book readers, not to mention the items once considered luxuries if not unknown, color TV, refrigerators, cuisinarts, and the complete spectrum of shoes including aerobic, dress shoes that are actually running shoes in disguise, hiking grade sandals, and cowboy boots. The Soviets were right that we capitalists are overwhelmed with choice. And our governments are immune? That federal budgets do not orbit around national space initiatives is inevitable when we have to figure out how to pay for the right of every citizen to MRI and CAT imaging, quintuple bypass surgery, post high school education, a privileged position along or at least a local on ramp to the information superhighway, child care, elderly care, the interstate highway system and an air traffic system protected by a Department of Homeland Security.

That bureaucracy grows faster than budgets, possibly faster than anything else, means that the budget for any one activity must, as the second law of thermodynamics warns, diffuse to below the limits of detectability. There are just not the billion dollar allocations readily available to do single space missions, and not through any fault of classical system performance. The world has changed, and the ability to do meaningful missions, if not hugely elaborate ones, on the existing more frugal resources allocated to them, is the future.

An example of this trend is the development of small geosynchronous communications satellites, carrying 4 to 8 transponders instead of the 30 or more of major GEO comsats. Examples include Orbital's Star series with over 20 already operational and the Israeli Amos, with other small GEOs in various stages of development at SSTL (GMP-D), OHB-led Small GEO ARTES 11. Notwithstanding that the cost per transponder is higher for smaller platforms, they are the fastest growing GEO segment. They are ideally suited for clients with smaller markets or who are pioneering new services, who require a low cost additional service, and to enhance service availability and reliability.

This trend is mirrored in remote sensing, where innovative companies are successfully providing commercial imagery from smaller platforms. Examples include the OrbView, GeoEye, RapidEye, and the Ofeq series of imaging satellites. The German SAR-Lupe series of satellites have demonstrated space-based radar imaging from small platforms, though at 720 kg they stretch the definition of a small spacecraft except when compared with other radar imagers like the Canadian Radarsat (2,750 kg) and Italian Cosmo-Skymed (1,900 kg).

Time

Every aspect of life runs on a faster clock now. Few scientific questions remain relevant over the 20 year time scale typical of major space missions. Iridium was a good idea when it was conceived, but by the time the global telecomms network was launched, terrestrial cellular was serving most of its market. Politically, our leaders need results fast enough to help them be reelected. We get our music, books, and news instantly, and it is logical to expect the same from our space activity.

Rapid development is the focus of the Responsive Space initiative, developing spacecraft, ground and launch technologies and systems which can address emerging, often unexpected, requirements for communications and remote sensing. The TacSat series of satellites are designed to develop rapid response capability. But rapid development is also critical for educational applications compatible with engineering programs of a few years, and for technology demonstrations. The Los Alamos ALEXIS satellite was developed to test in orbit, within a few years, 2 new remote sensing technologies for eventual integration into major military systems.

Constituency

To sustain public or private sector support from an ever more distracted populous means reaching deeper into the public for support, but also for ideas. TV was envisioned as a means to bring classical culture (orchestral concerts and opera) to the public, and the early PC makers had no clue what the public would do with them—storing recipes was not, counter to their misguided albeit well-intended hopes, a killer app. Now we have the iPhone and the app store, with tens of thousands of reasons you cannot live without the technology in your hand while waiting for a bus, out with friends or facing a deadline in the office.

Space will survive without participation of the general public, along side high energy astrophysics, Egyptology and Orchid cultivation. But given space's potential to make human life better, in managing information, population, transportation, projecting life off our planet, and discovering if anyone else is out there, it is imperative we integrate the ideas of billions of humans, and that means putting the technology in the hands of non experts. Small satellites have driven the internationalization of space, the development of new applications of spacecraft including education, servicing, constellations, and technology development and demonstration, and participation in

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

space by students and amateurs which contribute to the growth in the space constituency. Companies like Pumpkin and SpaceQuest provide kits from which small satellites can be built. Amsat, the amateur radio satellite corporation, has a nearly 50 year history of launching small satellites built by amateurs.

25.3.4 Why Small is Different

Small Satellite Engineering

As satellites become smaller, their design considerations radically change. Their structural elements tend to be oversized compared with strength considerations just to make them physically manageable, and to be able to attach components to them. Structural mass is relatively insignificant, so aluminum has remained the widespread material, the advantages of composites not being worth their practical complications.

Coupled with their small size, the temperature gradient across a small satellite is typically not significant. Consequently, they do not employ heat pipes and other mechanisms and flight modes to equalize the heat load across the spacecraft. Because of their very large surface area to volume ratio, small satellites cannot easily manage internal temperature with electric heating and thermostats, hence these elements are largely absent, relying instead on coatings and insulation to minimize dark/light temperature variation. For these reasons, they typically do not require more complex finite element thermal and structural modeling. Also because their surface area is large and volume for containing payload elements small, physically smaller spacecraft tend not to need deployable solar panels to enhance collection area.

Probably the most significant engineering change caused by smaller scale is the reduced parts count. Fewer parts means the probability of a component failure is small. Redundancy, the first line defense against failure in major space systems, has limited or no role in smaller spacecraft. And because they tend to execute shorter duration missions at lower orbits, their choice of parts is not as restricted by the need for radiation hardness nor long life. More modern components typically have lower power consumption, smaller mass and volume, lower cost, and are simpler to work with, further lowering complexity which ultimately is the enemy of reliability. Despite their often being developed by students or amateur teams with limited training and experience, very small satellite reliability is comparable with major space systems thanks to their simplicity. It is easier to make a reliable flashlight than a reliable jet transport aircraft for carrying 400 passengers from Boston to Rome nonstop. Hence, the quality of the components used in a flashlight need not compare with those in a jet transport aircraft.

Limited to smaller instruments, microsatellites collect less data. And free to use the most modern, energy efficient, highest performance components, they have disproportional data processing horsepower which can be used to compress data. Thus the amount of information they need to downlink is vastly lower than major systems. Coupled with their typically low orbit altitudes, their transmit power levels can be very low, and yet directional antennas are rarely used. Smaller observing apertures imply lower spacial resolution, so tight pointing requirements are rarely necessary, simplifying the ACS. Power systems producing a few Watts for a year resemble, very little; a 10 kW power system built to last 20 years in an intense radiation environment with a 99.9% end-of-life reliability.

The designer more accustomed to larger spacecraft has to learn that power losses previously considered insignificant must now be considered carefully. Power management is much more critical with a 2 to 20 W system power budget than at 20 kW.

The use of terrestrial components is often mentioned in connection with small satellites. This is true because of the more benign radiation environments of low orbits, their shorter lifetimes and the reduced parts count which allows use of potentially less reliable components to attain a given level of systems reliability. However, the use of terrestrial components carries with it other potential complications. A part which has proven reliable on Earth, but which requires modifications to operate on orbit, can be compromised. Each terrestrial part must be analyzed for its suitability in space, considering among other factors outgassing, zero gravity, thermal excursions, radiation tolerance, fault tolerance and correction, mounting for higher g-loads, magnetic properties, and autonomy. Typically, selected terrestrial components will be tested to ensure their compatibility with the space environment, to the extent possible on a small mission budget. Ultimately, the use of terrestrial components is made on a case by case basis trading off the performance, cost, and schedule advantages they offer with the resources required for their qualification.

The design process in any complex system is highly iterative, but a typical starting point for a small spacecraft is the payload power requirement since, in general, the spacecraft will scale with the power required. It is critical to work with payload suppliers to scrutinize the uses of power and minimize them, through more efficient design and through limiting instrument duty cycle. And it is critical to maximize power production, choosing orbits and attitudes well suited to production of solar energy.

Once a power budget is defined, the spacecraft size can be estimated based on surface area requirements and solar panel efficiency. Battery volume and mass will derive directly from the power requirement in umbra. Using these anchor points the designer can proceed to a trial mass budget based usually on assumption of an aluminum structure, the mass of the electric power system, and specifications of the payload and other components which might be available from vendors (e.g., radios, attitude control sensors). Often, the small satellite launch accommodation is a fixed mass and volume, and there is no savings derived from minimizing scale below the size of the accommodation. Thus, some inefficiency in structural or component mass may be appropriate to minimize cost, development time, and risk and complexity.

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

25.3.5 Small Satellite Project Management

Every on-board system, and the methods for their design and development are affected by scale; the team that designs and develops a small satellite is also different. It is smaller, 6, 10, or 15 people, versus the often thousands of people who must somehow be managed in contributing to a major program. It is the size of the team more than any other parameter which determines that the product can be managed as a microspace project. An organization of 15 or fewer people creating a system in 2 years or less is fundamentally different than a large program:

- Every person in the program communicates with every other team member. Optimizations, in use of on-board resources (power, mass, volume, viewing angle, computing power, downlink bandwidth), and in use of personnel, are global instead of limited to a subsystem development team which works to a set of interfaces fixed during the systems engineering and planning process

- Changes are made dynamically since every person on the team can assess the impact of a change on the subsystems within their responsibility, contrasted with a large program in which requirements must be respected at almost any cost, because it is too difficult to communicate potential changes to each program subelement, and too time consuming for them to continuously assess change impacts. Moreover, most if not all the team members in a small program understand the product at the system level and can understand the impacts of changes on the overall system viability and performance

- This ability to manage changes is at the heart of the ability to use non-standard parts. Each team member participates in the process of ensuring that a novel component will meet their particular requirements. As with changes, inquiring each member of a team of hundreds or thousands, even if each one did understand the overall system, would be impossible, especially given the much larger number of such requests which would arise if the program did not adhere to a standardized parts criterion set and list

- Documentation is reduced to only that which is produced in the process of engineering. Most of the information on how the system is designed and integrated is shared among the team members, rather than codified in writing and diagrams. In fact, despite the documentation created in larger systems, reliability and manufacturability still hinges critically on staff communications and continuity. But in a team of a dozen people engaged for less than 2 years, the turnover in personnel is zero or very small compared with a team of hundreds or thousands engaged for one to two decades

- The web-like architecture of a small team is protected from loss of information associated with the loss of one of its members because each person participates in the work of several others. Job descriptions are broad and dynamic. An analogy is a basketball team which despite having particular roles described for each player, can dynamically reorganize as the players and the ball move around the court to respond to the immediate state of play

- Rather than relying on formal reviews, the first line of error checking is enabled by the sharing of responsibilities. The lead communications engineer works closely with another team member who might be the lead on the power or information management system. This second person is in the best position to review the detailed design of the communications lead and discover errors or weak points. They're also ideally positioned to maintain the information on that systems design, philosophy, and implementation logic in the case of the loss of the lead. In that sense, every position has a backup who can step in, also in case the workload in a related area becomes excessive

Instead of formal reviews, which greatly increase the workload on the small team and may thus force exceeding the team size managable as a small project. Technical interchange meetings specific to subsystems, and at the systems level without delving into subsystems details, are held around a conference table with a few (2 or 3) outside reviewers and only those team members directly concerned using actual design documents, drawings, spreadsheets, simulations and calculations. Management should encourage these and rigorously maintain lists of action items which arise from them, as well as from working experience during the system development.

The dynamic, small group is a true team in the sense of basketball, football, soccer, or formula-1 racing, every person directly in communication with all the others and prepared to take on whatever roll the demands of the day require. This contrasts strongly with the more rigid structure required of large organizations, airlines, armies, and universities, for example, which are made of largely independent factions each focused on a part of the bigger mission of the organization.

Thus, it is key to the successful management of a microspace program to limit the scope and complexity of the project. Several notable failures, either on orbit or program cancellation due to apparent financial overruns, which in fact result when the small team becomes overwhelmed, are the result of trying to do a big job with 15 or fewer people.

In a world of finite resources, which is the world in which spacecraft are developed, money spent for one purpose is unavailable for all other purposes. The key to reliability is a diverse team including seasoned and fresh perspectives, tightly engaged in the project's detailed development, under experienced management supervision.

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

As money gets diverted for clean rooms, for special parts, for formal and largely useless reviews, the team is put under pressure—to do more for less.

A much larger useless consumer of resources is being conservative, which is anything but. Specification of a longer operating lifetime conserves nothing, but spends time in the impossible pursuit of proof that a system will endure. Lacking that proof, we are forced to create redundant architectures, reasoning (lacking proof or even analytical indication) that they ought to last longer than single string. But given their added cost and complexity, their more than doubled parts count, the probability of a design or operations error, the probability of a component failure, since there are more of them, while at the same time increasing the spacecraft cost and mass, which increases launch cost, resulting in a larger loss in case of the possibly greater likelihood of failure, the probability of loss can be greatly increased thanks to redundancy, and when the failure occurs, the amount of the loss will be greater. Which explains why redundancy is largely absent in highly reliable consumer products and even with only a few exceptions, in our automobiles, which are highly reliable, reasonably priced, and capable of doing great harm to us in case of a failure.

Specification of higher electrical power, to ensure adequate supply at end of life, higher radiation tolerance, wider thermal operating range, more g-force ruggedness —do these make systems more reliable? The opposite. They consume precious resources to create capabilities which are never used. They add cost, complexity, bureaucracy, documentation, and lead time. They decrease the choices available to the designer. They make solar panels unnecessarily large. If a deployable array becomes necessary to create additional power, and it fails to open or is too flexible to allow the attitude control system to function properly, the excess of power that would theoretically have been available at the end of a mission whose lifetime was spec'ed way beyond the actual desires of its clients' attention span, can hardly be considered a wise investment.

Not only the members of the technical team have to have the flexibility to assume various roles. Management must also be active in systems engineering beginning in requirements definition to contain complexity against the natural pressures, to add features, to get more return on the project investment, even to add "reliability" through increased complexity, which in fact lowers reliability by taxing already scarce resources. Every requirement needs to be scrutinized and minimized. If the program is not comfortably achieved by 10 or 12 people, and thus expanded to a team of about 15 including some margin for surges in workload, then the program is at best a hybrid with more conventional space programs and must be managed more formally, which will immediately double the team size and require different systems and management architectures, more documentation, more bureacracy, more meetings, and a lot more money and time. If pressed to stuff what is not a small program into a small program package—of limited time, money and team size with too much complexity to easily handle in a team of a dozen people, I advise: a) attempting to renegotiate requirements and restructure the program and the spacecraft so that they two are in harmony and failing that and b) resigning management of the program. Beware avarice and greed. I have never regretted a tough negotiation to create a project do-able by the small team, but I will always regret the attempts to go along to get along resulting in at best a burned out team and at worst a failed mission. No money is saved doing a failed mission at a bargain price.

The facilities used by the team should also be considered in the program's management. Small satellites are generally not built in cleanrooms because of the additional work and time imposed on the team for maintenance of a clean environment. But also because the components, design and manufacturing methods used in a small satellite should be robust. Trimming structural mass to an absolute minimum costs a lot of time and money, results in a fragile structure more costly to test, more likely to fail in test, which will create large temperature gradients among spacecraft elements in orbit, one much more difficult to protect over the development lifetime, and in any case is probably not helpful for a satellite of very small mass. Saving 400 kg in a 2000 kg spacecraft is, if possible significant. Saving 400 g in a 2 kg spacecraft, if it were even possible, probably offers no substantive advantage. The cost to launch those 400 g is not measurable. For all these reasons the small spacecraft is built to work in an industrial or academic laboratory benchtop environment, and its systems should function at 1 atmosphere and 1 g so that they can be tested with a minimum of special facilities.

Carry this concept forward. The attitude control system should be testable easily. The ground station should be developed alongside the spacecraft so the two can talk to each other, both to speed development avoiding last minute disconnects the first time the spacecraft "sees" the real ground station previously having been only subjected to a simulation, and because when launched there will already be many hours of experience in controlling the spacecraft through that link, greatly enhancing reliability. The power system should function in Earth Sun when the spacecraft is rolled out on a cart into a parking lot. These sorts of practices blur the line between development and test so that before even entering into formal integration and test, the team is confident in the functionality of the system.

Test

Rather than relying on detailed analysis, the small program emphasis is on testing and it is imperative on management to promptly execute the analysis, design, development, and integration phases to conserve time, personnel, and financial resources for testing. Most subsystems are only tested at the benchtop level, not in thermal vacuum chambers and other special facilities, because the entire microspacecraft is smaller and less complex than the typical conventional spacecraft subsys-

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

tem. Thus the manager accepts the risk that a particular subsystem might fail in systems test despite being bench-top tested, in exchange for the savings in time and money realized by moving to test sooner. The cost of a subsystems failure in systems test is in any case not great, as de-integration of a small, simple spacecraft to unearth a failed component is not so difficult.

A famous scientist once said that an expert is a person who has made every possible error in a particular field. Similarly a reliable system is one which has already failed in every way it is capable of. Small satellite teams are less bureaucratic and should be less afraid to bring to light and discuss errors, rather than attempting to push them down the line. Not to mention that there is no line for them to be pushed down. The team stays with their spacecraft from concept through launch.

Thus the natural objective of the project team—in development, in integration, and test, is to find and figure out failures. Reliability is the result of having used the system so much in so many situations that we have found and fixed every possible way it could fail and then to have used each failure to improve reliability. It is ironic that the ISO series of quality standards has become too complex and bureaucratic for certification of very small teams and organizations, yet the core of the ISO philosophy, to bring failures to light, learn from them, and continuously improve the product as a result, is epitomized by these small teams and their smaller, less complex products.

Small missions are short duration and, in fact, all but a few operational satellites long outlive their usefulness on orbit. Reliability 10 years after launch has very little value compared with the working on day one and for the first few weeks or maybe months. Hence, risk wearing it out. The canonical philosophical argument is whether, after 6, 12, or 18 months of integration and test operations, to replace the old rechargeable batteries with new ones. The answer is simple NO! The probability that batteries that have been working fine for a year will fail in the near term is small. New batteries are unknown and their only advantage is the possibility of adding a year at the end of the mission life, maybe. Versus the real risks of the disconnecting and reconnecting, disassembling and reassembling, and the gremlins they can release, problems which might be introduced. Plus, there is the possibility of damage, an emergency order for a replacement part that may or may not arrive and be as reliable, too late for adequate testing (and probably no money left either). And sometimes damage is not realized—electrostatic discharge, a momentary short circuit that weakens a component. Who knows if these new batteries are really better—at least the old ones have proven reliable, they have heritage.

Nothing works the first time, and often not the second, so we have to find those early failures. The result is the 1,000 hour test in which the spacecraft must operate for 1,000 hours in complete autonomy, with no tweeks by anybody. After all, we are expecting it to operate in space for tens of thousands of hours. A 1000 hour test takes time, especially since most spacecraft fail the first, second, even third attempt. Which tells us that the test is doing its job and worth the time which it requires.

Thermal vacuum and vibration testing is widely misunderstood as a simulation of the space environment. This is only partially possible on Earth, expensive and unnecessary. Simulation of the space environment is done in computer models, and the role of testing is to verify the fidelity of those models. Certainly the tests are designed to approximate the types and levels of stress the system will see in space, and it must demonstrate in test that the system continues to function or in the case of launch loads will function after. But the key is to test that our models of how it works, models of structural rigidity, of thermal behavior, of battery charge / discharge, are accurate over the full range of flight environments. Those include not just temperature and vacuum and vibration, but also the environment of having to survive and perform functions even when not in contact with the ground and to not require reset so often that no productive work gets done.

The system should never be asked to do what it was not designed to do. Exceeding a temperature limit in test is to willfully weaken the product. That too is not conservative. Neither is discharging the batteries to zero more than maybe once to ensure the power system can recover from that peculiar state. Do not do things because they feel good—do things because they are rationally designed to ensure operation on orbit, meaning to uncover problems that might arise in the operational space environment. Which also includes being operated by people other than those who built the spacecraft, a very common cause of failure usually not considered in the test plan.

The Role of Analysis

Just as it is the role of test to verify analysis, it is the role of analysis only to support functionality. It is relatively easy to wage a war against reviews, overspecification and paperwork because the development crews don't like them. But who fights against over-analysis. The client is used to seeing a 4,000 node thermal model, and sitting in front of a computer playing with fancy software that builds models beats working even on a good day. We all want to change the world while plugged into an iPod while staring at a 27" hi-res monitor. You are in fact changing the world—for the worse. Because once again, resources are being spent without contributing to mission success, hence they are decreasing that probability.

Small satellites are called that because they are, dimensionally, small, and coupled with their usually heavy construction, normally of aluminum, they don't support a thermal gradient. They are a short circuit. And in LEO, they alternate rapidly between sunlight and umbra, which tends to minimize temperature excursions. They do not have enough battery power to heat themselves up much. And yet, we are complicit with those who would bureaucratize these programs to death in creating complex models of their thermal, and also mechanical behavior.

There is no substitute for the thinking process. Create a small spreadsheet to look at the gross fluxes in and out of heat. Scan the design for specific components that might be stressed. You would probably miss them in the 4,000 node model anyway. Getting an intuitive feel for the thermal behavior is an absolute must starting point. If that behavior is confirmed in test, that job is finished—leaving room for in depth study and design of any critical points, of which there are likely none. And small thermal problems are usually eradicated with a little extra aluminum and thermally conductive RTV at the site. Don't fix what ain't broke.

Before moving on to mission examples, two last cautions.

1. **Beware efficiency.** Nothing consumers buy is selected for efficiency, whatever the ads and consumer reports say. All cars are highly inefficient compared with buses and trains, not to mention walking and bicycling. But we use them to the exclusion of all other transportation modes. Single family houses with the best heat pumps and insulation are much less efficient than multi-family dwellings. The most eco-friendly farmer in Vermont is sucking down much more energy than the average New Yorker. So whither goeth efficiency?

   It is sacrificed to utility. Nobody will reward you for maximizing the number of bits for each dollar paid for a mission. The user wants a certain capability in space, or scientific result, for a certain amount of money. Leave maximum bits per dollar or minimum dollars per pound to big missions with economies of scale. And that means? Don't add capability because you can, because the electric power appears to be available and there's mass margin. That is the siren of efficiency beckoning you onto the shoals. You might need those margins, you might need the money spent on them later if something goes wrong. New features are always more trouble than we thought they would be. That is Murphy's Law. *Alles nicht so einfach*: It's never quite so simple as it seems. Sometimes the best thing you can do for mission success is to turn off the lights, lock the door, go home and say "it's enough." That is a valid and necessary and difficult management function—to take responsibility to not do more and instead conserve resources for another day.

2. **There are no cost-optimized missions**
   Space missions are mostly not done for the sake of their mission objectives. The list of agendas imposed on the mission is, while individual to each mission, at the same time shared by all space activity. They include:

   - Spreading work through multiple political districts to garner wide support
   - International cooperation which protects missions from national whimsy
   - Education and the infection of the next generation with a zest for space
   - The use of, or development of new technologies, some of which must be shown to spin off to terrestrial application
   - Completion on a timeline compatible with re-election of government supporters
   - Making money: on the part of agencies, contractors, universities, service providers
   - Television: something about the mission must play to the media. Otherwise it is perceived as a very expensive tree felled in a distant forest
   - National security and technological superiority —or the projection of that possibility
   - "Made in"—use of domestic suppliers despite cost and performance inferiority
   - Use of a particular launch vehicle, suitable or not
   - Compliance with myriad federal regulations for safety, contracting, and program execution including purchasing
   - The career aspirations of the program's leadership
   - Documentation and other costs associating with removing blame for any failure from the project's management or sponsors

Rarely are any of these multiple optimizations mentioned when the call goes out for a minimum cost mission. Cost suppression is naturally assumed to be the goal of all human activity at least in the US, the birthplace of Walmart, Swag and Free Downloads. But all those other agendas are always present and consume a much larger proportion of the budget of a small mission than a large one. If you build the program team and plan for minimum cost, you will find yourself helplessly understaffed, behind schedule and over budget trying to also accommodate the *sub rosa* requirements which are the true motivations for your mission.

25.3.6 SmallSat Mission Examples

There are patterns repeated in all microspace missions:

- A small number of payloads. These are not omnibus missions—they at root are single purpose machines built to do one thing well, simplifying their design, their requirements for thermal control, for pointing, and power. Every system benefits from not having to support a large number of payload elements and operational modes

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

- A small development team. Only possible if the mission scope is limited, a group whose members can all meet conveniently and often without consuming much of their workday, and can easily share the full spectrum of responsibilities to balance workload among everyone is critical. Self management is only possible for teams up to 12, maximum 20 people, above which size some form of stratification has to be instituted, which blocks the free flow of information and the ability of individuals to help where help is needed. The group members of a small team may be specialists but they are also generalists. Ideally, device builders also write their own software, and are as comfortable soldering as analyzing circuit stability or thermal profiles
- Short development time: to stretch over a decade or more requires detailed documentation, the ability to hand the project off through a series of generations of staffing and departments, multiple adjustments, and additions to the mission and technical requirements, all of which are counter to achievement of low cost. Just keeping a program management function in place for a few thousand days can cost more than an entire small spacecraft program
- Low cost launch: A launch that costs $20M to $50M is not compatible with a spacecraft developed for 1% of that amount. Logic dictates expanding the investment in the spacecraft. Even if 10× or 100× the amount is ultimately spent on the payload, the mission cost including ground segment and launch less than doubles, while the perceived utility and reliability are increased (by some metric) much more
- Low cost operations: The expression "low cost satellite" says it all—we focus on the cost of the item to be placed in orbit, often to the exclusion of considering operational costs. The two major ones being launch ops and on-orbit ops. Sending 15 people to Kourou or Kwajalein for a planned two weeks which grows, possibly because of problems with the weather, your or another payload, the rocket or other contingencies affecting the launch site, to two months, can cost more than the spacecraft. If it needs to be repacked, shipped home, re-assembled, stored, retested, reshipped, re-re-assembled and another campaign of several weeks accomplished, the development cost can be lost in the noise, not to mention the many decisions agonized over to save a few dollars early on. The design should allow one person to handle launch ops, then you can send two, maximum, to have a backup

More expensive still can be on-orbit costs. The first month everyone will want to be in on every contact, every meeting, every decision that might affect mission success. There is no alternative, so you can assume virtually the full crew on the payroll, and if the ground station is remote, all their travel, for that month. But then if you are unlucky, the damned thing will run on orbit year after year after year, sending back all sorts of data, beloved by a constituency you can't afford to say no to. A couple of employees staffing a ground station, plus the station's care and feeding for a few years is comparable in cost to the development program. Can't say no? Maybe, but if the spacecraft is designed like modern consumer devices (iPhone) the customer can and should operate it. The money is spent, but if it's someone else's money you don't care. Your experts are on call if there are problems. But design the software to be idiot proof, so that the clients don't break it. Cuisinarts, chain saws, automobiles, hairdryers—are all potentially lethal. Their manufacturers make sure the users and the appliance both survive their interactions. Your design should do no less.

Spacecraft are with few exceptions custom built for particular missions and each one follows its own path from concept to design to development and test to launch and operation. But looking at one or two individual cases in detail one can see how many of the ideas about small missions translate into real programs. This is done in the associated web article.



25.3.7 The Future

How can we extrapolate to the future a five decade history of special cases? One way is to expect a continuing series of special cases, small experiments, niche missions, which do not fit the classical space model of 10+ year time scale development cycles, proven and stabile technologies, great complexity, and high performance coupled with astronomical pricing. In fact, as the pace of technology and science accelerates, many questions cannot wait decades for answers. Responses to new threats have to be addressed this year, not in 20 years.

Another is to look at technological progress. Today's satellite of a few kilograms can accomplish that which required several hundred kilograms a decade or two ago. Thus we can look at missions now done by medium sized spacecraft with high resolution remote sensing, for example.

But many missions, particularly remote sensing, require large apertures, and it may be in that single area that many future missions will be determined. At this writing, the idea of creating a synthetic small aperture from a constellation or cloud of small satellites is much discussed but not yet realized. The obstacles are largely in the realm of communications and information—the exact position and orientation of each of the micro or nanosatellites constituting the network. And with the explosive progress in information processing and communications characteristic of today's terrestrial networks, it seems a safe prediction that these issues will be resolved for these orbital networks. Using a large number of small, low cost satellites will provide numerous advantages for remote sensing missions, optical, radar, and lidar. Arrays can be large along and across orbit track, providing high resolution and wide field of view.

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

Design Guidelines for Small Spacecraft

Power scales everything. **Do** minimize it:

- After 6 months is full power needed all the time
- Scrutinize all circuits for bleed resistors, voltage dividers, other
- Designs that don't matter on the ground but waste power
- Does everything need to be left on all the time for "stability"
  - A thermostated heater might save many electrons
- Chose an orbit and attitude that makes power production efficient
- High efficiency (GaAs) solar panels are almost always worth their cost
- **Do not** test to simulate space and launch. Test to test your models that predict behavior in space and launch
- **Do** pass a 1,000 hour operational test with no cheating (helping the spacecraft keep working)
- **Do not** waste money on proving long lifetime and other intangibles. Spend it on test and on adequate staffing and facilities—tangibles which absolutely add to mission success probability
- **Do** test all deployables in Earth atmosphere and gravity on the bench top
- **Do not** take every box through thermal vac and vibe. Test them on benchtop, integrate, and take the whole spacecraft through formal test
- **Do** test the spacecraft rolled out in the parking lot on a sunny day, radios, solar panels, everything
- Assume clean rooms **do not** exist and go from there. Use them at the end of the mission for PR photos and to keep people from touching (breaking) your satellite
- **Do** create and use the FIST: Fully Integrated System Test should be formalized so that on every rebuild you can assure that the spacecraft is still what it used to be
- **Do not** switch out anything late in the program —particularly to have "fresh" batteries
- **Do** only those tests necessary to prove it is what you designed and works right
- **Do not** test beyond reasonably expected environmental conditions or exposure durations
- **Do** use advanced technologies if you can be reasonably confident they will work in your application and environment when they simplify your design. Complexity is more dangerous than a well thought out application of a terrestrial part
- **Do not** assume that a modified terrestrial part will have the reliability of the unmodified part in terrestrial use for which it was designed. Once you change it, it's a new part
- **Do** overstaff. You always need more people than you thought, and it is impossible to train new staff when you are up to your ears in alligators, and in a small program one or two extra people can be the difference
- **Do not** assume that your job is done with successful turn on on-orbit. If the customer then breaks it, it's still your fault. Stay with the operations team forever, or almost forever
- **Do** build LEDs into the spacecraft to show you immediately what is on and what is off. Avoid running down your batteries to 0 charge not realizing the spacecraft was on. Leave them there forever
- **Do not** accept any "free" features, extra capabilities, payloads. if it's not in the contract, it's not on the spacecraft
- **Do** use your own GS to avoid interfacing with huge bureaucracies that operate huge ground stations with huge gain that you hugely want. Temptation leads to ruination. With your own GS you are able to operate and test the satellite every day of its development life in its real environment with its real ground station
- **Do not** create documentation for meetings, and do not waste time preparing for and traveling to and from meetings
- **Do** instead save all communications, design and other documentation developed in the course of engineering. Plus use photography and videos to document the as built item and how it goes together and comes apart

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

Field Guide to Small Spacecraft

To seem a good steward of someone else's money, to appear modern, to hope to be able to actually keep track of what's going on in a program, to avoid dealing with the real complexity some missions entail, the labels Small, Low Cost, and Micro are attached to many spacecraft programs. No real spacecraft program is without compromise but here are some features characteristic of small missions:

- One simple, well defined mission, not everything to everybody
- Cost limits are serious enough that the program manager can redefine requirements and mission goals to stay within them
- System designers interact with and negotiate with mission clients in definition of all requirements and the way in which the mission might be realized
- Resulting in a program reasonably accomplished by 12 people in 24 months
- 12 people all of whom communicate and share work without restriction
- Availability of a launch accommodation of cost similar to that of the spacecraft
- Risk is related mostly to design and fabrication errors, not parts failures
- The only required documentation is that produced in the engineering and test process, plus an interface requirements document
- Reviews are small, focused exchanges of technical information among specialists and one or two systems engineers and managers
- A mission failure will not result in: 1) a phone call from the President of your country; 2) anyone losing their job or being unpromotable for the remainder of their career
- No magic: no technologies or devices that are only hoped to exist or function are critical to the development program nor to the mission's success

The networks are highly robust to failure since the loss of any one or even a few members would cost a small percentage of system performance. This also makes them much less vulnerable to attack. The systems can be easily upgraded by launch of more members, or by launch of members with different capabilities—other sensors or more capability in downlinking, for example.

It is risky to predict based on a notion that the things we do in space will remain constant, just the boxes will shrink. After all, the Big Three space applications is now the Big Six or Seven. This was an error in thinking of the computer industry, which was able to transition from mainframe to minimainframe, but did not understand the importance of the internet, gaming and other forms of digital entertainment, social networking, and particularly communications and virtual presence, which is now available in bits and pieces with Skype and Google's Street View, which will likely expand to viewing and real time presence being integrated into a more powerful form of virtual presence.

The old computing applications are still important—complex calculations for science and engineering, spreadsheets, word processing, and page layout (the latter 3 completely unknown in the 1970s). But the newest and smallest platforms, like the iPad, lack even a keyboard because most of the newer applications are so completely different.

What might be these new applications to which micro and nanospace are ideally suited, as an iPad is suited to video presence, music, and image sharing?

One that is already well established is responsive space—the ability to launch relatively simple space capability on demand. More rapid changes on the ground demand more responsive space systems, and given the development time for major programs, responsiveness will be addressed by small systems and missions. Another is robotic presence. Conventional space was scaled for human spaceflight—the Saturn V and the Shuttle. But much of the science of today and tomorrow is beyond the reach, financially, physiologically, and physically, of humans—landing on the moons of other planets, for instance. The robots which will do it can be as small as technology permits, and their size and mass will continue to shrink. The future of space science may increasingly be in very small, albeit costly and complex, space missions which have little resemblance to the mainframes of yesterday and today.

Looking at one other unexpected development in the computing world, from which space draws so many technologies, in communications, in information processing, but also in for example power management as computing becomes increasingly personal and portable and drives development of new batteries, for example, smaller, less complex satellites put space in the hands of the non-experts who do not "know" how space is used. It was these non-experts who created video games, an industry larger than the film industry.

Non-experts write and read digital books, create and watch Youtube videos. Similarly, there is reason to believe that the next space killer app will be one thought of by students working on a CubeSat who see it not as a space platform but as an entertainment or advertising platform, as a way to do zero gravity manufacturing or agriculture. The non space community is about 100,000 times larger than the space community, and if space is accessible to it, 100,000 times more likely to invent an application we too accustomed to what we think we know that space can do for us can imagine.

We space experts like to say that space is that ultimate frontier, but the integration of the rest of humanity into the space community, possible through small, low cost, simple space systems, may be the route into that new frontier. Our job may be not to find the future of space, but to make it possible for the rest of Earth's people to do that job for us.

Table 25-2◀, Fig. 25-13◀, Eq. 25-5◀

25.4 CubeSats

Robert Twiggs and Benjamin Malphrus,
Morehead State University

The CubeSat was developed to allow people with little or no experience in the design of space missions to start with an open mind to incorporate new ideas and concepts into designs and missions that have no historical restrictions. CubeSat programs are often university student projects—the challenge therefore, is to find ways to accomplish the mission at the lowest cost. Trying these new ideas and concepts inherently includes a risk of mission failure. Failure here is an acceptable option because without trying new things, ideas, processes, and techniques, and without pushing the boundaries, new innovations would not occur. The excitement and challenge of having an opportunity for the new students to do "real" things in space exploration is a motivational experience that cannot be achieved by lectures in the classroom and represents education at its best.

25.4.1 History of CubeSats

I (Twiggs) developed the concept of the CubeSat at the Department of Aeronautics and Astronautics at Stanford University in collaboration with Jordi Puig-Suari at the Aerospace Department in California State Polytechnic University in late 1999. The CubeSat concept originated with the spacecraft OPAL (Orbiting Picosat Automated Launcher), a 23 kg microsatellite developed by students at Stanford University and The Aerospace Corporation in El Segundo, CA to demonstrate the validity and functionality of picosatellites and the concept of launching picosatellites and other small satellites on-orbit from a larger satellite system. As defined in Sec. 2.1.8, picosatellites have a weight between 0.1 kg and 1.0 kg. OPAL is shown in Fig. 25-14, with four launcher tubes for deploying picosatellites. One of the picosatellites is shown being inserted into the launcher tube in Fig. 25-15.

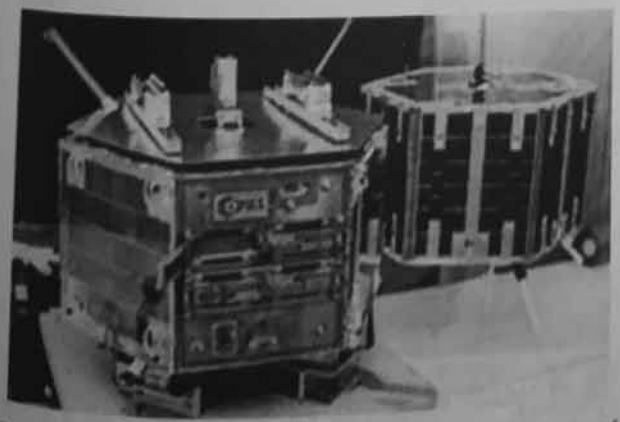


Fig. 25-14. OPAL & SAPHIRE Microsatellites. (Courtesy of Stanford University)



Fig. 25-15. Picosatellite Loaded into OPAL. (Courtesy of Stanford University)

The satellites developed by students within university programs during the 1980s and 1990s were all nanosatellites (1 kg–10 kg size) and microsatellites (10 kg–50 kg size). The feasibility of independently funding launch opportunities for these nanosatellites and microsatellites was limited as the costs typically were in the $250,000 range—a price point well beyond the resources available to most university programs. At that time, the only available option was to collaborate with government organizations that would provide the launch. The OPAL satellite was launched in early 2000 by the US Air Force Space Test Program (STP) with sponsorship from the Defense Advanced Research Projects Agency (DARPA) for The Aerospace Corporation picosatellites.

The OPAL mission represented a significant milestone achievement in the evolution of small satellites by accomplishing two milestones: 1) providing proof of viability of the concept of the picosatellite and 2) initiating the introduction and development of an innovative orbital deployment system. The picosatellite launcher concept used for the OPAL mission, in particular, represented a major advancement that would enable the technological evolution of small satellites—setting the stage for the development of the CubeSat form factor and the P-POD orbital deployer system. OPAL demonstrated a new capability with the design of an orbital deployer that could launch numerous very small satellites contained within the launcher tube, simplifying the mechanical interface to the upper stage of the launch vehicle and greatly simplifying the satellite ejection system. While the OPAL mission was extremely successful and established the validity of a picosatellite orbital deployer, we sought to find a lower cost means of launching student-built satellites. The stage was set for the development of the CubeSat form factor and its evolution toward an engineering standard.

CubeSat Engineering Design Standard

The primary intent of the development of the CubeSat was to provide a standard set of dimensions for the picosatellite's external physical structure. The goal was to make it compatible with a standardized orbital deployer, which then could be adapted to interface with a wide

Table 25-2◀, Fig. 25-15, Eq. 25-5◀

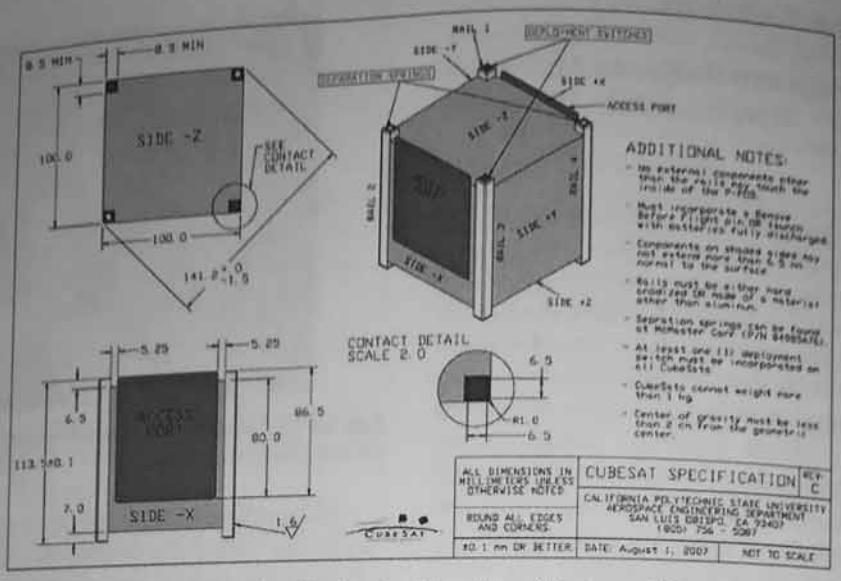


Fig. 25-16. Standard for CubeSat Mechanical Structure. (Courtesy of Cal Poly, SLO)

variety of launch vehicles. Unlike the development of most modern engineering standards, we (Twiggs and Puig-Suari) did not consult with other universities or with the commercial satellite industry to establish this standard because most other university satellite programs and commercial ventures were concentrating on larger satellites rather than smaller ones. There were discussions in the late 1990s within the AMSAT community in the United Kingdom centered on building a small amateur satellite but there were never any attempts to develop a standardized design.

The concept of a design standard for a picosatellite and associated launcher would lead to a large number of picosatellites being launched on a single launch vehicle. We envisioned launch vehicles accommodating several launcher tubes, each containing a few picosatellites. This feature, combined with the system's adaptability to essentially any launch vehicle, would greatly increase the opportunities for flights of university picosatellites.

The final concept of the CubeSat structural standard we developed and currently adopted by the small satellite community is shown in Fig. 25-16. They believed that if one organization could provide the integration of the launcher with the launch vehicle through a carefully orchestrated interface process with the launch services provider, then it seemed possible to acquire launch opportunities for university programs that would be affordable (less than $50,000 per 1 kg satellite).

The physical standard that resulted is based on a $10 \times 10 \times 10$ cm cube that has a volume of exactly one liter and has a mass of 1.33 kg or less [Munakata et al., 2009]. Initially, a standard of 1 kg was defined by Twiggs, et al., but NASA and Cal Poly have recently adopted a standard of 1.33 kg. The standard $10 \times 10 \times 10$ cm CubeSat is referred to as a one unit "1U" CubeSat and is scalable in 1U increments. The current widely used orbital deployer can accommodate up to a 3U CubeSat, allowing satellite developers the flexibility of producing 1U, 2U, or 3U CubeSats. 2U and 3U CubeSats are required to have the same mass and approximate centers of gravity, (CG) that is 2.66 kg and CG in the center of the structure for a 2U and 3.99 kg and CG in the center of the structure for a 3U system. This standardization requirement facilitates the use of a single orbital deployer design for a variety of combinations of 1–3U Cubes. This scalability permits inherent flexibility for developers to incorporate a wide variety of payloads and has led to tremendous appeal of the form factor to both university and commercial projects. The commonly used 1U, 2U, and 3U systems are illustrated in Fig. 25-17.

Standardized Orbital Deployer—the P-POD

The standardization of the CubeSat design facilitates a wide array of designs to be launched and deployed using a common deployment system. The most widely

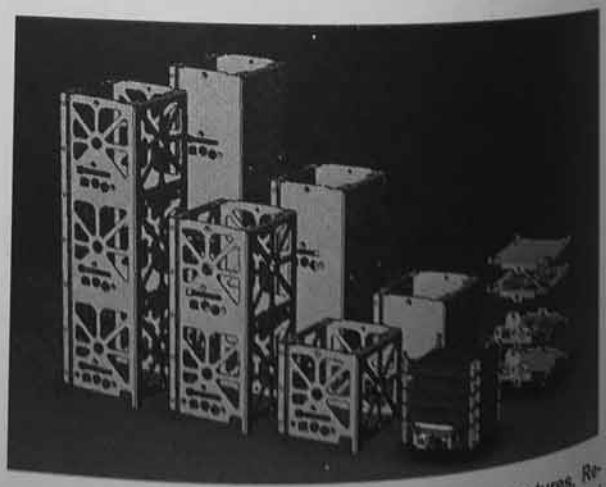


Fig. 25-17. Typical 3U, 2U, and 1U CubeSat Structures, Respectively. The drawing on the far right illustrates a typical stacked configuration for the subsystem boards. (Courtesy of Pumpkin, Inc.)

Table 25-2◀, Fig. 25-17, Eq. 25-5◀

used deployment system is the Poly-Picosatellite Orbital Deployer (P-POD). The Cal Poly P-POD system shown in Fig. 25-18 has been used to deploy over 90% of all CubeSats launched to date, and 100% of all CubeSats launched since 2007. The P-POD is a tubular, spring-loaded mechanism easily integrated into any launch vehicle. P-PODs are integrated into the upper stage of a launch vehicle and transport the CubeSats into orbit, deploying them after a signal is received from the launch vehicle. The P-POD is capable of carrying three standard CubeSats. It is made of anodized aluminum and has a mass of roughly 3 kg. Currently, there are two designs for the P-Pod—the Mk III single tube that holds three CubeSats, and a double tube that holds six CubeSats. Both have a square tube design that produces a reliable linear path resulting in a low satellite spin rate. The P-POD is a robust design that has significant flight heritage and is currently the deployer of choice of US launch service providers.

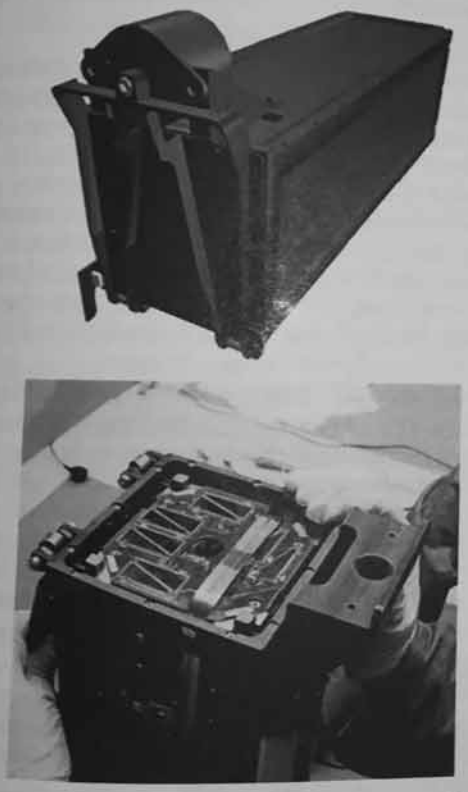


Fig. 25-18. P-POD Top (Courtesy of Cal Poly–SLO) **and CubeSat Installed in P-POD on Bottom.** (Courtesy of Kentucky Space)

Alternative deployers/separation systems to the P-POD have also been developed—notably the X-POD, the T-POD, and the SPL. The X-POD (eXperimental Push Out Deployer) is a CubeSat deployer designed and built at the University of Toronto Institute for Aerospace Studies/Space Flight Laboratory in Toronto, Canada. It was designed to be custom built for each satellite and may be tailored to satellites of different sizes ranging from a single CubeSat to larger nanosatellites. The X-POD version has an independent release mechanism for the spring deployer that also provides automatic feedback to indicate deployment has taken place. The T-POD (Tokyo Pico-satellite Orbital Deployer) developed by Tokyo Institute of Technology, Japan, was originally designed to deploy a single 1U CubeSat and was first used successfully in 2003. A 2U configuration of the T-POD was developed and used in 2006 by the same group. The SPL (Single Pico-Satellite Launcher) manufactured by Astrofein (Germany) is designed to deploy a single 1U CubeSat. The SPL's innovation will allow the separation velocity to be defined by the customer (1 m/s is the default velocity). The alternative CubeSat deployers each have innovations that appeal to specific CubeSat developers although the P-POD continues to be orbital deployer of preference.

25.4.2 The CubeSat Philosophy from a University Student Perspective

The CubeSat philosophy was to develop a design for a small satellite that could be built by students in a short 1–2 year period of time and launched at a low cost. For a university-level engineering program, the total cost of the project needed to be less than $100,000. The missions would be based on one of two concepts of operation: testing or providing flight heritage for new technologies, or supporting one or two scientific instruments as their primary mission payload. With these goals in mind, a CubeSat design can:

- Have minimal development costs by keeping the satellite small
  - Fewer solar cells – (high-efficiency, flight qualified cells are costly)
  - Limited number and types of payloads
  - Designed and completed in less time
  - Requires smaller and less expensive test facilities
  - Easier to transport than larger satellites
- Reduce launch costs
  - Less launch capacity required (lighter is cheaper)
  - Integrator acts as an intermediary with the launch provider and purchases launch capacity in quantity
- Protect the primary payload
  - Provides more reliable protection for the primary payload
  - Easier to attach the multiple individual adapters to the launch vehicle
- Reduce design choices
  - Fixed envelope simplifies the design choices for the developer
  - The payload must fit the bus, rather than the bus fit the payload

Table 25-2◀, Fig. 25-18, Eq. 25-5◀

- Develop a market for multiple vendors
  - Initially no commercial off-the-shelf (COTS) components were available for purchase—developers had to design or adapt components for their satellites
  - Multiple vendors now provide subsystems, components, and even development kits for CubeSat missions
  - Competition has reduced the price point of components
- Attract new vendors
  - Increase in availability of commercial parts lead to an increase in startup companies
  - Most initial component designs came from university environments—a free exchange of information between universities was encouraged

Most CubeSats have missions related to educating and training university students. The CubeSat form factor, however, is now widely used by the commercial space industry and defense departments worldwide. CubeSats have proven to be a cost effective form factor for getting payloads into orbit quickly and inexpensively.

25.4.3 Building CubeSats

For an inexperienced group of university students, starting a space program to undertake the design, manufacture, launch, and operation of a satellite can be as intimidating as a novice trying to climb Mount Everest. Some of the major considerations are provided here as a supplement to the details in other chapters.

Design

In designing a CubeSat (as with all satellites), the entire mission should be kept in mind. It is better to build, launch, and operate a Sputnik than to have worked on a highly complex satellite that never leaves the lab. The best approach for learning development at the university level is to keep it simple, get it launched and operated.

The design process leading to the development of a small satellite based on the CubeSat standard greatly reduces the scope of the design space that the satellite can encompass because of its size and weight. The total systems engineering process can be learned in a short period of time and then later applied to larger, more complex space systems. Some design considerations are constrained, regardless of the payload. The primary design constraints are:

1. The physical dimensions and weight are defined
2. With body mounted solar cells, the power generation capability is limited
3. The spacecraft orbit will be determined by the primary spacecraft orbit
4. The spacecraft will be protected in the launcher during launch
5. The communications systems' frequencies and data rates are limited

Environment—Design Differences

There are a number of special design considerations that are unique to designing a student built CubeSat. How are they the same or different from conventional larger satellites? Design differences between conventional space mission approaches and the CubeSat space mission approach are explored below.

Conventional Space Mission Approach

The conventional design approach to building spacecraft is to use components that are both space proven and radiation hardened. This leads to a significantly higher cost for design and production of the radiation-hardened components, since there is not a great demand.

Additionally, since it is expensive to undertake equivalent radiation testing or to find flight opportunities where components can be tested, there is a limit to the types of components that can be used for conventional space missions.

CubeSat Space Mission Approach

In most cases, the use of space proven and radiation-hardened components for university student CubeSats designs are not an option owing to their significant costs. Designing a CubeSat without this restriction leads to the expectation of a short operational life of one year or less. As a result, the designer can take advantage of new components being developed for personal electronics and trends like the iPad. Having this wide range of usable components does not mean that any component is acceptable for flight use, but that the trade space for judicious selection of components is broader.

The designer still needs to undertake trade studies before selecting of components to find those that may be more radiation-resistant due to the base technology used in their manufacture. Since the designer is now challenged to use other options to mitigate effects of radiation, options may involve using schemes to correct software when corrupted by single event upsets, for example, incorporating a watch dog timer to restart the processor if and when disruptions occur. Power systems can be designed to monitor for unexpected levels of current draw which occur in some components during latch up, then cycle the power to reset the device. Redundancy in critical components is also an option but is limited given the mass and volume constraints.

Orbit—Design Differences

Conventional Space Mission Approach

The mission requirements dictate the type of orbit required for commercial, scientific, and government (civilian and defense) space missions. This requirement along with the cost of the mission generally means that the spacecraft will be the primary payload, defining most CubeSats as secondary payloads. The launch vehicle provider then provides the optimum launch for the primary payload.

Table 25-2◀, Fig. 25-18◀, Eq. 25-5◀

CubeSat Space Mission Approach

Often, the university student CubeSat designer must begin the design of the space mission without any knowledge of the specific orbit in which the CubeSat will reside. However, the designer can select a range of orbits that will satisfy the CubeSat mission. This way, when launch opportunities arise the developers can adapt to a wide range of possible orbits thereby increasing the probability of being accepted as a secondary payload.

This approach for selecting the desired orbit can and has extended the launch time for CubeSats by many years. However, with the increased interest in CubeSat use by the science and government communities, there now appears to be more opportunities for university CubeSat launches.

Structures—Design Differences

Conventional Space Mission Approach

Large conventional spacecraft structures are generally custom-designed for each specific payload. These structures take a wide variety of approaches for example in size, payload accommodation, power generation, thermal control, and antenna placement.

CubeSat Space Mission Approach

The CubeSat structure, unlike conventional spacecraft, is already defined for the designer. See Fig. 25-16. The 1U, 2U, and 3U structures are standardized to fit in the Cal Poly P-POD launcher shown previously in Fig. 25-18. Structures conforming to this standard can either be purchased from a CubeSat component vendor or manufactured by the university student designer. These structures are generally made with aluminum sheet metal or machined from aluminum stock using the T-7075 material.

One advised requirement of the final structure made of aluminum is to have it treated with an Alodine finish. This coating prevents the surface from oxidizing and prevents non uniform charging of the structure in the plasma associated with low-Earth orbits. A new design trend is using 3D printing using special plastics, as shown in Fig. 25-19.

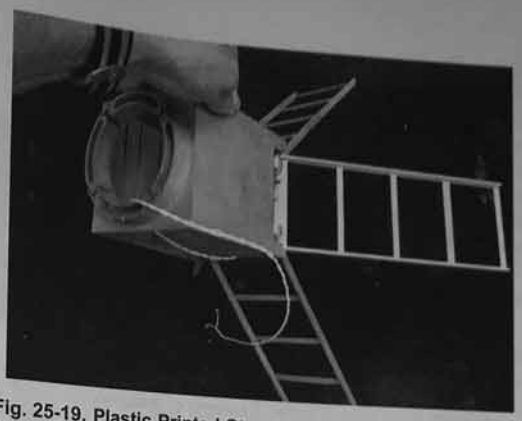


Fig. 25-19. Plastic Printed Structures for CubeSats. (Courtesy of RAMPART Team)

Electrical Power System—Design Differences

Conventional Space Mission Approach

The electronic power systems for larger commercial spacecraft may require regulated voltages of 28 V DC or more. The power requirements may range from several hundred watts for microsatellites to kilowatt levels for large satellites. These power requirements dictate the use of large solar panels that can produce hundreds of volts needed for battery charging. In turn, these systems increase the complexity of handling large voltages and power.

CubeSat Space Mission Approach

The electrical power system (EPS) represents one of the most common system failures in CubeSats that have been developed by university space teams. The need for lower bus voltages and power requirements of the CubeSat reduces this complexity. The power system can be as simple as using primary batteries to operate the CubeSat for a short time after launch or can be a more complex design using secondary (chargeable) batteries that are charged directly by solar panels. These systems typically do not incorporate charge control regulators.

Battery technology for CubeSats is currently limited to Nickel Cadmium (NiCd), Nickel Metal Hydride (NiMH), and Lithium Ion (LiIon). The LiIon technology is the best current technology because it has the highest capacity weight ratio. However, it requires precise charge regulation that, if not properly controlled, can cause battery heating and battery explosion problems. The range of regulated voltages for a CubeSat may be from 3-5V DC. Unregulated voltage for two LiIon cells in a series may be 7.2V. To charge these two cells would require a voltage of 8-10V from the solar panels.

The NiCd is an older, but proven technology that is inexpensive and reliable but has a memory capacity characteristic that requires reconditioning after use to maintain long life capacity.

Selection of the solar cells is also limited by the currently available technologies. Each technology has ranges of efficiencies and output voltages. When using large solar cells commonly used on the larger satellites, a typical CubeSat panel produces less than 5 V. Another approach being used by CubeSats is to use smaller solar cells so that enough cells can be wired in series to achieve a desired voltage. These two types of solar cells are shown in Fig. 25-20.

Peak power tracking is a new method being employed for CubeSat solar panel control to optimize the incident sunlight angle and the power generated. This method uses active control to vary the load on the solar panels in order to keep them at the maximum power point on the I-V curve.

With new applications for the CubeSats constantly emerging, there is an ever increasing demand for more peak and average power from the solar panels. This demand has now far exceeded the strategy of using only body mounted cells. Deployable solar panel systems are now being used to increase the power available to the bus by both increasing the collection area of the solar panel

Table 25-2◀, Fig. 25-19, Eq. 25-5◀

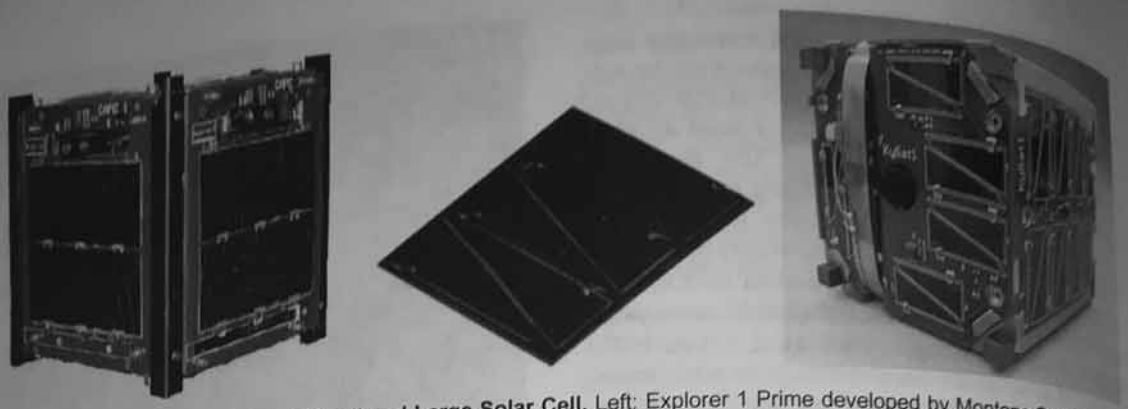


Fig. 25-20. Examples of the Use of Small and Large Solar Cell. Left: Explorer 1 Prime developed by Montana State University uses large solar cells; Center: Example of inexpensive small solar cells (Triangular Advanced Solar Cells made by SpectroLab and mounted on a PCB board solar panel); Right: Ky-Sat 1 developed by Kentucky Space Program uses small triangular solar cells. (Courtesy of Cal Poly & Kentucky Space)

system and by optimizing the incident solar angle. A variety of these designs are under development now with several commercially available (illustrated in Figs. 25-21 and 25-22).

Attitude Determination & Control—Design Differences

Conventional Space Mission Approach

Attitude Determination and Control (ADC) systems for larger conventional spacecraft are a requirement for solar panel control, power generation, and pointing of communication antennas, and possibly the payload.

CubeSat Space Mission Approach

ADC systems were not typically included in the early CubeSats. It was enough in some of these early missions just to prove a minimal system. Those CubeSats that needed to enhance their communications capability by having a known orientation would use passive magnetic stabilization. Passive magnetic stabilization is a simple method of using permanent magnets along the one axis of the spacecraft to achieve some alignment of the spacecraft with the Earth's magnetic field. Some of the early CubeSats used the same techniques as the AMSAT small satellite builders by also including hysteresis rods transverse to the magnets to dampen out possible induced rotations.

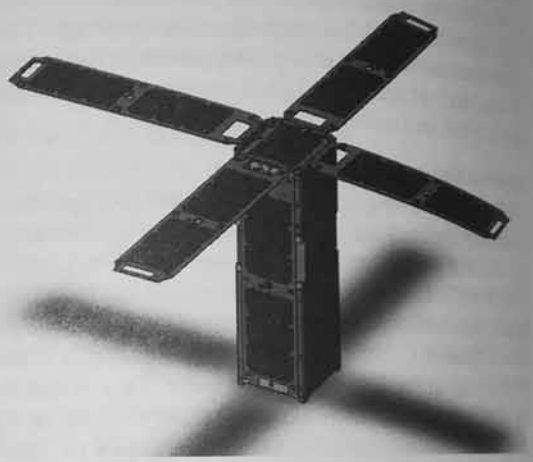


Fig. 25-21. New Solar Panel Designs for CubeSats. (Courtesy of Pumpkin, Inc.)

Strategies for ADC systems employed by early commercial satellites are now being used by the CubeSat designers. Beyond the use of passive magnetic stabilization, some of the simpler systems use gravity gradient booms, and magnetic torquer coils for limited stabilization. There are several vendors that have developed active torquer coils and 3-axis wheels that are now commercially available to control the attitude of the CubeSats.

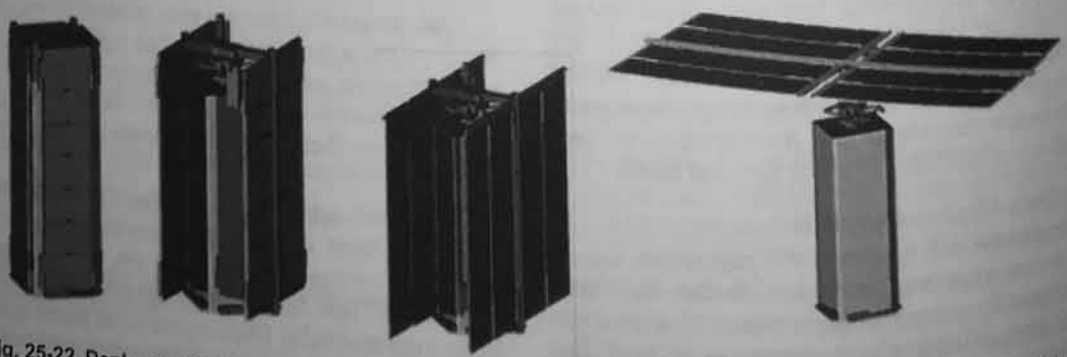


Fig. 25-22. Deployable CubeSat Solar Panels—Stage 1, 2, 3, and Fully Deployed, Respectively. (Courtesy of Tethers Unlimited, Inc.)

Table 25-2◀, Fig. 25-22, Eq. 25-5◀

A full suite of CubeSat ADC systems are now being developed. Many of these systems use new MEMS technologies which have not been available in the past. They are especially applicable to the CubeSat form factor given its small size requirements and limited development budgets.

Command and Control—Design Differences

Conventional Space Mission Approach

Command and control systems for larger commercial spacecraft generally require larger amounts of high processing power than those employed in CubeSats. The size and power consumption of these components lead to increased overall weight.

CubeSat Space Mission Approach

The overall structure of software systems written for CubeSat command and control is similar to that used in the commercial spacecraft. Fortunately, a wide variety of commercial off-the-shelf (COTS) components are now available.

The command and control processors available from the large market for cell phones, PDAs, iPads and other portable devices give the CubeSat designer an extensive choice. Processors like the Texas Instrument MSP-430 series processors are commonly used. Its power consumption in normal operating frequency is generally so low that it is negligible in the overall power budget. When used in the quiescent mode, it uses microwatts of power.

Other processors like the ARM are capable of significant data processing on board. There is also a trend toward multiple core processors which will allow true simultaneous processing. With the ability of the university CubeSat programs to take more risks than previously acceptable, many of these new processors can be space-qualified for future commercial and scientific missions.

Communications—Design Differences

Conventional Space Mission Approach

Most conventional spacecraft require communications systems that can transfer large volumes of data in the Gb/s range, particularly in the downlink modes. To transfer this information requires high power and wide bandwidths. To be able to achieve desired link margins, high gain directional antennas are required on the ground. Orienting the spacecraft antennas toward the ground assets requires accurate attitude stabilization. To accomplish this effectively, a 3-axis attitude control system with pointing accuracies of 1 deg or less is needed. All of these requirements increase the size of the spacecraft and the overall cost of the mission.

CubeSat Space Mission Approach

Communication systems for CubeSats in the past have been flown with simple or no attitude control. This, of course, limits the bandwidth and the link margin, resulting in low data rate transfers of 1,200 bps to 9,600 bps.

New CubeSat missions push the limits of data that needs to be transferred. This requires high data rates, which require wider bandwidths and higher transmission power to maintain the link margin between the CubeSat and the ground stations. Achieving the necessary bandwidth requires going to higher frequencies in the L, S, and even X radio bands. New, higher performing radios for small satellites are now being developed that can increase the data rate and add greater flexibility to missions. One example is the Software Defined Radio or SDR. SDRs will reduce the fixed hardware designs now used on the CubeSats to digital controls and have the capability of after-launch frequency and modulation scheme modification, facilitated by uploading new software.

Applying for and acquiring appropriate licenses for communications frequencies is generally a new experience for the CubeSat developer. Since the frequency spectrum is limited and a well used and controlled commodity, the developers are often perplexed by the difficulty in negotiating the licensing route. CubeSats used for education purposes have been able to qualify to use the amateur radio bands. There are many restrictions, however, on how these bands meet the amateur operating rules and can be used. Amateur radio bands typically used by the CubeSat community are in the 2 m (144 MHz) and 70 cm (420 MHz) frequency allocations. Occasionally, the ISM (Industrial, Scientific, and Medical) radio bands are used for higher data transfer needs, particularly the S-band 2.400–2.500 GHz ISM allocation. Required permits and licenses needed for radio transmission in these bands are discussed in Sec. 25.4.4.

Thermal—Design Differences

Conventional Space Mission Approach

Conventional spacecraft require precise thermal control for the main spacecraft bus electronics and in many cases, very accurate control of the payload. However, there are more means of thermal control associated with larger spacecraft than on (the smaller) CubeSats. The large spacecraft may contain active thermal control systems such as radiators for heat dissipation, as well as protective insulation due to ample available room on the spacecraft.

CubeSat Space Mission Approach

Thermal control of CubeSat systems is equally important to the success of these missions. However, it generally does not require an active control system as on the larger spacecraft. Components that meet the Military Standard temperature range can be used in CubeSats if the developer is careful to consider conscious thermal design elements. Using innovative thermal design strategies, the CubeSat internal components can be kept within –30 C to +10 C providing a suitable operating range.

The most important thermal design considerations are to make sure that internal heat generating components have conductive thermal paths to the CubeSat frame for distribution of the heat. Control of the CubeSat thermal conditions when the internal heat is conducted to the exterior surface of the CubeSat is made possible through external thermal surface preparations.

Table 25-2◀, Fig. 25-22◀, Eq. 25-5◀

Modeling and thermal analysis of the CubeSat is required to manage the internal heat generated and the direct solar heat on the external surfaces. Thermal cycling as the spacecraft structures move in and out of eclipse must be considered during these analyses. Most launched CubeSats have operated in the nominal –20 C to +30 C environment of space without extensive internal or external thermal systems.

Propulsion—Design Differences

Conventional Space Mission Approach

Considering conventional spacecraft missions, the requirement for propulsion can vary widely. Propulsion systems may be required for orbit changes or orbit maintenance. Propulsion systems generally are quite complex in their use of high I_{sp} type thrusters for even attitude control requirements.

CubeSat Space Mission Approach

The first generation of CubeSats generally had no propulsion systems for orbit maintenance and orbit changes.* To date, only one CubeSat (CanX-2 launched in April 2008) has flown a propulsion system. CanX-2 successfully implemented a sulfur hexafluoride (SF6) liquefied cold gas system. A picosatellite developed by the Aerospace Corporation consistent with the size of a CubeSat called the MEMS Pico Sat Inspector (MEPSI) also successfully flew a cold gas system. MEPSI was deployed from the Space Shuttle Discovery during the STS-116/12A.1 mission on December 20, 2006. MEPSI was the first in a series of tests in the development of small, autonomous satellites that are used to observe larger spacecraft—another application of picosatellites with great potential.

Numerous space missions are now being developed that have cold gas and miniaturized systems similar to those used in larger satellites. Other more exotic systems are being developed for use on CubeSats that will feature unique and innovative propulsion systems. Larger, more expensive satellite missions cannot afford the risk associated with these unconventional propulsion systems being developed by the university CubeSat developers.

An example of innovative propulsion systems for CubeSats includes cold/warm gas propulsion utilizing MEMS-based engine nozzles and exotic bi-propellant systems. Figure 25-23 shows a 3D printed complex propulsion system being used on a 2U CubeSat for orbit altitude augmentation using a heated Freon gas.† Figure 25-24 shows a 3D printed bi-propellant system that uses part of the structure as one of the burned components. Using this 3D printing technology, these structures are very inexpensive to manufacture and can have very complex shapes.

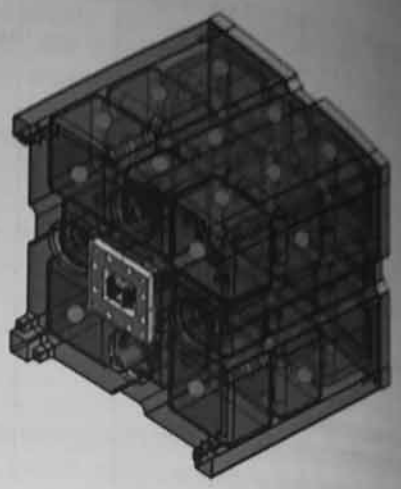


Fig. 25-23. Heated Freon Gas Propulsion. (Courtesy of Adam Huang, University of Arkansas)

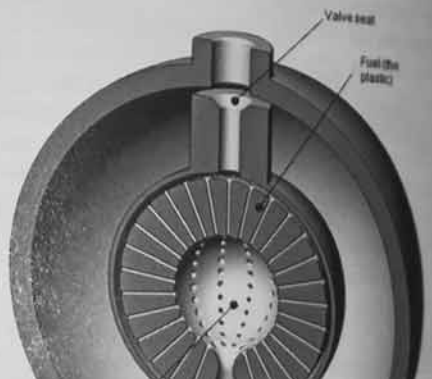


Fig. 25-24. 3D Printed Integrated Propulsion. (Courtesy of Planetary Systems, Inc.)

* ION—the Illinois Observing NanoSatellite featured a low-thrust electric propulsion system but failed to achieve orbit during the 2006 Dnepr launch from Baiknour, Kasakhstan.

† The RAMPART CUBESAT (Rapid prototyped MEMS Propulsion and Radiation Test CUBEflow-SATellite) incorporates a MEMS-based cold gas thruster. The RAMPART team includes Gilbert Moore (Project Starshine), Walter Holemans (Planetary Systems Corporation), Adam Huang, John Lee, and Matthew McMullen (Department of Mechanical Engineering, The University of Arkansas), Jim White (Colorado Satellite Services), Robert Twiggs, Benjamin Malphrus, and Nathan Fite (Space Science Center, Morehead State University), David Klumpar, Ehson Mosleh, and Keith Mashburn, (Space Science and Engineering Laboratory, Montana State University), David Wilt and James Lake (US Air Force Research Laboratory/Space Vehicles Directorate), Stewart Davis (CRP USA, LLC), Wes Bradley (Analytical Graphics, Inc.), 2Lt. Thomas Chiasson (USAF), Jay Heberle (Universal Space Network), and Pat Patterson (Space Dynamics Laboratory)

Other CubeSat propulsion systems under development take a more conventional approach, but with a twist to accommodate the CubeSat form factor. MIT scientists are working on a miniature ion propulsion system that will be ready for implementation on CubeSat systems in 2013. The MIT propulsion system works slightly differently than a conventional ion drive. In a typical ion engine, electricity charges a liquid propellant in a special chamber on board the spacecraft, and then disperses the propellant. In the MIT engine design, the propellant is pre-charged on the ground and encapsulated in small needle-like structures. When an electric charge passes through the needles, the propellant is expelled. Since each needle is effectively its own engine, the entire system is modular and customizable.*

These innovative propulsion systems will greatly enhance the capabilities of CubeSats by allowing new orbital configurations and by allowing the creation of stable pointing platforms for Earth observing (i.e., nadir pointing) and astronomical observations.

Launch Integration—Design Differences

Conventional Space Mission Approach

The developers of the larger primary spacecraft have direct contact with the launch vehicle provider. All tests on the spacecraft are performed to meet the launch environment for that particular launch vehicle. All of the launch vehicle's performance characteristics are based on putting the primary spacecraft into the desired orbit. The primary spacecraft mission essentially covers 100% of the launch costs.

CubeSat Space Mission Approach

CubeSats are considered secondary payloads for the expendable launch vehicles. The acceptance of secondary payloads is generally at the discretion of the primary payload provider since they have assumed the total cost of the launch. The development of the P-POD has alleviated much of the risk to the primary satellite by providing significant protection for the primary payload through total containment of the CubeSat, minimizing the possibility of accidental release and subsequent damage to the primary payload.

Therefore, the cost for the CubeSat launch is based on the additional effort required to accommodate this small payload. The university CubeSat developer may only have the option to accept or decline the orbit being provided since they have no control over which orbit the primary payload owner may pick. The developer must design their systems to be compatible with the launch environment of all possible launch vehicles that will offer secondary launches for CubeSats. This variable can affect the cost from a design perspective.

For university CubeSat developers, the launch will be managed by an intermediate integration organization. This greatly simplifies the launch process for the CubeSat community. With integration services provided by experienced groups like Cal Poly and SRI International, the developer need not be subjected to the complication of working directly with the launch provider.

When the CubeSat is delivered to the integration service organization, final testing is conducted in the CubeSat launcher. The next time the developer is able to take control of the satellite is after the CubeSat is on orbit.

Manufacturing and Environmental Testing Facilities—Design Differences

Conventional Space Mission Approach

The facilities required to test and flight-verify large spacecraft are extensive and expensive. In many cases, large spacecraft cannot be tested in flight configuration. This then requires the final environmental testing to be conducted on individual components. Large spacecraft require thermal cycling, vibration testing, vacuum testing, and acoustical testing. Large thermal-vacuum chambers that can accommodate large spacecraft are expensive to maintain and operate. Large vibration shakers for spacecraft require special facility structures to support the significant weight of the spacecraft and extreme vibration. Care must be taken, for example, during the design of these facilities that mechanical resonances are not induced in the building or nearby structures.

CubeSat Space Mission Approach

CubeSats require the same type of testing as large spacecraft, but the cost associated with smaller thermal-vacuum chambers and vibration systems is much less. Acoustical testing is normally not required for CubeSats since the acoustics only affect large fragile structures such as deployable solar panels. Testing equipment for the CubeSats can be as simple as well-controlled home refrigerators, cooking ovens, and glass bell jars. Achieving vibration levels required for certain launch vehicles, however, can be challenging as vibration systems (shaker tables)—even smaller ones—are not commonly available.

Project Management—Design Differences

Conventional Space Mission Approach

Large conventional space missions have grown to have legions of technicians and engineers working on a single program. The spacecraft design and implementation requires elaborate documentation of requirements. Once these documents are completed, technician groups work in relative isolation to implement their tasks. Engineering groups are formed to handle design, quality control, testing, launch, and operations. The larger the number of groups and individuals in these groups the less effective the intergroup communication.

Because of the extreme costs of these missions, the total mission is developed with a primary objective of being *risk averse*. This leads to over design and use of older components with flight heritage. This process stifles innovation.

Table 25-2◀, Fig. 25-24◀, Eq. 25-5◀

* Paulo Lozano, the H.N. Slater Assistant Professor of Aeronautics and Astronautics at MIT.

Table 25-3. Representative CubeSat Missions Flown from 2003–2011. See the website for a more complete and up to date listing of launched CubeSat missions.

Name	Type	Organization	Mission	Launch Date	Rocket
QuakeSat	3U	Stanford University/Quake Finder, USA	Earthquake detection	6/30/2003	Rokot/Briz-KM
CUTE-I (Oscar 55)	1U	Tokyo Institute of Technology, Japan	Separation system demonstration, Educational, Technology testing & Amateur radio	6/30/2003	Rokot/Briz-KM
CP-3	1U	California Polytechnic University, USA	Educational, Technology testing & Amateur radio	4/17/2007	Dnepr-1
MAST	1U	Tethers Unlimited, USA	Tether experiments—operated from 4/20/2007 to 5/30/2007	4/17/2007	Dnepr-1
COMPASS-1	1U	FH Aachen–University of Applied Science, Germany	Demonstration of commercial off-the-shelf components and taking photos	4/28/2008	PSLV-CA
CanX-2	3U	University of Toronto, Canada	Technology demonstrator for formation flying	4/28/2008	PSLV-C9
K-Sat	1U	Kagoshima University	Technology Demonstrator and Earth Remote Sensing of Water Vapor	5/20/2010	H-IIA 202
RAX	3U	University of Michigan and SRI	Bi-static RADAR measurement of plasma formations in Earth's ionosphere	11/20/2010	Minotaur

CubeSat Space Mission Approach

University CubeSat programs can accommodate implementing risk into their design. This leads to designs that use newer technologies that need not have flight heritage and can be much less expensive. Many university teams that have flown successful missions have had fewer than 10 students doing most of the work. Small team size leads to good communication and facilitates changes in systems design and implementation. Although the university teams do not produce optimum documentation and designs are not replicated, advances in technology and education are prominent.

25.4.4 Permits and Licenses

Conventional Space Mission Approach

Large spacecraft missions conform to communication requirements assigned to the agency operating the spacecraft. Launch location to the geostationary orbit is limited and controlled by international agreements.

CubeSat Space Mission Approach

The CubeSat developer must get the proper satellite radio operations permissions. Note this is permission, not a license. Permission here refers to those using the amateur radio bands. International Amateur Radio Union (IARU) coordinates amateur frequency allocations worldwide. In addition, for US universities, the developer must notify the US Federal Communications Commission of the agreed use of the frequencies through the IARU and provide launch information and other mission parameters. The procedures for these coordination efforts are found on the Cal Poly website at CubeSat.org or the amateur satellite website at AMSAT.org. This is a requirement that must be completed by the developer and is not provided by the launch integration organization.

If the CubeSat contains a sensing device such as a camera that can view the Earth in any frequency of the electromagnetic spectrum, the US Department of Commerce requires (under Title II of the Land Remote Sensing Policy Act of 1992, Public Law 102-555, 15 U.S.C. § 5601 et seq.) the satellite owner to obtain and have a license for Earth remote sensing. The US NOAA manages the application process and must grant the license and approval before the CubeSat is launched.

There may also be restrictions on collaborating with other universities and organizations outside the US. The US Department of State manages and enforces the International Arms and Traffic Regulations (ITAR) that need to be complied with regarding discussions, exchanges of information, provision of services, and collaboration with any non-US personnel or organizations. This may include non-US citizens and international students at the developer's university. Most universities have legal council that can advise the developers on the university policy and ITAR restrictions.

All launch providers at present require certification of the de-orbit time for the CubeSat to be 25 years or less. The orbit life varies significantly with orbit altitude. Since university CubeSats are typically designed for an operational life of one year or less, orbits less than 500 km are most desirable. At this altitude, de-orbit times generally do not require specific de-orbiting systems. Higher orbits may require the implementation of a system or device to increase the spacecraft's cross sectional areas, (and therefore atmospheric drag) at the end of the operational life to meet the de-orbit requirement.

25.4.5 Past CubeSat Missions

The first CubeSats were launched on a Eurockot (a retasked UR-100N ICBM) from Plesetsk, Russia on June 30, 2003. The first CubeSats—6 of which were included on this first launch—represented a wide variety of projects ranging from university student projects to commercial aerospace programs. While most of the launch campaigns have been successful, two notable launch anomalies destroyed a number of CubeSats en route to orbit. Fourteen CubeSats were lost during a launch failure of a Dnepr rocket in July 2006, and an experimental launch of the Space-X Falcon 1 in 2008 destroyed 2 NASA CubeSats.

Seventy CubeSats have flown since the launch of the first group in 2003. A representative list of CubeSat launches is shown in Table 25-3. Of the 70 achieving orbit, most have been successful to some degree. Reflective of the challenges associated with engineering systems for the harsh space environment, approximately 6 CubeSats have never been contacted and only limited contact has been possible with another 5 CubeSats. Table 25-3 chronicles the success rate of CubeSats flown between 2003 and 2011.

It is estimated that 100–150 CubeSats are under development with nearly 60 in preparation for launch in the next few years. In fact, the Von Karman Institute (VKI) is undertaking a project to develop 50 additional CubeSats with world-wide collaborators. The VKI project, called QB50 is an ambitious project to investigate a relatively unexplored region of the Earth's atmosphere (see Sec. 25.4.10 for more detail). The CubeSat form factor is rapidly becoming the de-facto standard for university class missions, for governments with beginning space programs, and now for specific applications within the aerospace industry, NASA, and ESA.

25.4.6 CubeSat Missions—Applications

CubeSat systems are devoted to an incredible variety of missions from bus to component testing, to complete science missions (including astrobiology and ionosphere research). CubeSat developers have demonstrated the versatility and capability of the CubeSat using payload systems based on miniaturized electronics, micro-electro-mechanical systems (MEMS technologies) and innovations in sensor and microprocessor and micro-controller systems. To date, CubeSat missions have focused on the following applications:

- Technology Demonstration
- Earth Remote Sensing
- Ionospheric and Auroral Research
- Effects of the Space Environment on Biological Systems (Astrobiology)
- Radiation Effects on Space Technologies
- Astrophysics

Although numerous CubeSat missions have produced interesting and important science results, a few examples are selected here to serve as illustrations—notably QuakeSat, GeneSat, RAX, O/OREOS, and the proposed Firefly and IAPETUS.

QuakeSat, a 3U CubeSat developed at Stanford University and launched in 2003 was a highly successful mission whose science objective was to study earthquake precursor phenomena from space. The science mission was accomplished based on measuring extremely low frequency magnetic signals in LEO. A magnetometer was deployed on a 0.7 m boom to isolate the sensor from magnetic fields generated by the spacecraft electronics. QuakeSat innovations included deployable solar cells to improve bus power. Roughly 500 MB of data were returned during the mission, leading the team to develop a larger and more sophisticated follow-on satellite.

Another example of an extremely successful CubeSat science mission is GeneSat. GeneSat was a result of collaboration between NASA Ames, industry partners, and universities including Stanford. The satellite consisted of a satellite bus, confined to 1U while the payload was contained in a 2U structure—a configuration that is fairly typical of 3U CubeSat systems. GeneSat's mission objective was to develop a miniature life support system that could fit into a triple CubeSat and could deliver nutrients and perform assays for genetic changes in biological samples, in this case E. Coli. A pressurized sealed vessel and an integrated analytical fluidics card assembly, which included a media pump, valves, microchannels, filters, membranes, and wells to maintain the biological viability of the microorganisms, comprised the payload system. On-board optical sensors were used to detect genetic changes. An environmental system was required to maintain a specific temperature within 0.5 C during the experimental phase, which was nominally 96 hours. The satellite monitored the temperatures external and internal to the CubeSat, as well as the radiation environment. Significant amounts of science data were transmitted down at S-band. GeneSat was a significant experiment, being NASA's first CubeSat mission and the first biological experiment conducted with a CubeSat. Owing to the success of GeneSat, NASA launched two follow up missions, PharmaSat and O/OREOS. O/OREOS, shown in Figure 25-25, is a triple CubeSat that observes how microorganisms and organic molecules adapt to the stresses of the space environment. The GeneSat

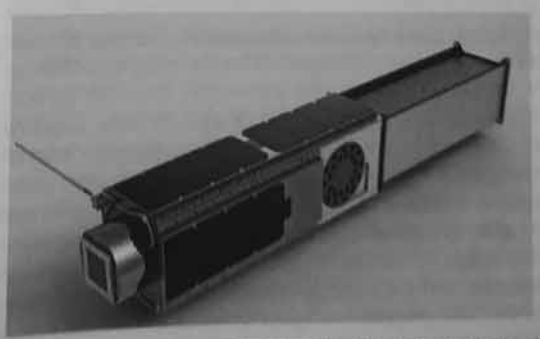


Fig. 25-25. NASA Ames O/OREOS 3U CubeSat. (Courtesy of NASA Ames)

heritage biological CubeSat has incredible potential to return significant science in the field of astrobiology.

SwissCube, Fig. 25-26, is an example of a CubeSat that incorporates a significant science mission and innovative technologies. Developed in Switzerland and led by the École Polytechnique Fédérale de Lausanne (EPFL) Space Center, SwissCube was launched in September 2009. SwissCube was Switzerland's first satellite, making a case for the importance of the CubeSat form factor which provides a low cost entry point for countries to develop space programs. SwissCube images atmospheric airglow, a photoluminescence of the atmosphere at night. Airglow arises from light emitted at 762 nm (peak wavelength) from an oxygen layer 85 km to 100 km above the Earth. To image the Earth airglow, SwissCube must be pointed toward the limb of the Earth and requires attitude determination accurate to better than 1 deg. To obtain the required performance given the constraints of the CubeSat form factor, the developers chose to use MEMS components. The ADXRS614 form Analog Device, a MEMS-based gyroscope, was selected for SwissCube because of its low power consumption and high sensitivity for low rotation rates. The gyroscopes are needed during eclipses, accounting for 30% of each orbit of the satellite. Three gyroscopes were mounted at 90 deg to each other—one on the main ADCS printed circuit board and two on smaller boards mounted perpendicularly to the ADCS board. While the spacecraft has performed nominally, no scientific results or images of Earth airglow have been published in the open literature to date.

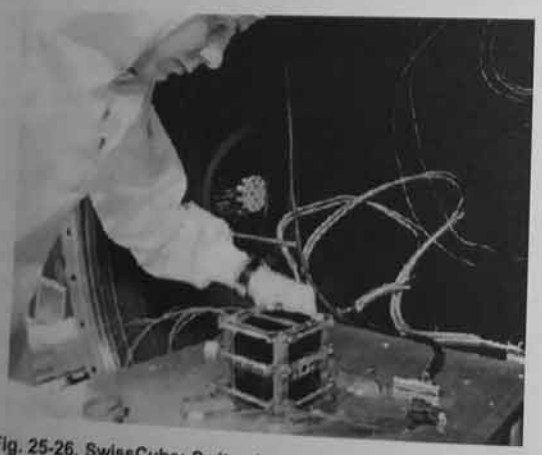


Fig. 25-26. SwissCube: Switzerland's First Satellite. (Courtesy of École Polytechnique Fédérale de Lausanne Space Center)

Figure 25-27 shows the Radio Aurora Explorer (RAX), which is a 3U CubeSat whose purpose is to investigate the causes of turbulence in the Earth's ionosphere, a portion of the upper atmosphere that is ionized by solar radiation. RAX is a result of collaboration between the University of Michigan and Space Research International (SRI). RAX was launched in November 2010. The project's mission is to observe small-scale plasma turbulent structures that occur in response to intense electrical currents in the space environment. The

Table 25-3◀, Fig. 25-27◀, Eq. 25-5◀



Fig. 25-27. Radio Aurora Explorer While Under Development RAX was Launched on November 20, 2010. (Courtesy of the University of Michigan)

structures can attenuate and distort communication and navigation signals affecting a variety of commercial and defense systems. RAX receives signals from megawatt ground-based radar transmitters to investigate these auroral structures.

NASA's "Firefly" Mission, Fig. 25-28, is a CubeSat designed to study terrestrial Gamma Ray Flashes (TGFs) produced by lightning. TGFs are produced by beams of energetic electrons, which are accelerated in the intense electric fields generated by thunderstorms. Prior to work performed with NASA's Compton Gamma Ray Observatory, scientists believed gamma rays could only be generated by energetic extraterrestrial sources like the Sun, supernovae, and black holes. TGF electron beams are more powerful than any others produced in near-Earth space, and understanding the underlying physics of these phenomena will lend insight into physical processes that may occur on other planets or in astrophysical environments, as well as in the Sun's corona. Firefly is a collaborative effort by NASA Goddard Space Flight Center; Universities Space Research Association (USRA), Columbia, Md.; Siena College; University of Maryland Eastern Shore, Princess Anne, Md.; and the Hawk Institute for Space Sciences, in Pocomoke City, Maryland which is scheduled to fly in 2011.

25.4.7 Launch Vehicles Currently Used and Proposed

The CubeSat standard has served to increase access to space, particularly for university-class missions. The number of CubeSat launch opportunities, however, has not increased at a rate necessary to meet demand of the numerous small satellite developers that have appeared since 2001. The number of CubeSat developers has nearly doubled in size every 18 months since 2001 [Crook, 2009]. Commercial launch opportunities, at any cost, are not widely available in the US for CubeSats. Although some launch service providers have been established in Russia, Europe, and India for CubeSats, US based CubeSat developers face additional challenges due to the fact that all spaceflight



Fig. 25-28. NASA's Proposed of NASA)

hardware built in the US is and the International Trade ITAR is a set of US governm export and import of defens All hardware that is intend space is controlled by ITAR for US developers on non- the ITAR regulations pose notable examples of US Cu US launch platforms. Tab prominent launch vehicles

Several companies and launch opportunities in clu motras, Innovative Soluti are companies that offer 2003].

Table 25-4. Launch Vehicl

Launch Vehicle	Cou
Rokot-KM	
Kosmos-3M	
Dnepr-1	
PSLV (3)	
PSLV-CA	
H-2A-202	
M-5 (2)	
Minotaur-1	
Minotaur-4 HAPS	
Falcon-1	
Falcon-9	

25.4.8 Space Operati

Mission Operations

The mission operati programs differs signi missions. Mission oper



Fig. 25-28. NASA's Proposed Firefly 3U CubeSat. (Courtesy of NASA)

hardware built in the US is subject to US commerce law and the International Trade in Arms Regulations (ITAR). ITAR is a set of US government regulations that control the export and import of defense-related articles and services. All hardware that is intended by its developers to fly in space is controlled by ITAR regulations, posing challenges for US developers on non-US launch vehicles. Although the ITAR regulations pose a particular challenge, there are notable examples of US CubeSats that have flown on non-US launch platforms. Table 25-4 lists some of the more prominent launch vehicles used for CubeSat deployment.

Several companies and research institutes offer regular launch opportunities in clusters of several cubes. ISC Kosmotras, Innovative Solutions in Space (ISIS), and Enroot are companies that offer such services [EUROCKOT, 2003].

Table 25-4. Launch Vehicle Used for CubeSat.

Launch Vehicle	Country of Origin	Date 1st Used
Rokot-KM	Russia	2003
Kosmos-3M	Russia	2005
Dnepr-1	Russia	2006
PSLV (3)	India	2008
PSLV-CA	India	2009
H-2A-202	Japan	2010
M-5 (2)	US	2008
Minotaur-1	US	2006
Minotaur-4 HAPS	US	2010
Falcon-1	US	2008
Falcon-9	US	2010

25.4.8 Space Operations and Ground Stations

Mission Operations

The mission operation aspect of university CubeSat programs differs significantly from commercial space missions. Mission operations, as defined for commercial satellite missions in Chap. 29, is a collection of activities, plans, procedures, policies, and tasks required to complete the flight phase objectives of the mission. Mission operations are intricate and complicated tasks that involve mission control, data transfer, spacecraft and payload operations, data processing and archiving, and navigation and orbit control. The latter task is essentially the only activity not typical of CubeSat mission operations—otherwise CubeSat missions involve all aspects of mission ops associated with major space systems. However, given the level of financial resources, staffing, and experience that CubeSat operators have, mission ops for these missions is generally much less formal and intricate.

CubeSat developers and operators, analogous to large satellite operators, are generally required by the program sponsors or launch services provider to develop a Mission Operations Plan (MOP). The features and characteristics of MOPs are described in detail in Chap. 29. Differences between MOPs for large satellites and CubeSats are a function of the greater complexity and expense of larger satellites. Major differences relate to the fact that major spacecraft often have:

- More complicated payloads that require control operations
- Propulsion or station-keeping systems
- Requirements for precision pointing or orientation
- Expendables that need to be monitored during flight
- Higher data rates—more complicated data processing and transfer requirements
- Requirements for real-time (or near-real time) command
- More sophisticated subsystems to monitor and control
- Significantly longer mission durations
- Greater requirements for operational risk management
- Extensive operational risk processes and policies
- The ability to analyze and troubleshoot on-orbit anomalies

It is instructive to have the student developers create a viable MOP. While much engineering experience is gained in the development phase of CubeSat missions, invaluable experience can be gained by students during the operational phases as well. A high level overview of the development and operational phases that are typical of CubeSat missions is provided in Table 25-5. The simpler mission operations requirements allow the programs to be performed at low cost and most often with student operators. The orbits available to CubeSat missions, combined with relatively relaxed requirements for data transfer, facilitate the use of amateur radio equipment for ground segment systems. All of these considerations combined allow for missions that are relatively inexpensive and can be managed, operated, and implemented by university graduate and undergraduate students.

Table 25-4◀, Fig. 25-28, Eq. 25-5◀

Table 25-5. Typical Mission Operation Phases and Modes.

Satellite: CubeSat X	Pre-Launch			On Orbit			
Phase	Final Prep & Delivery	Integration and Testing	Launch	LEOP	VHF/UHF operation	S-Band Ops	E/PO Ops*
Control Nodes	N/A	N/A	N/A	Ground station 1	Ground station 1 & 2	High gain ground station	Public Schools, Amateur Radio Operators
Pass Scheduling	N/A	N/A	N/A	2 × daily	1 × daily	2 × weekly	Variable
Landmark Start	L – 8 months	L – 4 months	Beginning of launch window	L + 1 week (After bus and systems payload check out)	L + 2 months	L + 4 months	L + 12 months
Landmark End	L – 4 months	L – 2 months	Launch	L + 5 weeks	L + 4 months	L + 12 months	L + 24 months
Time Frame	4 months	2 months	L- 0 days	1 month	2 months	8 months	12–24 months

* E/PO Ops: Education/Public Outreach Operations

CubeSat Earth Stations

Ground stations designed to operate CubeSats are consistent with the low cost COTS approach applied to the CubeSat spacecraft development. Most CubeSats utilize Earth stations that are based on Amateur Radio designs. Since up- and down-link frequencies are almost always in the amateur radio bands, the ground station equipment employed to track and command the satellites is based on amateur radio technology. The typical UHF/VHF ground stations consist of a simple UHF/VHF Yagi array used in conjunction with a positioner, back end receivers (similar to the ones on-board the satellite) and tracking software (Ham Radio Deluxe and Nova are widely used) to track and control the satellite. In-house software is often developed to support commanding the spacecraft, decoding the beacon, and to acquire data and telemetry. The Earth station systems are responsible for commanding all available actions including all transmissions from the satellite such as telemetry, commanding the payload instruments, and controlling the various satellite subsystems. A system-level schematic diagram of a typical CubeSat VHF/UHF Earth station is provided in Fig. 25-29, and Fig. 25-30 shows the Kentucky Space VHF/UHF CubeSat Earth Stations at the University of Kentucky.

GENSO

The CubeSat community has undertaken the development of a world wide network of university and amateur VHF/UHF Earth stations that, theoretically, will be used to support all CubeSat missions. The Global Educational

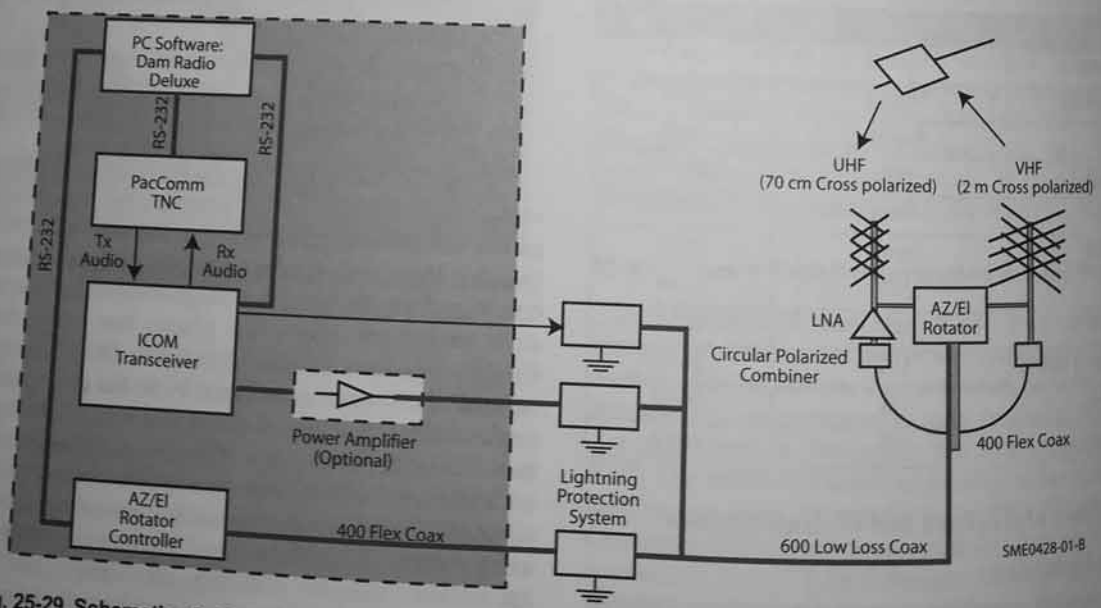


Fig. 25-29. Schematic of a Typical VHF/UHF CubeSat Earth Station Systems. (Courtesy of Morehead State University Space Science Center)

Table 25-5, Fig. 25-29, Eq. 25-5◀

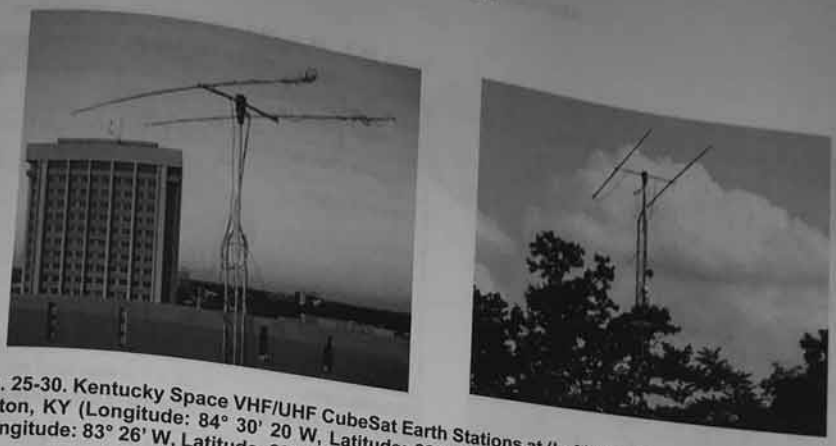


Fig. 25-30. Kentucky Space VHF/UHF CubeSat Earth Stations at (Left) the University of Kentucky, Lexington, KY (Longitude: 84° 30' 20 W, Latitude: 38° 02' 15"N) and (Right) Morehead State University (Longitude: 83° 26' W, Latitude: 38° 11' N). (Courtesy of Morehead State University Space Science Center)

Network for Satellite Operations (GENSO) is a project initiated and supported by the International Space Education Board (ISEB). ISEB consists of the Education Departments of the Canadian Space Agency, the European Space Agency, the Japan Aerospace Exploration Agency, and NASA. The initiative is managed by the Education Office of ESA and the AMSAT Radio Amateur Community. Teams of volunteers from the US, Europe, and Japan are working on the GENSO network, which has several objectives. The primary goal of the GENSO network is to increase the amount of data that can be down-linked from a particular satellite by increasing the available pass times. The idea is to automate existing ground stations to allow unattended operation by remote users and support the development of new ground stations in desirable locations around the world [Melville, 2006]. Other specific objectives of the GENSO network are to:

1. Increase available pass times, leading to unparalleled near-global levels of access to educational small satellites
2. Provide remote access for operators to real-time mission data
3. Provide remote control of the participating ground stations
4. Optimize uplink effectiveness by modeling real-time link budgets and uplink station selection
5. Incorporate downlink error correction by comparing multiple data streams and redundancy checks
6. Create a global standard for educational ground segment software and mission control software interfaces
7. Develop standard GSE for educational satellite missions
8. Establish a common interface to apply for frequency allocation and coordination (in coordination with IARU)

Major advantages of the GENSO network are that satellite operators can control their satellite from anywhere in the world and data can be downloaded to hundreds of Earth stations geographically located all around the planet. The GENSO network should be scalable to potentially thousands of stations, effectively providing worldwide coverage for CubeSat operators [Shirville and Klofas, 2007]. As of early 2011, GENSO is in alpha testing.

While most CubeSat Ground Station Equipment (GSE) is based on VHF/UHF systems, higher gain and data rate systems have been developed and existing large Earth stations have expanded their capabilities to service CubeSat systems. S-band CubeSats, in particular, require larger, more directional ground stations to provide mission operations services. A notable example of a university based CubeSat Earth Station is the 21 m Space Tracking Antenna at Morehead State University in Morehead, KY in Fig. 25-31.



Fig. 25-31. The Morehead State University Space Science Center 21 m Space Tracking Antenna, Morehead, KY. (Latitude: 38° 11' 30.773" N, Longitude: 83° 26' 19.948" W.) (Courtesy of Morehead State University Space Science Center)

Table 25-5◀, Fig. 25-31, Eq. 25-5◀

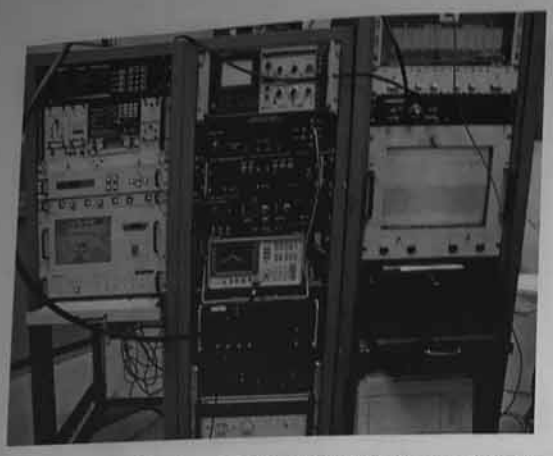


Fig. 25-32. The Morehead State University Space Science Center 21 M Space Tracking Antenna, Control Room. (Lower Equipment Room 2010) (Courtesy of Morehead State University Space Science Center)

The MSU 21 m Space Tracking Antenna is engaged in ongoing research programs in radio astronomy and is also capable of operating in satellite ground station mode, providing telemetry, tracking, and command (TT&C) services for a wide variety of satellite systems from its control room shown in Fig. 25-32. The 21 m is used as a test bed for advanced RF systems developed by the faculty, students, and collaborators.

This system has the capability of tracking fast moving, low power transmitting small satellites in LEO, as well as satellites at geostationary, lunar, and potentially Earth-Sun Lagrangian orbits. The 21 m is currently configured for operation at L-band and K_u-band. Feeds are being implemented at S-band and there are near term plans for the development of a C-band system among others.

The instrument also serves as an active laboratory for students engaged in research and training in space science, electrical and mechanical engineering, telecommunications electronics, astrophysics, and space systems operation. It is largely operated by undergraduate students who work in the associated laboratories to achieve hands-on training in RF systems and techniques.

The basic performance characteristics (aperture, dynamics, and radio frequency) are provided below in Table 25-6 for the current operating frequency regimes (L-band, S-Band, and K_u-Bands). The 21 meter system has a drive system that is capable of tracking LEO satellites at an altitude of 350 nautical miles and a maximum elevation angle of 77 deg. While the 21 m Space Tracking Antenna at MSU is atypical of university-class CubeSat Earth stations, this facility and several others (i.e., SRI Earth stations) have become available for high data rate and ultimately more distant (beyond LEO) CubeSat missions. These high gain quasi-dedicated CubeSat Earth stations will provide communications capabilities enabling CubeSats to achieve significant science well beyond LEO.

Table 25-6, Fig. 25-32, Eq. 25-5◀

Table 25-6. 21 m Performance Characteristics.

Function	Performance
Antenna Diameter	21 meter
Receive Polarization	RHCP, LHCP, VERT, HORZ
Travel Range	AZ ±275 deg from due South (180 deg) EL –1 to 91 deg POL ±90 deg
Velocity	AZ Axis = 3 deg/sec EL Axis = 3 deg/sec POL Axis = 1 deg/sec
Acceleration	AZ = 1.0 deg/sec² min EL = 0.5 deg/sec² min
Display Resolution	AZ/EL = 0.001 deg POL = 0.01 deg
Encoder Resolution	AZ/EL = 0.0003 deg (20 bit)
Tracking Accuracy	≤ 5% Received 3 dB Beamwidth (0.028 deg RMS L-band) (0.005 deg RMS Ku-Band)
Pointing Accuracy	≤ 0.01 deg rms

25.4.9 New Trends in Manufacturing

The requirement of the structure is to always fit a small predefined envelop and remain viable through launch vibration and shock. Once through this launch environment, the structure is now subjected to a whole new set of environments with no vibration, no more shock and relatively benign mechanical environment. Taking a lesson from the toy industry, once in space we can transform and morph the satellite into a size and shape that better suits the new space mechanical and thermal environments.

A new, novel manufacturing of complex structures is being explored. A new 2U CubeSat being developed with this process is called RAMPART, shown in Fig. 25-33. A new material called XTend has been found to have desirable properties for making a structure with a 3D printer. It has low out-gassing and high strength characteristics. It is initially being used to make parts for race cars and electric motorcycles. It is also possible to electroplate this material to assure that it has the conducting faraday cage properties for mitigating spacecraft charging that occurs in the plasma of LEO.

There are now designs that offer multiple CubeSat launches by combining many P-PODS like the Naval Post Graduate School CubeSat Launcher in Fig. 25-34. There is also a trend to P-POD type launchers for larger spacecraft based on the CubeSat such as the one shown in Fig. 25-35.

25.4.10 The Future of CubeSats of the Next Decade

What can we predict to be the future use of CubeSats for the next decade? From my (Twiggs) point of vision, I had no concept of the changes that the CubeSat would

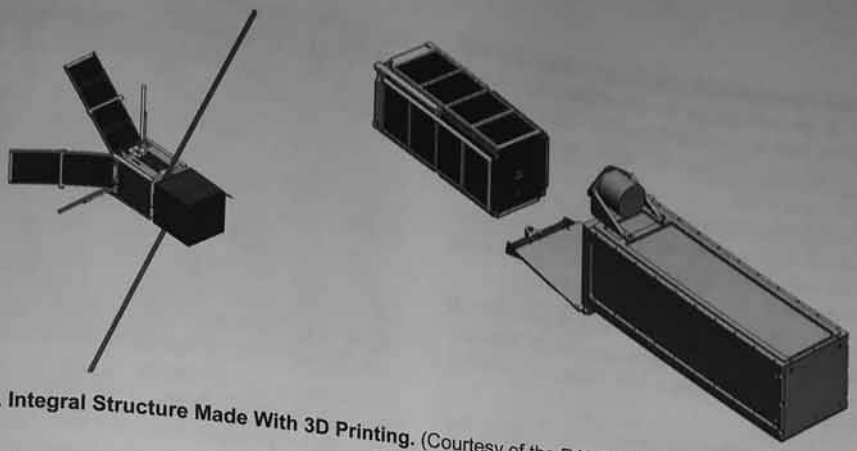


Fig. 25-33. Integral Structure Made With 3D Printing. (Courtesy of the RAMPART Team)

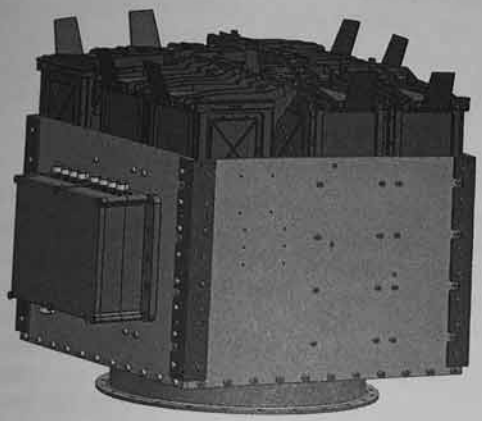


Fig. 25-34. NPS CubeSat Launcher. (Courtesy of Naval Post Graduate School)

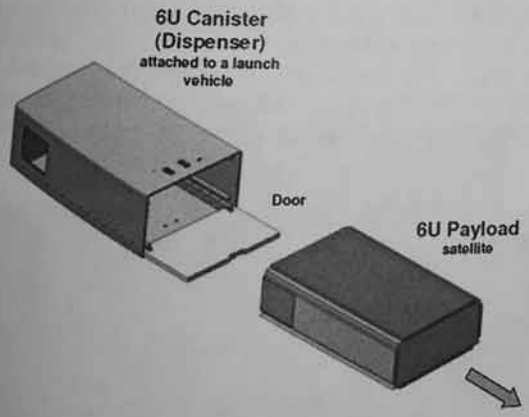


Fig. 25-35. The 6U Launcher. (Courtesy of Planetary Systems, Inc.)

have on the small satellite space community. I can only speculate on what the future will bring, but let's take a look at what has happened in the last 2 years then maybe we can project that into the future.

An assessment of the current state of CubeSat technology facilitates extrapolation toward a prediction of future trends and technological advancements. Recently, the small satellite community has seen dramatic improvement in the performance and capabilities of subsystems, including higher bus power, improved deployable systems, improved processor architecture, and higher bandwidth/throughput communication systems. These systems have tremendously enhanced the capabilities of CubeSats as evidenced by the large number of commercial aerospace ventures now flying them.

Future advances, particularly in the areas of flight dynamics, advanced power, processor systems, and propulsion will facilitate a wide variety of science and commercial payloads. These advances are driven by the current revolution in micro-electronics, especially in the area of MEMS devices. Passive spacecraft stabilization systems have been refined and active stabilization and orientation systems relying on momentum wheels and magneto-torquers are now entering the commercial market. Horizon sensors, Sun sensors and star trackers are also now becoming commercially available. MEMS-based and other propulsion systems are being developed that will be used for spacecraft orientation and orbit adjustment.

These payloads will facilitate experiments in imaging (likely in the optical and infrared spectra), astrophysics (i.e., the study of highly energetic astronomical phenomena), and space physics (i.e., the study of aurora, magnetic fields, radiation belts, and the plasma electrical environment of the Earth-Sun system). Studies of the Earth in regard to its natural resources and aspects of climate change will also be facilitated by these advancements. Future missions will send CubeSat systems to Earth-geosynchronous orbit, to the moon, and ultimately to Mars orbit. These future missions will require advances in CubeSat systems being developed in laboratories world wide today, so the following points are important to consider:

1. Almost all major areas of satellite use have now come to realize that much can be done with small satellites, especially taking advantage of the new developments in the commercial areas for personal devices like PDAs.

2. The launch community is finally realizing that there is an advantage to providing secondary launches and those that have the large primary spacecraft are willing to share space on the launch.
3. The initial generation of students that had opportunities to work on the small satellites and CubeSats are now in the industry and have a growing influence.
4. NASA has the ElaNa program to provide launch opportunities for university programs. In Europe, the Vega launch program has planned CubeSat launches.
5. Major science is now being performed using the CubeSat concept with the Astrobiology work at NASA Ames and the NSF RAX project from the University of Michigan.

Building the Talent to Do CubeSats

CubeSats in education have only been available to the undergraduate and graduate university community thus far, but there is a fast growing trend to move space training projects to the pre-college level of students. The CanSat balloon launch and low altitude rocket launch are becoming popular programs in the US and have a large following in Japan and Europe. The CanSat, generally confined to the size of a soda drink can (another standard) is being built with all of the technology used in the CubeSats. The students in these pre-college programs will soon be building CubeSats that will be operating in LEO.

The CricketSat is an even more fundamental space project. This small device is launched on party balloons and provides educational experience in small space missions by recording simple measurements of temperatures as these balloons ascend. This exposure builds talent to start CanSat projects earlier in education.

Expanding the Use of Space

Since projects that deal with space have a major fascination for us all, they are now being used as one of the major focal points in education to inspire young students to see the value in STEM (Science, Technology, Engineering and, Math) subjects in their education. These space projects not only provide the use of important education subjects, but also give students empowering hands-on experience. This is especially motivating to the students because they can take personal satisfaction in their physical creations.

The success of the CubeSat operating in space has now demonstrated that it does not take a major organization with legions of engineers and scientists to do space missions. So now we are seeing the transition from just making things work in space to doing real science missions. The CubeSat has provided the space platform that has raised the confidence for the "novice" to look at new types of space missions.

New Uses of Space for Education and Science

How can the attributes of the space environment be used to an advantage for new science? Consider the attributes of an ultraclean vacuum and new exposure to space particle and spectral radiation. In addition, the orbiting spacecraft can provide microgravity for unique research opportunities.

A new program has just begun with NASA under a Space Act Agreement with NanoRacks LLC that will provide CubeSat sized experiments called CubeLabs to be flown on the ISS. The CubeLab experiment is installed in a rack on the Japanese Experiment Module and provided with power and a communications link. Data generated in the CubeLab can then be transferred to a NASA website for the experimenter. This new service gives students an opportunity to do a space mission for experiments in a microgravity environment inside a CubeLab. Now, the opportunity for students to do important science in space missions will enable them to make valuable contributions to science, improve their education through increased returns on their efforts, better prepare them for university engineering and science education and, ultimately, create a new generation of highly trained engineers and scientists.

Creating New Devices for Use in Space

The CubeSat concept is expandable to larger sized cubes, but the real challenge is to do useful space missions with smaller and less expensive space devices. The cost of launching spacecraft into space is based on size and weight. Now that there are more science and government applications for the CubeSat technology, launching a single 1U CubeSat has increased from the initial university launch cost of $40,000 to $80,000 or more. Decreasing that cost can only be done by decreasing the size and weight of the spacecraft and providing means of combining many devices into one launch package.

To that end, a new, smaller version of the CubeSat has been developed at Morehead State University called the PocketQub. The PocketQub is a 5 cm cube compared to the 10 cm CubeSat cube. The concept is to combine eight of these PocketQubs into a spacecraft package the size

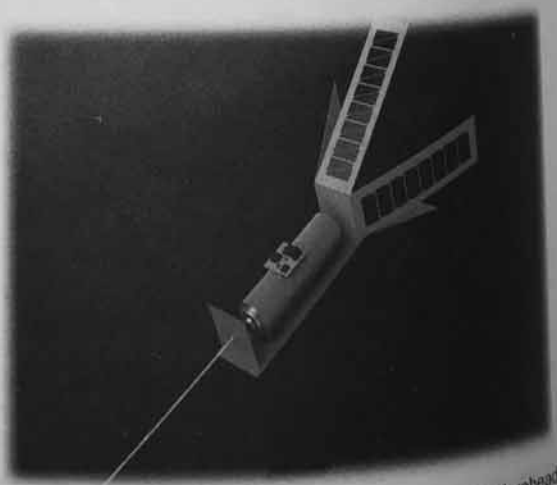


Fig. 25-36. PocketQub Illustration. (Courtesy of Morehead State University Space Science Center)



Fig. 25-37. Pocke
Morehead State U

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25-37. The lau
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Fig. 25-38.
(Courtesy

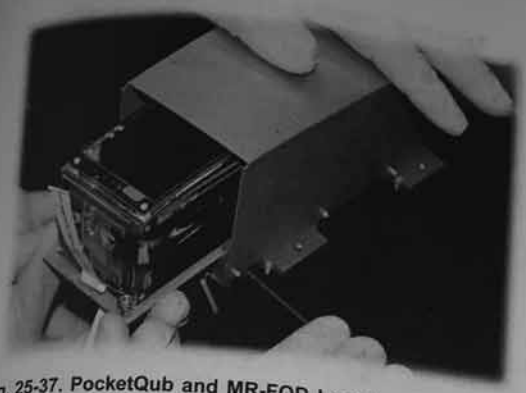


Fig. 25-37. PocketQub and MR-FOD Launcher. (Courtesy of Morehead State University Space Science Center)

and weight of a CubeSat. This should reduce the cost of the launch to 1/8 that of a CubeSat. The PocketQub and a PocketQub launcher are shown in Figs. 25-36 and 25-37. The launcher was developed as a multi PocketQub launcher that could hold 1–3 PocketQubs and reside inside a NanoSatellite from the University of Rome as shown in Fig. 25-38.

Going further, a new variation of the CricketSat is being developed called the Earth Odyssey Moonbeam. The EOM is a version of the CricketSat with a processor to generate Morse code transmission using the 5 mW 433 MHz transmitters used in car remote door openers. The size of the EOM would be such that 24 of them could be contained in the volume of a 1U CubeSat. These EOMs would operate with a small solar panel on each side used to charge an ultracapacitor. The EOM could be multiplexed so that a cloud of them could make space measurements when launched from an altitude below the ISS and have a short orbital life time.

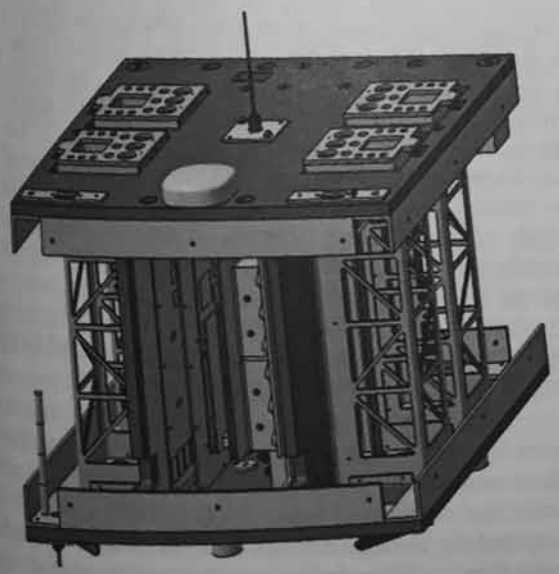


Fig. 25-38. EduSat Nanosatellite with PocketQub Launchers. (Courtesy of the University of Rome)

25.4.11 List of CubeSat Builders and Component Suppliers

Pumpkin, Inc. 750 Naples Street San Francisco, CA 94112 USA www.pumpkininc.com/	**Innovative Solutions in Space** Molengraaffsingel 12-14 2629 JD, Delft The Netherlands www.isispace.nl
Clyde Space Ltd Glasgow G20 0SP Scotland www.clyde-space.com	**GomSpace ApS** Niels Jernes Vej 10 9220 Aalborg East, Denmark www.gomspace.com
Surrey Satellite Technology Ltd Guildford GU2 7YE United Kingdom www.sstl.co.uk	**Astronautical Development, LLC** Morehead, KY 40351 USA www.astrodev.com
Sequoia Space CRA 9 N 62–44 (OFICINA 404), Bogotá, Colombia http://sequoiaspace.com	

25.5 Differences Between International Approaches to Space

Brian Harvey, *Spaceflight Author*

The purpose of this section is to set international space programs in a comparative perspective before we look at how different countries approach their space endeavors and their approaches to development before lessons are drawn. As will be seen, the bipolar space development world of USA vs USSR has given way to a more tiered picture. The United States space program still dominates the world picture by size, funding, and ambition; followed by a second tier of Russia and China, both able to launch human flights; a third tier of space nations with unmanned, yet relatively large programs (e.g., Japan, Europe, India); and a fourth tier of emerging nations, like Iran. Some countries have followed approaches quite different from the United States, taking a statist route in support of economic and social development (e.g,. India), others emphasizing science (e.g., Europe). Perhaps the least understood is Russia, with its structure of institutes, design bureaus, long production runs, mass production, and non-obsolescence punctuated by occasional but impressive innovation. Its experience is doubly important, for it forms the model for the ambitious space program of China.

25.5.1 Comparative Perspective

We use 3 reference points in our comparative perspective: the historical timeline, launch rate, and budget.

Using the launch of a first satellite as a marker, world space programs developed in 3 waves. The first, historic wave was that of the old rivals, the Soviet Union (1957)

Table 25-7. Waves of Space Development by First Satellite Launch.

Superpowers	
Soviet Union	1957
United States	1958
Middle-Range Powers	
France	1965
Japan	1970
China	1970
Britain	1971
India	1980
Emerging Nations	
Israel	1988
(Brazil)	(1997)
(North Korea)	(1998)
Iran	2009
(South Korea)	(2009)

and then the United States (1958), the space super powers. The second wave was that of what were then considered the middle ranking space powers: France, Japan, China, Britain, and India (1965–1980). This group has mutated recently. China with its human flight capability and future ambitions may now be considered among the super powers, while the British launcher was cancelled and the French program evolved into the European Space Agency (ESA). The third group is what we may call the "emerging space powers" of Israel and Iran, countries which have successfully launched satellites and 3 countries which have attempted to do so but so far without success: Brazil, North Korea, and South Korea. All are developing small, specialized, highly focused space programs to meet their particular needs and political priorities. These 3 waves are captured in Table 25-7: The launch rate has broadly followed the pattern of historic evolution. The first country in space, the Soviet Union, has launched more rockets than any other. Although the early years of space exploration saw more American than Soviet launches, from the late 1960s the Soviet launch rate was higher, climbing to an average of up to 2 launches a week in the 1980s. With the collapse of the Soviet Union, its launch rate fell by between 2/3 and 3/4, but in only 2 years at the end of the 1990s did it slip below that of the United States, a position from which it quickly recovered. It is still not generally realized that Russia has been, by the measurement of annual launches, the leading space faring nation throughout this century so far. From the 1970s, with the evolution of the Ariane rocket, Europe has generally found itself in third place, behind the two superpowers. Table 25-8 captures the historic figures for successful launches.

What may be more useful, though, is to look at the figures for recent years, which takes out of the equation, the high historic Soviet rates and puts the emerging powers in a clearer perspective. Here, if we look at successful

Table 25-8. Total Space Launches, 1957–2009.

Soviet Union/Russia[1]	2,882
United States	1,375
Europe [2]	193
China	121
Japan	74
India	24
Israel	5
Iran	1

Defined as successful launches and attributed to the country of the launch site or launcher.
[1] Includes Ukraine and the Zenit 3 Sea Launch
[2] Includes pre European Space Agency launches by France, Britain

launches for 2006–2009, we see how Russia continues to dominate but that China has moved into third place (Table 25-9). China has overtaken Europe (despite temporarily slipping behind Europe in 2009 because of a number of launcher delays). Otherwise, the proportions of launches has been stable over the past decade, with relatively low launch rates from Japan, India, and the emerging space powers.

Table 25-9. Total Space Launches, 2006–2009.

Russia	118
United States	63
China	32
Europe	24
Japan	12
India	8
Israel	1
Iran	1

Disaggregating these figures, they reflect quite different areas of interest, endeavor, and funding. United States missions, despite being half those of Russia, cover the broadest scope and range: military, human, science, applications, deep space. The most important elements in the Russian program are human-related missions in support of the International Space Station and commercial launches which generate scarce revenue. Likewise, European launches are dominated by commercial launches on the Ariane 5. The Chinese and Japanese programs cover a broad range, from Lunar to human-related applications, while those of India and Israel are very specialized (applications and military reconnaissance, respectively).

Using our third reference point, budget and levels of spending, a different picture emerges. The United States dominate world space spending, more than all the other countries combined (Table 25-10). Such a budget has enabled the United States to have the most comprehensive space program in the world, covering almost every possible field of exploration and application. Next comes Europe, which is a combination of the European Space

Table 25-10. Estimated World Space Spending, 2008, in FY08$M.

United States	44,000
Europe	10,000
Russia	4,000
China	4,000
Japan	2,500
India	800

Based on [Pirard, 2009].

Agency and national programs. Figures after that are problematical, due to issues of exchange rate measurement, the basis on which budgets are calculated, and the measurement of labor costs. The Russian figure does not include its vast, obsolete but workable infrastructure, whose costs were paid off many years ago. The country in this group to watch is India, whose spending level was historically quite low but which has accelerated more than any other nation in recent years. Most of these figures primarily reflect the level of government investment and may not include the value of commercially sensitive foreign earnings. These are significant in the case of Europe (the main role of Ariane 5 is as a commercial launcher) and Russia, which transformed its program from the most state-dependent to the most commercially driven in the world in only 10 years.

It is also important to disaggregate these figures. Half the American spending is military, divided between the various branches of the military. By contrast, the proportion of military spending in Europe is low and limited to national programs (the European Space Agency is entirely civilian). The proportions within the other countries are not known precisely.

Assembling these 3 elements together—historic evolution, launch rates, and funding—we now get a composite picture of 4 tiers: the United States, the largest in scope, funding, size, and ambition; the other 2 superpowers of Russia and China; more modest but well established programs (e.g., Europe, Japan, India); and the emerging nations (e.g., Iran).

25.5.2 Philosophy and Approach

Space programs fulfill comparable, but subtly different roles, in different countries. The twin driving forces in the 2 original space programs were science (the International Geophysical Year, 1957–1958) and a bundle of amorphous but real objectives which include national prestige, technology as a national economic driver, and an ambition to be first. This, as we know, developed into a full-blown competition, "the space race," supplemented by a series of military programs and purposes. Both superpowers developed what may be called comprehensive programs that covered the entire gamut of space activity: human, scientific, applications, military, and deep space. In the end, only the United States had the resources to sustain this pace in the long run and the Soviet Union retreated from a number of areas (e.g., deep space exploration, weather satellites, and space science).

Other countries, or groups of countries, lacking resources of such scale, have had to be selective in the fields of space exploration that they chose to develop. The European Space Agency, for example, has a long-term, multi-annual science program, which is the only mandatory element to which all its members must subscribe, leaving member states free to choose to which other programs they wish to contribute (e.g., launchers, human spaceflight, applications). In Japan, the National Space Development Agency (NASDA, which, together with ISAS, was the precursor of JAXA.) focused on applications important for the country's technological development, for example, in the field of telecommunications. China focused on those fields most important for its economic priorities, namely telecommunications (telephone, television), weather warning (*Feng Yun* satellites), remote sensing (*Zi Yuan* satellites), and navigation (*Beidou* system). Israel's program is almost entirely military, designed to provide photo reconnaissance intelligence on its neighbors.

Perhaps the most remarkable example is India. Here, the great visionary of the Indian space program, Vikram Sarabhai, saw the space program as a purposeful instrument of economic and social development. In his most famous declaration, he said:

> **We do not have the fantasy of competing with economically advanced nations in the exploration of the Moon or the planets or manned space-flight. But we are convinced that if we are to play a meaningful role nationally and in the community of nations, we must be second to none in the application of advanced technologies to the real problems of society which we find in our country. The application of sophisticated technologies and methods of analysis of our problems is not to be confused with embarking on grandiose schemes whose primary impact is for show rather than for progress measured in hard economic and social terms (Address at Thumba, 3rd February 1968).**

Accordingly, the space program was used to bring educational television to the villages; to provide weather warning for monsoons and floods; to provide the telephone infrastructure; and for remote sensing (e.g., detection of water, crop mapping, urban planning). It was focused on the needs of the poor, who got television before anyone else;* the rural areas; and those in most social need (e.g., farmers, women). It is still highly focused around practical applications, a recent example being telemedicine. Only in recent years has the Indian space program expanded beyond this highly targeted remit into wider fields. Even

Table 25-10, Fig. 25-38◀, Eq. 25-5◀

then, its Chadrayan Lunar mission used an adapted Earth remote sensing satellite and its human spaceflight program is likely to focus on Earth observations.

Space programs have generally been most successful when part of a national, state-led vision of development. Although the United States would be officially ideologically averse to such a role, in practice such considerations were far from absent in the period of the early space race, when space development was linked to the need for improved educational performance and foreign policy imperatives. Even today, it is implicitly linked to scientific leadership across multiple fields. Most countries have benefitted not only from individuals who have had a strong vision of space-led development, not just India but also Japan (Hideo Itokawa), China (Tsien Hsue Shen), and of course Russia (Konstantin Tsiolkovsky, Sergei Korolev) but also from governments which have a clear idea as to how space programs fit into national development. Here, the most striking contrast is between France and Britain in 1960. Most observers assumed that Britain, based on its technological and industrial assets, would be the European leader in space research. Instead, the Fifth Republic in France made spaceflight an integral part of a program of state-led economic planning: the development of a series of high-tech industries (e.g., railways, aircraft), industrial decentralization (e.g., Toulouse), and the development of a workforce of highly skilled engineers and scientists. By contrast, Britain is, 50 years later, still inching its way to its first space policy, but has meantime lost most of its capacity and fallen behind France, Germany, and Italy in its economic indicators.

25.5.3 Approaches to Organization

In the institutional construction of space programs, these countries have many elements in common. These are:

- The setting of high-level objectives at state or government level;
- In some countries, a national advisory body to advise on high-level objectives and to set middle-level targets and objectives (e.g., Japan);
- The entrusting of their execution to state agencies (NASA in the United States, CNES in France, JAXA in Japan, ISRO in India, the new UKSA in Britain).

In practice in most countries, the space agency has become much more than an instrument of government, but has refined objectives, policies, and priorities over periods much longer than the electoral cycle, creating a long-term dynamic for development.

Our original modeling of how governments approached space program development dates to the 1960s. Essentially it revolved around the following paradigm, which was that in the United States, Congress set high-level objectives while NASA supervised the development of programs through open competition by leading manufacturers (many drawn from the aviation world). Early American space achievements were celebrated as a triumph of the best competition of private industry, a point once noted by Frank Borman as he mused in *Apollo 8*, awaiting liftoff ("We're sitting on a million parts, all of which must function perfectly—and all built by the lowest bidder") [Young, Silock, and Dunne, 1969]. By contrast, so the model went, the Soviet system was a command-and-control top-down one in which the Soviet leadership set objectives and assigned tasks to state industries, which, as the outcome of the Moon race proved, were clearly not up to the task.

Although it is a well known, accepted narrative, in reality, this modeling is erroneous. It transpired that the old Soviet Union was full of rival design bureaus which competed for contracts from a leadership whose decisions were often the outcome of the interplay of many parties, industries, the military, scientists, designers, and politicians. Examination of the outcome of the Moon race told us that the Soviet Union did have the technology to get cosmonauts to the Moon, but was let down by its failure to manage wasteful rivalries between the bureaus. By contrast, NASA exhibited the skills of command-and-control hitherto ascribed to the socialist economies and was infinitely more successful at handling the aerospace companies (and their political interlocutors).

The design bureau did mark out a fundamental difference between the American and Russian system, one which persists to this day. It is also relevant to the up-and-coming Chinese space program which was, institutionally at least, designed along Soviet lines in the 1950s and 1960s. The Soviet/Russian approach to missions and spacecraft design is based on a 3-stage architecture not well understood in the west:

- The institute, where scientists present concepts for missions to government for adoption;
- The experimental design bureau (OKB in Russian), which turns them into practical effect and builds and tests the first hardware;
- The production organization or factory, which produces or mass-produces the item in question.

The Russian scientific community is dominated by several thousand institutes, a marked difference from the American system, where equivalent learning takes place in universities (continental Europe still has a strong tradition of institutes, of which the Russian institutes were historically an outgrowth). These institutes, which range from the large to the tiny, are broadly part of the system supervised under the Academy of Sciences [Graham and Dezhina, 2008]. Although criticized abroad for separating science from the teaching community, they have produced high-quality outputs and served state developmental needs well. There is an argument that institute-

* Television was first distributed to the poor in the villages via comsats and dishes to transmit educational TV (e.g., farming, literacy, hygiene). Only later was there national TV for everyone.

Table 25-10◀, Fig. 25-38◀, Eq. 25-5◀

based development enables a high level of theoretical work to be undertaken, which can be the basis of quality engineering, science, and their application. Russian rocket engine development, probably the most advanced in the world, was based on institutes that date to the 1920s, with, literally, strong institutional memories.

The critical element and lynchpin of this trio, though, is the OKB, the design bureau. Here, the old Soviet space program developed a series of specialized OKBs, each normally associated with its leading designer, like Sergei Korolev (OKB-1, now Energiya), Dmitri Kozlov (OKB-1 §3, now Progress in Samara), OKB-7 Arsenal (military satellites, St Petersburg), Mikhail Rechetnev (OKB-10, communication satellites, Krasnoyarsk), Vladimir Chalomei (OKB-52, now the Khrunichev company), Semyon Lavochkin (OBK-301), Valentin Glushko (Gas Dynamics Laboratory, rocket engines), Mikhail Yangel (OKB-586, now Yuzhnoye, Dnepropetrovsk, Zenit rockets).

The design bureau, which has no direct American comparator, was often given premises of former private factories confiscated by the Bolsheviks, with a leading engineer put in charge (a "general designer") and it was expected to deliver the goods under the system of socialist competition developed by Stalin. There were high levels of incentives for delivery (financial rewards for the workforce, the prospect of new assignments, influence in shaping future missions and requirements, and political access).

Any new project or spacecraft design was expected to run through the spectrum of *institute—OKB development—production*, with government approval at each point. The thoroughness of the approval process incentivized the OKBs to develop ever newer "modifications" of their existing designs, which, being merely "modifications", did not have to seek fresh round of governmental approval (hence many had the letter "M" after their designator). This is most evident in the design of the piloted spacecraft, *Soyuz,* which dates to 1960 and which has since evolved into the *Soyuz T, Soyuz TM, Soyuz TMA* and the new *Soyuz TMA-0M*. Although the concept of the satellite "bus" is seen as a western invention, it has its roots in the Soviet system where a standard design was used as the basis of a whole series of subsequent adaptations and evolutions. The original design for a photo reconnaissance satellite, the *Zenit*, dates to 1958, but became the bus that was adapted for human flight (*Vostok, Voskhod*), a series of photo reconnaissance satellites (*Zenit*), biological cabins (*Bion*) and still flies today for micro-gravity research (*Foton*). The Cosmos program, the main scientific program begun in 1962, was built around a standard bus (called the DS, or *Dnepropetrovsky Sputnik* after the city where it was built), with an initial batch of 18 ordered, later replaced by the DS U series, with 30 missions. At a time when most western satellites were still hand-crafted for individual missions, the Soviet Union had already introduced the concept of the standard bus.

It was a system adapted with considerable resourcefulness. In 1972, the Soviet Union began a program of astronomical missions called "the great observatories." Rather than construct purpose-built spacecraft, existing models were adapted like a Venera (Venus) spacecraft for the *Astron* observatory, a Mars probe for the *Granat* observatory, Progress for the *Gamma* observatory, and a space station module for the *Kvant* observatory. The total number of fundamental spacecraft designs in the Soviet or Russian program is surprisingly small, with significant cost benefits.

An important feature of the Soviet system was the system of financial control and accountability (or the lack of it). Financial transactions in the sense understood in western capitalist countries were minimal and took the form of bookkeeping movements rather than cash, and if a project were designated a matter of "supreme national importance," resources could be commandeered almost at will. The downside of the lack of financial control was that the old program tolerated a relatively high failure rate. The first soft-landing on the Moon in 1966 was preceded by 10 failures: the chief designer could not be called before any congressional committee for wasting money, and no one could calculate how much was lost anyway (though he could and did receive angry phone calls from the General Secretary). There was no incentive for building spacecraft to fly long missions. At a time when American communications satellites had on-orbit lifetimes of 10 years, their Russian comparators were averaging 25 months: it was simpler to build and launch a replacement. Only in recent years, under the new pressures of a capitalist Russia, has a sustained effort been made to lengthen the mission durations of applications satellites.

Russian spacecraft were optimized around short missions. Several science missions were mistakenly interpreted as failures because the spacecraft transmitted only a few weeks or months, while American equivalents would broadcast for years—but in the Russian case, they would be replaced on orbit by another, also designed to return a large quantity of data in a short period of time. On the other hand, little expense might be spared in the development of new systems. Early Lunar, Earth orbiting, Venus, and Mars probes were sophisticated, with high-quality instruments, extensive instrument suites and related systems (e.g., cameras). Testing regimes were extensive and demanding, the Venera landers for Venus being a case in point. As a result, missions generated a high scientific return.

The absence of commercial imperatives meant that many of the drivers evident in the development of American and other space programs were absent within the Russian and Soviet system: it was easier to design a sound, basic system and evolve it over time. Short periods of often impressive innovation (e.g., closed cycle engine operation) have been followed by lengthy periods of evolution and operations. The R-7 launcher, for example, was in concept in 1948 and finally designed in 1953. It has now been launched 1,751 times and its most recent iteration is the new model developed for launch at the European Space Agency base in Guyana. It is, arguably, entirely obsolete, but also reliable and profitable. The

Table 25-10◀, Fig. 25-38◀, Eq. 25-5◀

design flaws were ironed out of the system not years but decades ago. Similarly, the *Nauka* large science module, due for launch to the International Space Station in 2012–2013, is based on the *Almaz* space station design of 1964. This particular module was built as a backup to the first module of the International Space Station, *Zarya*, in 1998, and will have sat in the hangar for the better part of 15 years before launch—a situation unimaginable in the United States. But again, it is a system that brings considerable gains in cost, and reliability.

Returning to the R-7, it was and is a system designed for durability and simplicity—durability to cope with climatic conditions en route to and at the launch site, simplicity so as to enable it to be operated by successive launch teams of relatively low-skilled workers (until recently, normally military), "problems that can be fixed with a household wrench." Liftoff takes place when engine thrust exceeds launcher weight, thereby freeing the launch clamps held in place by nothing more than a series of balanced weights. Foreign observers are amazed at how launches take place in all weathers, with temperatures reported as low as –50 °C. Indian officials coming to Baikonour to watch an Indian satellite launched in freezing dense fog automatically assumed that it had been canceled, only to be told that the faint red glow in the distance was their rocket taking off. Relative simplicity of function means that the last delay to a piloted Soyuz launch was in 1971 (there have been delays on the new *Soyuz 2* launch system, but due exclusively to electronic problems arising from the introduction of digital systems). This approach produces considerable gains in schedule and cost. At the same time, it would be a mistake to regard the approach or workmanship as crude. When a group of German rocket enthusiasts inspected an R-7 on the pad at Baikonour, the first thing that struck them was the quality of workmanship: "every rivet was perfect," they noted.

The Chinese space program, as noted earlier, was institutionally based on the Soviet system. At this stage, we do not have sufficient information to model the Chinese approach nowadays and the degree to which it has adhered to this paradigm. The Chinese follow a similar Academy of Sciences system and we know that there are state design bureaus in both Beijing and Shanghai, sometimes in rivalry with one another. Universities, though, may play a more important role than institutes, for many of the scientific papers arising from the program come from universities, while new spacecraft technologies are developed in the technological universities. An important area in which China differs from the old Soviet program is its stricter accounting and bookkeeping systems. The Chinese space program has a multi-year but finite amount of funding and the message "we can't afford failures" has been drummed into the program at all levels from the senior management. Quality control has been an obsession, even at a cost of the speed of the program (a feature of the human space program, for example, has been its slow pace). The best example was the unmanned *Shenzhou 3* mission, which after several unsatisfactory months of pad testing at the launch site in north west China was sent all the way back to Beijing on the railway for a fresh round of quality control tests. The delays cost 6 months, but the mission was successful. The emphasis on quality control has paid off, for there have been no failures on the *Long March* launcher since 1996 (albeit some final stage malfunctions).

Perhaps the most "statist" space program is actually the Indian. From its establishment in 1947, the Indian state was built around strong concepts of state planning, accompanied by a wave of institution-building (Sarabhai alone set up 47 institutes) and a policy of indigenization. This is understandable in a historical context, for India had been robbed of its scientific and industrial heritage during the colonial period, industry filling principally a production function and being denied opportunities for its own research and development. An important priority in the Indian space program has been to learn methods and approaches developed in other, more advanced countries, and then to apply them by constructing them at home. This is most evident in the communications satellite program, INSAT, which began as foreign built and foreign launched. The next series introduced domestic-designed and built components. The next series was entirely domestically built, but still foreign launched. With the new GSLV launcher, INSAT is domestically built and domestically launched. The statist approach is in evidence at the ISRO Satellite Center in Bengaluru, where satellites are designed and built under one roof, the center having over 2,400 staff. Although Small and Medium Size Enterprises (SMEs) are involved in supplying parts of Indian space hardware, the main development is carried out by a single cradle-to-grave state center of the type we always associated (we now know wrongly) with the Soviet Union.

By contrast, the space programs of Japan and Europe have followed what is a more recognizably American approach to manufacturing and development. The relationships between Japanese governments and their large industrial corporations have traditionally been a close one, and it is little surprise that contracts for launchers, satellites, hardware, and facilities have gone to the large corporations like Mitsubishi, Mitsubishi Electrical Co., Nissan, and Ishakawajima Heavy Industries. In Europe, the picture is more complex. The introduction of the single market in 1992 was accompanied by a set of procurement rules designed to promote free competition, open the market to new entrants, and level the playing field for smaller players and SMEs. At the same time, ESA is committed to the principle of the *juste retour* (fair return) whereby each country receives contracts broadly comparable to the budget it provides for ESA in the first place, something which clearly puts France in a strong position. In reality, *juste retour* is becoming harder and harder to calculate on a country-by-country basis because of the rapid pace of consolidation of high-tech industries in Europe, where the traditional national company has given way to multi-national companies with branches and affiliates scattered across the member states. In recent years in fact, several

Table 25-10◀, Fig. 25-38◀, Eq. 25-5◀

new, smaller companies have won contracts both for ESA and national space programs (e.g., OHB Kayser Threde).

25.5.4 Conclusions and Lessons Arising

Overall, approaches to space programs and operations have more features in common than there are differences. A certain convergence is hardly surprising, granted the level of international cooperation that has been present since the International Geophysical Year (1957–1958), the intensity of bilateral and multilateral cooperation, and the influence of high profile projects which require inter-agency and inter-company working (e.g., International Space Station, recent unmanned Mars missions). India and China are two countries that stick out from the convergent model: China, because we know less about its internal philosophy, approaches, and organization; and India, because of its unique approach of state-led development and indigenization for economic and social purposes. China may be less different than we imagine and as it reaches its goal of full parity with the superpowers, we may know more.

The traditional distinction in development, logistics, and manufacturing was always the bipolar one of USA vs USSR. The lengthy exposition of the Soviet/Russian program here shows that the differences were not what we thought they were and that competition for design, contracts, and manufacturing by the leading contenders was as real in both (though better handled at political level in the United States). The Soviet/Russian system, though, does present us with a number of learning points. These points do not argue that their system is intrinsically "better" but does argue that it has merits and should in any case be understood. It is a program which produced high-quality engineering, products, and scientific returns, often at remarkably low cost; was sufficiently flexible to survive and rebound from the country's economic collapse around it; and leaves Russia still the world's leading space faring nation by number of annual launches. Many of its distinctive features derive from its particular and now archaic form of bookkeeping, commercial imperatives (or the lack of them), academic traditions, and peristaltic approach to innovation, all of which had up and downsides.

Our learning points are:

- The role of institutes in concept and mission design, important for theory, applications, standards of excellence, scientific drive, and institutional memory
- The use of the key feature of the experimental design bureaus, the OKBs
- Development of a relatively small number of spacecraft (and rocket) designs and buses, which are modified and adapted for a broad range of purposes over lengthy periods, in some cases over 60 years, with considerable cost savings
- A focus on simplicity, durability, reliability, the avoidance of over design, and operation by people of all skill levels
- The merits of using several spacecraft on short duration missions, rather than smaller numbers (or a single number) of larger, longer duration spacecraft vulnerable to single point failures

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26 Launch Vehicles

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26.1 Launch Vehicle Selection
Target Orbit and Injection Accuracy; Spacecraft Launch Mass and Margin Policy; Key Dimensions and Volume; Trade Study A—Target Orbit, On-Board Propulsion, Mass, and Volume; Budget, Schedule Requirements, and Risk Tolerance; Launch Vehicle Capability, Availability, and Reliability; Contacting Launch Service Providers; Trade Study B—Cost, Schedule, Risk, Availability, Capability, and Reliability; Trade Study C—Revisit the Options in the Process Flow; Selecting Candidate Vehicles and Assessing Launch Environments; Proceeding With Detailed Spacecraft Mission and Design

26.2 History Prior to 2010

26.3 Basic Mechanics of Launch

26.4 Launch Environments
Coupled Loads; Shock; Vibration and Acoustics; Electromagnetic Compatibility (EMC); Ascent Venting

26.5 Available Launch Vehicles
USA; International

"...launch remains a high-risk element affecting mission success"

—NASA Policy Directive 8610.23C "Launch Vehicle Technical Oversight Policy," August [2006].

"The only natural predator of a spacecraft is a launch vehicle."

—Major General Nathan Lindsay, USAF (retired)

This chapter provides guidance on the acquisition and management of a launch. The mechanics of flying a vehicle to orbit are addressed only insofar as they bear on the conditions and challenges that the spacecraft customer should anticipate. The scientific principles and technical challenges of flying a vehicle to orbit are a lot of fun to discuss, as we will see in Chap. 27. Most of the spacecraft customers with whom I've interacted, view launch vehicles with awe, pride, and attraction, only until they realize that they've got a mission to fly, unexpected risks and expenses, and a budget that's tight in every conceivable way. There will only be a few times, like the moment captured in Fig. 26-1, when you look at the launch vehicle as a cool rocket. Most of the time, it will be a very expensive nuisance, and one of the greatest risks to the success of your mission.

This chapter makes considerable use of standards and policies published by NASA and the US Department of Defense (DoD), for the reason that the documents cited are publicly available and represent good guidance for use in varying forms throughout the commercial, government and international communities. Much of the launch business takes benefit from the standardization inherent in a well-established commercial industry, such as telecommunications. The documents cited here are also mostly written at a high level, not something you need a PhD to understand, but they provide adequate understanding to a technical or project manager of how the subject matter relates to their mission. In-depth treatments of each topic are available in a wide variety of other published sources.



Fig. 26-1. Orbital's Stargazer L-1011 Launch Aircraft Arriving with a Pegasus XL. After a Ferry Flight to the US Army's Reagan Test Site in the Republic of the Marshall Islands. (Photo courtesy of: NASA/Ivy Springer)

The commercial launch vehicle community makes selective use of government standards, being mostly free of the contract levied standards that were features of the pre-Commercial Space Launch Act era. The ULA Delta II pictured in Fig. 26-2 is a commercially designed and managed evolution of a vehicle that was originally developed under Government standards. NASA and military standards today are more likely to serve as the framework of terms and guidance to approach some of the technical problems that will be described here, rather than demanding rigid compliance to the standard itself.



Fig. 26-2. A United Launch Alliance (ULA) Delta II Lifts Off from Cape Canaveral Air Force Station, Florida. (Photo courtesy of NASA)

Get Smad This is equally true for International Commercial Launch Services, though with slightly different heritage government standards and policies. The European Cooperation for Space Standardization (ECSS), which is supported by the European Space Agency (ESA) and other commercial entities, offers available guidance in a variety of areas, including launch. Similarly, the AIAA routinely partners with the ANSI and the IEEE to publish technical guidance that may be useful in planning the launch phase of a mission. JPL, APL and the Aerospace Corporation have public accessible guidance policies as well. Ultimately, when seeking technical guidance for your launch phase, there is reliable assistance available from these reputable institutes.

This chapter addresses a subset of significant historical developments over the last two decades. It covers the selection process, launch environments, and features that one may encounter. It also features a listing of available launch vehicles (e.g., Soyuz, as shown in Fig. 26-3), along with information about their use, performance, and cost data.

26.1 Launch Vehicle Selection

The launch vehicle selection process focuses on balancing performance, availability, risk, and cost. If circumstances are such that you have many choices available to carry your spacecraft to orbit, choose the most mass delivered to orbit as you can afford. *Performance* for a launch vehicle universally means separated mass delivered to orbit. In this context, performance is a pretty big part of the intermediate and final iterations in the selection process. The physics and economics of this are straightforward, and favor seeking the most delivered mass at the least cost you can find among credible launch service providers. *Credibility* in this context is, and will remain, a subjectively-defined term depending on the user's risk posture.

The following subsections are intended to help you develop your own idea of credible selection factors. Figure 26-4 is offered as guidance for the overall process flow, but with the acknowledgment that circumstances will vary with each spacecraft customer, many times rendering some of the specific guidance and recommended trade studies inconsequential.



Fig. 26-3. Night Launch of a Samara Space Center (TsSKB-Progress) Soyuz Vehicle from Baikonur Cosmodrome, Kazakhstan. (Photo courtesy of NASA)

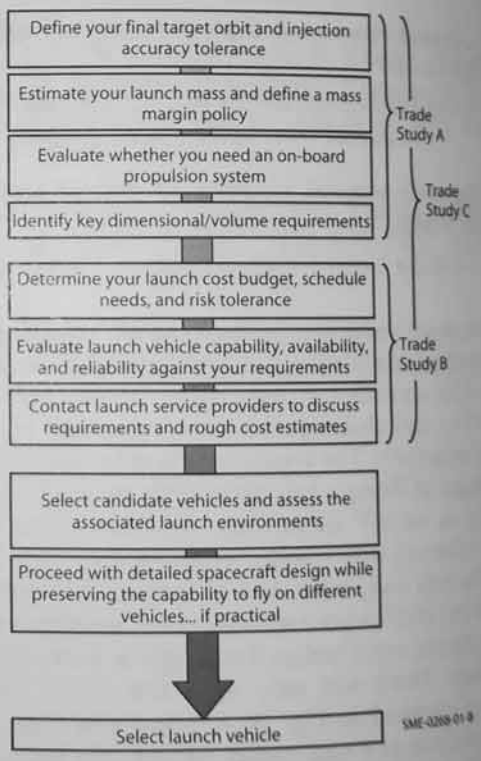


Fig. 26-4. Launch Vehicle Selection Process Flow.

26.1.1 Target Orbit and Injection Accuracy

The process flow in Fig. 26-4 begins with the desired target orbit, likely to be the first known quantity dictated by your mission, be it communications, exploration, surveillance, educational, or astronomy. As with commercial

airlines, launch vehicle fleets are developed with a select number of missions in mind, and modification away from those target missions can become costly. Thus, selection of the target orbit becomes a reasonable first-cut at narrowing the field of candidate launch vehicles.

The importance of injection accuracy in your selection is dependent upon the mission you intend to execute, as well as whether or not your spacecraft will have an onboard propulsion capability. If so, how much of your velocity budget can you afford to reserve for vehicle injection errors? A modern commercial launch vehicle with digital avionics and a liquid upper stage engine can accomplish highly-precise interplanetary injections or deliver spacecraft to precise parking orbits to accomplish a subsequent navigation into a formation of low-altitude spacecraft. Table 26-1 shows some example injection accuracies that are typical of the industry. The values shown, published in the respective vehicle user guides, are also typically conservative statements of capability. Your actual performance will probably be better.

Table 26-1. Example Altitude and Inclination Injection Accuracies.

Vehicle	LEO Apse Altitude (km)	Inclination (deg)
Minotaur IV (HAPS)*	< ±18.5	±0.05
Dnepr	±4–10	±0.04–0.05
Rockot	±1.5%**	±0.05
Delta II (two-stage)	±9.3	±0.05

*Hydrazine Auxiliary Propulsion System
**e.g., ±6 km for 300 km altitude

The differences between the accuracies shown in Table 26-1 are often artifacts of convenience or heritage notation, rather than representing real differences in capability. For the US vehicles, ±10 nm and ±5 nm, respectively, are good round heritage figures. Launch vehicles with the capability to achieve precisely-timed liquid rocket engine shutdown commanded by their on-board guidance algorithms will all realize comparable injection accuracy capabilities. These capabilities are generally sufficient for even the most demanding spacecraft missions.

For launch vehicles with a solid rocket motor final stage (or *kick motor*), the injection accuracy problem is a harder one for the vehicle to manage. The contest for achieved accuracy between a solid kick motor (without any liquid rocket orbit adjustment or *trim* capability) and a liquid upper stage should always go to the liquid stage. Limited technology available today precludes executing a commanded shutdown of a solid rocket motor, which is unlikely to change in the next decade or so, because there's no obvious need. The solid motors in use today shut down when they run out of propellant. The consequence is that the solid stage will always deliver a fixed impulse, but due to manufacturing variations the delivered impulse can only be estimated with so much certainty. To compensate, some vehicle providers have developed velocity management algorithms in their onboard guidance software, while others have made use of small, low-thrust, liquid rocket orbit adjustment packages or *trim stages*. The Minotaur IV HAPS is an example of an optional capability that can be used for increased accuracy injection, or multiple orbit adjustments to service multiple payloads with an otherwise all-solid vehicle, which achieves the same accuracy as their larger liquid upper stage cousins. However, liquid trim stages are usually an option, not a baseline, and they come at increased cost (and reduction in delivered mass) that has to be balanced against the cost of using the spacecraft's own propulsion system to achieve the same result.

26.1.2 Spacecraft Launch Mass and Margin Policy

This discussion starts with reserving your mass margin; you're going to need it. The NASA GSFCs *GOLD Rules* [2009] provide guidance on the subject of mass margin, recommending 30% "dry mass" margin at the initiation of concept studies, stepping down in specific increments (e.g., 20% at PDR, 15% at CDR), plus a *three sigma* propellant margin (if the spacecraft carries a propulsion system) to cover worst case launch vehicle performance, worst case spacecraft mass properties, and other performance penalties, ESA [2005]. Guidance on the subject is comparable across the full design spectrum, only a bit more lenient with respect to dry mass margin, recommending only 20% during assessment studies.

Margin policy may differ according to circumstances, for example, GSFC mass margin policy may not be practical for very small spacecraft like the 6 kg PPOD (Fig. 26-5).

The use of fixed recurring structures and components allow a considerably smaller mass reserve to be maintained at various points in the design cycle. The message here is to think about and document as a mass margin policy early in the design phase, and then manage to it.

The term *dry mass* refers to the mass of your spacecraft prior to loading any propellants. Another term that you will encounter from the launch vehicle community is *separated mass*, which refers to the total mass that separates from the launch vehicle. This is significant because, in most cases, that separated mass will be the spacecraft plus part of the weight of the separation system provided by the launch vehicle. Part of the separation system flies away with you, usually the forward mating ring of a circular payload attach fitting. The mass of the metallic ring or other mating hardware is relatively insignificant to larger spacecraft, but can become significant to a mission in the smaller classes of service. The mass of the *flyaway* portion of the separation system has to be understood and counted as part of your dry mass budget and on-orbit mass properties. Elsewhere in this chapter, I will simply use the term *delivered mass* synonymous with separated mass.

Another term you will encounter during the selection process, one that will stay with you for the remainder of the mission, is *contract mass* or *contract performance*. This is the separated mass that the launch service provider contractually commits to deliver. When a launch service



Fig. 26-5. Educational Launch of a Nano-Satellite Mission (ELaNa-1) PPOD with Three CubeSats: Montana State University's E1P, Kentucky Space Consortium's KySat, and University of Colorado-Boulder's Hermes. (Photo courtesy of The CubeSat Program, California Polytechnic State University)

provider signs a contract that obligates them to deliver a certain mass margin, they will invariably reserve performance margin to protect themselves against:

- Production engine performance variations
- Production dry mass variations
- Vehicle Mass growth due to upgrades
- Other contributors that can reduce the delivered mass

Unless other arrangements have been made in the contract, this is not your margin to use, and most likely never will be. This might seem unfair, especially when your own management and your customer are criticizing you for inadequate spacecraft mass margins, while you're looking at many kilograms worth of unused propellant or meters per second of velocity margin on the launch side. However unfair it might seem, from a business standpoint, maintaining this margin could be critical for the launch service provider. If they were unable to meet their contractual performance commitment, they could lose money. If there was a failure, an insurance claim could be assessed against them. Purchasing a launch service is a business deal, pure and simple, what matters is what's in the binding contract. Case in point, if you are flying a mission where someone else (e.g., a government agency) contracts the launch service, you may find that the pre-negotiated contract performance does not always match the numbers that the contractor advertises in other forums. Keep in mind that the advertised performance is subject to change, the contracted performance is not.

The technical reasons why the vehicle manufacturers reserve considerable margin for themselves are straightforward. The hardware that will fly your mission was built on a production line. There are production variations in the masses of large metallic and composite structures, as well as the performance of rocket engines, that can carve away significant amounts of performance in a manner that may not become apparent until very late in the production and integration cycle. And, being built on a production line, occasional production errors and handling damage occur. If the hardware is repairable, it will likely be repaired and flown because of the economic penalties incurred by scrapping hardware late in the production cycle. Hardware repair nearly always incurs some mass penalty. Even the winds on launch day can carve out a few dozen meters per second or more if they are blowing the wrong way. I've personally witnessed a few launch counts that were delayed for exactly this reason until the winds became more favorable later in the day's launch window.

Launch vehicle missions are designed and analyzed to specific masses and orbital state vectors. The autopilot and guidance schemes are tailored to the flight profile that delivers your mass precisely to the target that you specified at the start of integration. So again, even though later in the mission integration cycle you find that injection and performance analyses show excess performance available (e.g., 100 m/s boost over and above even the provider's contractual service), unless previously negotiated, this is not yours to use. Once the mission profile is designed and the autopilot and guidance are coded into the software, that is it. Specify your targets early on, in a manner that the launch vehicle provider is aware of the flexibility your requirements may have, so if excess performance is available, a strategy to use it is in place. Case in point, an extra increment of velocity available in the last rocket engine burn, guidance can be programmed to continue to operate engine until fuel is depleted.

26.1.3 Key Dimensions and Volume

Closely behind delivered mass is payload fairing volume and accommodation for key dimensions. If you've already chosen your spacecraft bus structure, then you have a set of physical dimensions that are irreducible beyond a certain point. You can juggle solar panel placement, make antennas stowable/deployable, and add or subtract external instrument platforms as needed, yet the outer mold line of the bus structure normally can't be changed. If you have something like a rigid aeroshell for a planetary atmosphere braking pass or landing, then the capability to accommodate the diameter may become a driver on selection of launch vehicles. Similarly, optics usually aren't stowable, antennas maybe, but not lenses,

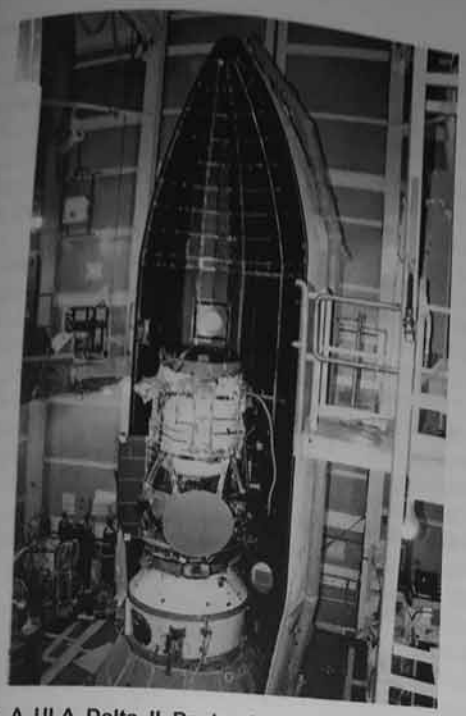


Fig. 26-6. A ULA Delta II Payload Fairing Being Installed Around NASA's Wide-field Infrared Survey Explorer (WISE). (Photo courtesy of NASA)

mirrors, and their support trusses. The WISE space telescope, shown in Fig. 26-6, posed unique packaging challenges (among many others) due to the use of solid hydrogen to cool the infrared detector elements.

All vehicle *user guides* or *mission planner guides* (the terms are interchangeable) will define payload *envelopes*. The terms *envelope* and *volume* are used similarly. I'll use envelope here. You will encounter terms like payload envelope, static envelope, and dynamic envelope. Each of these has a different meaning, and I won't dwell on a detailed explanation because the meanings in all user guides may differ. The physical mechanism behind the terminology as it relates to you is that the payload fairing, spacecraft structure, and vehicle structure are flexible in flight. They all move with respect to each other, reducing clearance. It's also true that many payload fairings have acoustic-dampening blankets installed on their interior which billow as the fairing volume vents during flight. It's necessary to know what the available volume offered includes, irrespective of what they call it, and what remaining incremental clearance losses may be assessed later in the integration process. If you are close to the maximum allowable dimensions, then the first precise analytical estimate of the actual minimum clearance won't be available until the first coupled load analysis cases are run. Prior to that, only historical/statistical estimates with reasonable uncertainty margins will be available. This is only important to the selection process if the spacecraft is approaching the limits of the advertised volume and dimensions. If it is, then this is where you need to read the user guide definition very carefully, and if there's any doubt about the interpretation, then ask the launch service provider for clarification.

The available volume considerations reflect back to the delivered mass evaluation for the reason that many launch service providers offer different-sized payload fairings as options to their customers. At the very smallest end of the performance spectrum, there won't be many options. However, this can change rapidly as soon as the vehicles become only a little larger. Not only different diameters, but also different lengths, become available. With larger payload fairings comes greater launch vehicle mass, and consequently less delivered mass performance available. There also comes greater costs. One example, albeit at the higher end of the performance spectrum, is provided by the $14M–$68M cost increase NASA, [2010] for moving from a 4 m diameter payload fairing to a 5 m fairing.

26.1.4 Trade Study A—Target Orbit, On-Board Propulsion, Mass, and Volume

At this point in the selection process, I would recommend that a trade study be considered involving the four variables discussed thus far.

- Target Orbit and Injection Accuracy
- On-board Propulsion System
- Spacecraft Launch Mass and Margin Policy
- Key Dimensions and Volume

The parameters of your intended spacecraft mission may be such that there is flexibility in one or more of the four. Consider that a mission to measure radiation or magnetic field strength may be tolerant of a wide range of orbit altitudes. The same mission may also be able to take advantage of a range of spacecraft bus sizes and capabilities, trading bus capability against mission life and instrument capability and sophistication to fit your overall budget. The limited size and deployable nature of the antennas or collectors involved makes volume unlikely to be a significant driver. So far, so good, and it appears you have a very large trade space available. However, back to the target orbit, you need to reach a certain orbital altitude, with apogee well above LEO (say, ten or twenty thousand kilometers minimum). This could be a limiting factor in finding a low-cost launch vehicle, for the reason that some of your more-affordable candidates are unable to loft you to the desired orbit altitude. You can do this mission without an on-board propulsion system, in fact you never really considered having one because of the heretofore needless expense and complexity for a mission that was well-tolerant of orbit injection errors. But now, with this trade study, that previous decision (perhaps made unconsciously) comes back under examination, with the finding that the addition of a small solid rocket motor to your spacecraft trades a limited increase in complexity for the gain of being able to use a significantly simpler, smaller, and more affordable launch vehicle.

Consider something of an alternative bounding case, the plight of the user attempting to fly an existing optical telescope with a fixed mirror size. This recommended trade study may be only minimally useful to you if your instrument is already fixed in size, such that the associated bus size offers no significant mass or dimensional trades. If you are such a user, then you may have already objected to the idea of any on-board propulsion system whatsoever because of the associated self-contamination hazards that it poses to your optics (and also probably wish that I'd stop raising the subject). No option for your own propulsion system, fixed orbital altitude due to astronomical viewing constraints, fixed telescope dimensions, and limited bus candidates combine to make this trade study a short and generally unproductive one in this example. Still, for the short time it takes to write a paragraph such as this one, the optical telescope mission designers can assure themselves, their management, and their customers, that they have examined the available trades at an appropriate point in their planning, albeit with no new options uncovered as a result.

26.1.5 Budget, Schedule Requirements, and Risk Tolerance

The question of cost is most closely linked to delivered mass. Reliable cost data in publicly released information is scarce. At best, this discussion represents only a rough estimate of the real launch service prices paid in the international market. US Government agencies rarely disclose the costs paid for launch services, as this data is considered competition sensitive under most circumstances. Launch service agreements between multinational companies and governments can be products of larger business and political negotiations with multiple objectives, and in any case are rarely subject to disclosure in anything other than the most cursory sense. The best advice here is to scour press releases and any published government reports (e.g., the US Federal Aviation Administration (FAA) publishes annual launch reports) involving candidate launch vehicles in order to gain some sense of the possible price range when you become seriously interested in pricing out your options.

You will see discussions in published reports mentioning *performance classes* of vehicles. Generically, vehicles that deliver less than approximately 1,800 kg to 18 km at 28.5 deg inclination (the latitude of Cape Canaveral Air Force Station, Florida) are considered to be in the *small class*. The *medium class*, long the domain of ULA's Delta II and TsSKB-Progress' larger Soyuz, is considered to span the performance range of 1,800 kg to 9,100 kg to the same orbit. Performance above 9,100 kg to this orbit is generally described as the *heavy class*.

Performance of US-flagged vehicles in the small class is bounded among vehicles flying today by Orbital Sciences Corporation's Pegasus XL and Space Exploration Technologies' Falcon 1. The Falcon 1e is an evolved higher-performance version of the original Falcon 1, and was advertised on the SpaceX [2010] website as "the world's lowest cost per flight to orbit of a production rocket." Falcon 1e launch price at this point was advertised as $10.9M for delivering just over 1,000 kg to a 185 km altitude circular orbit at 9.1 deg inclination if launched from the US Army's Reagan Test Site on Kwajalein Atoll in the Republic of the Marshall Islands. The Falcon family of launch vehicles is the only one as of this writing where the manufacturer openly advertises prices to the public.

By contrast, the most recent publicly released price point for the Pegasus XL was $36M for a contract awarded by NASA early in 2009. However, this figure was inclusive of additional integration, payload processing, telemetry and tracking support. This Pegasus configuration is advertised to deliver just over 450 kg from the airspace near the Reagan Test Site (achieving an 11.1 deg inclination) to a 200 km circular orbit.

The Taurus XL, Minotaur IV, and Athena vehicles define the upper end of the advertised small class vehicle performance. With nearly 1,600 kg to a 200 km-altitude circular orbit at 28.5 deg inclination for a Taurus XL if launched from CCAFS and just under 1,750 kg for the Minotaur IV to the same orbit from the same location. The Athena approaching 1,800 kg to LEO, reasonably assumed to be the same orbit as above, or nearly so. The Taurus XL purchase price for a reflight of NASA's Orbiting Carbon Observatory (OCO) is "approximately $70M" in 2010 dollars. This would be the same vehicle configuration as the Pegasus user guide performance estimate. Caution should be used in interpreting this price point, as this represents a reflight of a duplicate spacecraft/launch vehicle combination to one lost in early 2009. No reliable published reports for Minotaur IV or Athena prices were found in the published literature at the time this was written.

For medium and heavy class launch services, the published US and international price points are even more difficult to identify. The range of estimates available for medium-to-large geosynchronous telecommunications spacecraft launch services approximately spans $60M–$150M 2010 US dollars, or rather $100M 2010 US dollars with a 50% uncertainty based on variations in provider, economic strategies or multi-party negotiations.

You will also notice that the cost per unit mass delivered to orbit is not linear with the performance classes. Something to remember is that the larger classes provide more economically efficient bulk transport than do the smaller. This is largely because of fixed costs that are essentially similar for each vehicle, irrespective of size, or lesser demand, which reduces production demand, thus increasing the unit cost per delivered flight article.

Schedule considerations are equally varied. Interplanetary exploration missions with launch dates driven by planetary geometry are subject to negotiations with launch service providers (and their other customers) over the length of the launch window. A launch service contract will invariably specify an expected launch date with the same precision as the allowable mass of the customer spacecraft. Recall that all such transactions are business

transactions, so a spacecraft customer cannot rearrange their launch schedule without negotiations and possibly significant cost unless the contract grants them that freedom (an unlikely condition). Neither can the launch vehicle provider arbitrarily change launch dates without some manner of bi-lateral (usually multi-lateral) negotiations, unless the contract gives them that option.

In general, an established launch service provider with a busy commercial manifest is motivated to maintain a tight, consistent, and reliable schedule. Such a situation is driven by economics, where the ability to maintain a consistent schedule is equally attractive to the provider's own production infrastructure as it is to existing customers and future customers that consider using that provider. This example is of greatest relevance to launch vehicles servicing the international commercial geosynchronous telecommunications market, where a customer's on-orbit revenue is a serious, no-excuses element of the equation, and such is reflected by the contracts that govern the relationship.

For launch service providers that primarily service other markets, such as national security or scientific missions, the situation changes a bit from those that target commercial geosynchronous communications business. A provider with a busy manifest mixed between commercial, national security, and civil scientific missions will have a tougher time maintaining the tight and predictable schedule required of the providers that service mostly commercial clientele. The reason for this is not due to any failing of the launch service provider, but rather due to the nature of their customers. Both the reconnaissance missions undertaken for national security reasons and the civil scientific missions are very often "one of a kind" missions, with bus and instrument designs that are unique. A "one of a kind" mission means that more design and manufacturing issues have to be uniquely worked out along the way to being ready to be launched to go uniquely perform whatever it is that they do. They tend to experience schedule delays far more often than their commercial communications counterparts.

It's logical that there are very few commercial spacecraft, within reasonable limits, that can truly be said to be "one of a kind." "First of a kind" perhaps, but it doesn't make much sense to design a commercial spacecraft and not make any attempt to use it again in a similar application. The details of transponder numbers, frequencies, and signal processing can vary like options on a new automobile, but building a unique bus doesn't often make good business sense due to the resulting lost opportunity to amortize non-recurring costs over multiple builds (and hopefully customers). The situation is often very different for national security and scientific spacecraft, where the goal may be to take advantage of the highest technology available that meets their respective *risk tolerance* (i.e., the sharper the cutting edge, the greater the probability that the flight article might not work as advertised). The spacecraft bus is often uniquely designed around the instrument, and the same instrument would rarely be built twice because the cutting-edge technology selected at the start of the design will most likely be superseded by new development by the time the first spacecraft actually flies. Getting all the pieces to work together for a unique spacecraft is, therefore, more likely to take longer than allotted, with possible impact on the launch service provider and their other customers. The exact figures of launch delays and manifest changes due to problems encountered by the launch service providers, as distinct from the spacecraft customers, are not publicly available, but a consistent examination of published news reports over time should provide an acceptable basis for understanding the customer-driven schedule risks for each vehicle that may fall outside your ability to manage.

The preceding paragraphs are not intended to absolve the launch service providers of responsibility as a source of schedule risk, merely to address the most-frequent contributor first. Launch vehicles in the early part of their flight history are generally much more likely to sustain launch delays for reasons of their own launch readiness than are vehicles that have flown many times in the past. This, too, makes good sense given the preceding discussion of the pitfalls associated with being a unique or "first of a kind" spacecraft. Similar factors apply to launch vehicles. You should be particularly tolerant of launch schedule delays if you decide to fly on one of the first few flights of any launch vehicle. The delays associated with being the first, second, or third flights of a vehicle can often be measured in years.

This is the appropriate point in which to insert a discussion of your overall risk posture, or risk tolerance as it bears on the selection of a launch vehicle. Such factors as availability and reliability are yet to be discussed as selection criteria, but the issue of *availability* is usually addressed up front, in that if a vehicle is unavailable for your use, then you probably won't select them in the first place. *Reliability* is likewise addressed up front in the selection process, as much as possible, but the ultimate consequence won't be realized until your mission actually launches, although the reliability of other launches prior to yours on the same vehicle will have a bearing on your schedule risk, and indirectly, on your cost risk. Before you initiate contacts with the launch service providers, you should conduct an analytical examination of your own tolerance for cost, schedule, and technical risk (i.e., the risk of mission failure due to the launch vehicle).

A few examples are offered to help focus this assessment. Many low-cost scientific or educational spacecraft developed on strict cost budgets, and no launch priority, have a high tolerance for schedule delays and risk of mission failure, if it means the chance to fly. There's nothing wrong with this, and no reason to pretend otherwise if such is the case. Historically, even a first launch of a vehicle has a better than 50% chance of success, no provider in the world commits to such an enterprise with the expectation of failure, so selecting a launch on a first flight of a new or evolved vehicle may offer bargain prices if your schedule and failure risk tolerances are high enough.

Table 26-1◀, Fig. 26-6◀, Eq. 26-0

At another point on the cost-schedule-technical risk shape one might find the ubiquitous commercial telecommunications spacecraft that exists solely to provide revenue. Such enterprises are usually well-funded and able to manage significant cost risks, but the priority between accepting technical and schedule risks is more equal than one might think for such an expensive venture. These customers can and do purchase insurance against mission failure to help offset and manage their cost and mission success risks in a launch industry where a 95% launch success rate is considered pretty good. However, it's a much more difficult prospect to insure against losses of future revenue caused by not reaching orbit on schedule.

A third extreme point on such an imaginary three-dimensional cost-schedule-technical risk shape might be defined by those expensive and high-visibility missions undertaken with high national priority in national security or scientific endeavors. Such missions often seem to have phenomenal tolerance for cost overruns and schedule delays (interplanetary exploration missions excepted), driven by an equally extreme intolerance for failure. Despite the sometimes-extraordinary measures taken to assure success for such missions, they can and do experience launch failures, signaling that there is a recurring reliability limit worldwide in the launch industry. Those of us at all levels in the industry like to say things to each other to the effect that "we can't afford a failure." But the truth is that the industry and the customer base can no more afford a launch vehicle that "can't fail" than can maritime commerce and naval fleets afford a ship than "cannot sink." The analogy to ocean-going ships, including passenger ships, is useful to understand how there is no quantum leap in launch success rates that's likely to be realized in the coming decades. The launch industry in the higher performance classes is in a stable and mature pattern, as evidenced by the continued international use as more governments acquire space-based surveillance and communications capabilities while commercial communications providers are opening new geographic and service markets. Such a pattern does not encourage the investment necessary to achieve said quantum leap in performance or, most importantly here, reliability, over at least the next decade and probably well after.

These three examples are by no means intended to set boundaries on the possible extremes of your risk tolerance as a launch service customer. The reality of your situation probably exists somewhere within the bounds of the examples described, or you may chart new combinations. There's no wrong answer to the equation if you take the time to understand your situation before engaging your candidate launch providers.

26.1.6 Launch Vehicle Capability, Availability, and Reliability

Launch vehicle *capability* is the measure of the vehicle mass, injection, and other performance aspects against your requirements. Simply, can they meet your mission requirements? If not, consider that modifying an existing launch vehicle to add previously unavailable performance is a costly challenge. I have been a part of a couple such efforts, both highly successful and a lot of fun, but they should only be undertaken if circumstances both allow for management of some significant cost and schedule risks.

Vehicle *availability* comes in two broad themes. The first being access, where the vehicle is in production and the launch service is available for you to purchase. An alternative would be a used vehicle. A good example is decommissioned intercontinental ballistic missiles (ICBM), (e.g., Minotaur IV/V series of vehicles) although laden with government requirements, restrictions and general red tape. A series of meteorological missions, one of which is shown in Fig. 26-7, were successfully launched on decommissioned Titan II ICBMs in the late 1990s and early 2000s for the USAF and NOAA. A slightly different example of availability would be US spacecraft manufacturer access to international launch service providers, where the export of US-built spacecraft, even fully commercial ones, for launch on vehicles provided by other countries must pass scrutiny of the US Government, subject to a list of technology transfer, arms trafficking, and non-proliferation policies and treaties.



Fig. 26-7. National Oceanic and Atmospheric Administration's NOAA-M Meteorological Spacecraft Launched on a Lockheed Martin Titan II Space Launch Vehicle from Vandenberg Air Force Base, California. (Photo courtesy of NASA/William Hartenstein)

One specific subset of launch service availability is the influence of the launch site. Large liquid rocket engine vehicles require extensive launch site infrastructure, so they are generally confined to one or two launch

sites. Each launch site has limits on the range of launch and fly-out azimuths that directly limit the nature of missions flown from each. This is straightforward and reasonable, given that most launch ranges are reluctant to explain how empty stages or vehicle debris came to be dropped on populated areas. Large in-flight inclination changes can be accomplished, but are very expensive in terms of delivered mass performance, so they are avoided if possible. They cannot be avoided in a launch to geo-synchronous transfer orbit (GTO) or geo-synchronous orbit (GSO) of a telecommunications spacecraft from either Cape Canaveral Air Force Station in Florida, or Baikonur Cosmodrome in Kazakhstan.

A GTO launch on an Ariane 5 (from Guiana Space Center, Korou, French Guiana) or Sea Launch Zenit (from the equatorial Pacific Ocean) neatly avoids this penalty by launching very close to equator, thus eliminating the need for anything more than the most minor inclination changes. The fuel saved in this manner can offset differences in price by extending a commercial spacecraft's useful life, which means greater revenue potential for that spacecraft over time.

US launch ranges servicing polar or Sun-synchronous inclinations within the past decade only include Vandenberg Air Force Base in California, and the Kodiak Launch Complex on Kodiak Island, Alaska. International sites in Russia, China, India, and Japan also service launches to these inclinations.

On the East Coast of the US, NASA's Wallops Island Flight Facility in Virginia services launches to lower inclinations, approximately 38 deg to 60 deg, which encompasses the inclination of the International Space Station. The US Air Force's Cape Canaveral Air Force Station and adjacent NASA Kennedy Space Center are today, the only US launch range servicing missions to GTO/GSO. Taurus II, currently under development by Orbital, will initially fly from NASA's Wallops Flight Facility in Virginia and advertises a GTO capability with the addition of a solid rocket motor kick stage [OSC 2010b].

Some spacecraft customers have literally funded the construction of new launch sites to provide a capability for a specific vehicle to operate to a unique inclination. If you are not one of those customers, check the user guide and confirm that the vehicle of interest services the inclination you need with an existing launch complex. The same vehicle, normally, will not service the same inclination from more than one range, to do so would be very expensive.

The second broad theme under availability is launch scheduling or manifesting. If you have a commercial telecommunications spacecraft that needs to reach an operating orbit within a short window, then vehicles with existing manifests full of interplanetary and national security missions during that same period might be poor candidates to service your mission, because of the heavy revenue-production pressure under which you operate. And you would, no doubt, view launch delays in terms of horrific amounts of lost revenue per day.

If not under revenue-production pressure, most scientific and defense spacecraft are at least operating under tight budgets, an extra six months to a year stretch to the launch date can have an equally devastating effect as what the commercial communications customers encounter. A cost-capped science mission that spends an extra year on the ground has to compensate for the budget impacts somehow, and this usually means it comes out of the resources available to operate once on orbit, analyze, and publish data. The money has to come from somewhere.

Now, this is a good time to discuss the coupling between availability and another selection factor in this section, *reliability*. High reliability is somewhat neutral in affecting availability. This is because customers want to fly on highly-reliable fleets. They get their missions flown on schedule (mostly), but more customers mean greater pressure on the manifest, making it more difficult for a new customer to break in to the queue for a favorable launch date. Bidding wars can ensue, in extreme cases, increasing costs for the desired launch dates. Low reliability, however, has a deeply negative effect on availability. Two related mechanisms are behind this. First, launch failures have to be investigated and corrective actions taken to prevent recurrence. I've been part of eight failure investigations and none of them have been straightforward in either their execution or conclusions. Failure investigations and corrective actions take time, during which the vehicle is unavailable to service any customers on the manifest. This leads to the second effect, which is manifest backup (envision a traffic backup and you have a good grasp).

Launch vehicle reliability is a subject on which we could generate an entire book devoted to different predictive approaches and their merits. Such a treatment here would be relatively academic to your needs. The first bounds of the equation are historical in nature. New vehicles, those within the first three, six, or even ten launches, historically sustain the highest failure rates. Vehicles with several tens or even hundreds of prior launches generally sustain success rates around 95%, with a few (medium class and above) approaching 98% and very few examples sustaining 100% success over the life of the specific vehicle variant. (See Table 26-2.) Multiple government, private, and commercial websites are available with launch statistics for each vehicle in service today, and this will continue to be the case, so you can access good supporting data to research your own candidate providers with benefit of the most-recent launch trends. But be careful to consider one other aspect; we all fail. We hate it, we try to keep it to a minimum, but it will happen. The next question relevant to you is how long does it take us to come back into service?

Figure 26-8 illustrates an interesting relationship uncovered by Shao [2011] at Microcosm during the preparation of this chapter. I had not previously seen an attempt to correlate failure recovery time to any other predictive indicator. Figure 26-8 strongly suggests that there is such a correlation between the number of launches completed by expendable vehicle fleets and

Table 26-1◀, Fig. 26-7◀, Eq. 26-0

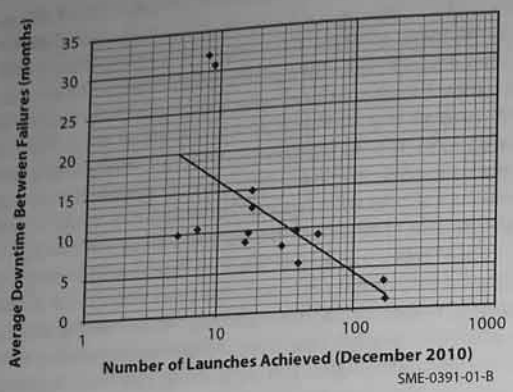


Fig. 26-8. Average Failure Downtime Versus Total Number of Launches Achieved.

their average downtime between failures. You will already have noted that the apparent relationship is logarithmic, and that the scatter decreases considerably after ten launches have been completed. The fourteen international fleets shown in the figure are all currently in service. NASA's Space Shuttle was eliminated from the dataset for the reason that Shuttle failures are investigated with a different set of rules than those applied to expendable vehicle failures. Fleets were also eliminated from the dataset if they had no failures (e.g., Atlas V), or if they had not yet successfully recovered from a failure and flown successfully (e.g., the Korean Space Launch Vehicle), or if they were no longer in service (e.g., Titan IV and Ariane 4).

The 2 apparent outliers in the dataset with fewer than 10 launches and average recovery times above thirty months bear additional explanation. The two fleets do not share nationality, but share other similarities in that they have been in service for many years, though at low flight rates. Both fleets have also been subject to multiple configuration changes since their inception, and of course, both remain in service today. The message here is that the relationship shown in Fig. 26-8, though useful, is subject to a great many factors that scatter the data.

It would have been interesting to observe the scatter among fleets with less than 5 total launches that had also recovered from an initial failure, yet none of the vehicles in service at the time of this writing had less than five flights and also had a successful failure recovery behind them. Note also that one fleet, the Russian Proton, with hundred of launches behind them, was not included due to the fact that their failure recovery history has not yet been studied in the depth required to estimate an average recovery time. However, upon cursory review I'm confident in asserting that such a study would not appreciably change the fit to the data presented here.

The way to use Fig. 26-8 to your benefit is to take it as a rough measure of the recovery time (hence launch manifest delay) required in the event of a failure prior to the flight of your mission. If you choose a vehicle with a

Table 26-2. Reliability Experience of Launch Vehicles as of December 31, 2010.

Launch Vehicle	No. of Successful Launches	Total No. of Launches	R
Atlas V	23	23	1.000
Delta II	163	165	0.988
Delta IV	15	15	1.000
Falcon 1	2	5	0.400
Falcon 9	2	2	1.000
Minotaur I	9	9	1.000
Minotaur IV	2	2	1.000
Pegasus XI	37	40	0.925
Space Shuttle	130	132	0.985
Taurus	6	8	0.750
Long March 2C/D	46	46	1.000
Long March 3A/B/C	36	38	0.947
Long March 4	22	22	1.000
Ariane 5	52	55	0.945
PSLV	17	18	0.944
GSLV	4	7	0.571
Shavit	6	9	0.667
H-IIA	17	18	0.944
H-IIB	2	2	1.000
Dnepr	15	16	0.938
Proton (since 1970)	321	348	0.922
Rockot	16	17	0.941
Soyuz	1654	1753	0.944
Zenit	28	30	0.933

flight history on the order of 20 missions, and that vehicle fails prior to your mission, you can reasonably expect the investigation and return to flight to take between 8 and 15 months. By 30 to 40 launches, that figure is closer to 6 to 10 months. This is a risk factor to consider in your vehicle selection.

Be aware that reliability will not appear uniform over the small, medium, and heavy performance classes. Table 26-2 provides a summary of demonstrated success records for vehicles that are currently in service today. The demonstrated launch vehicle reliability is lesser for the smaller performance class of vehicles than at the larger end of the spectrum. This is not due to any inattention or lack of commitment to success, and the observation is international in nature. The larger vehicles within the industry generally sustain higher flight rates over time, offering an opportunity to accumulate a successful record and average out the limited number of failures. There are other advantages accruing to those fleets that fly more often, notably that the production infrastructure has opportunity to develop a smoother rhythm, as well as higher flight rates offering an economically more-attractive basis on which to amortize the non-recurring costs of design upgrades intended to improve reliability.

Do not rule out the use of vehicles in the small class, only be sure to understand the recent history of each candidate vehicle and carefully consider whether it falls within your risk tolerance. This same should be done for any vehicle under consideration, regardless of class.

26.1.7 Contacting Launch Service Providers

The preceding sections should be considered the *homework* you have completed prior to contacting the candidate launch service providers. A smart buyer brings a credibility that encourages prompt answers with greater depth of explanation than what the obvious neophyte will receive. That said, you won't impress anyone at the launch service provider with your new-found knowledge of the industry, so don't make that one of your objectives. The objective should be to draw out a forthright, knowledgeable set of answers to the questions you've prepared, but more importantly to foster a conversation where they can probe the reasons behind the questions and even argue with your underlying conclusions based on your homework. Some of the arguments they'll offer will have utility to your overall mission requirements development, and some will not. The merits will come out in the discussion. They are in business to make money, and they do that by first gaining your business and then selling even more optional services to you, like any good business operation. Yet, the prospect of your repeat business is a major consideration on their part that helps govern their side of the engagement. No one wants an unsatisfied customer.

Managing your requirements is a key aspect of engaging the launch service providers. The need to actively manage your requirements is self-evident, as reinforced by the process flow diagram offered in Sec. 26.1.1. Also self-evident, is the need to disclose all of your expected requirements, along with the reasons behind them, to facilitate an honest partnering to see how they can best be met, or if they can't (or are too expensive), then to understand the details of why and what that means to you both. I encountered a spacecraft team once that neglected to specify their real sensitivity to moisture in a cooling purge, preferring instead to introduce an auxiliary drying system as part of their on-pad purge regulator equipment. The reason they gave for not identifying the requirement was that they didn't feel the launch service provider could meet it. Subsequently it was discovered, by chance, that the system they brought to the pad unintentionally contained materials that became hazardous to their spacecraft and the surrounding workers once outside of a carefully controlled lab environment. All turned out well in the end, though it reinforced the need to be honest about what you really think you need. Of course, I'll also advise ruthless reconsideration of those requirements that add significant complexity or become cost drivers. The two approaches, full disclosure and ruthless reconsideration, are not incompatible, rather, they are inseparable in the management of your requirements.

Now is a good point at which to discuss expansion and contraction of requirements in the wake of the first detailed interactions with the launch service providers. If you have no requirements for such things as instrument purges, propellant servicing/defueling (required by the range if you have hazardous liquid propellants), exotic gasses, battery charging, spacecraft telemetry and commanding prior to separation, special cleanliness measures, or just about anything else beyond delivery of an inactive spacecraft to a point in orbit, then you can skip the next few paragraphs.

Most launch vehicles will offer a standardized set of umbilical wire channels for spacecraft use, as described in each fleet's user's guide. These are intended to service the common tasks that most of their spacecraft customers perform such as charging or conditioning their batteries, remotely initiating spacecraft prelaunch and test sequences, and having the spacecraft sense separation by means of *breakwires* that open upon separation of an in-flight umbilical disconnect. The wire gauge, number of channels, and provisions for spacecraft ground support equipment will be standardized to meet the average needs of the greatest proportion of their historical customers. They can accommodate any number of requirements beyond this limited standard electrical capability, but doing so will carry added costs.

Likewise, any launch service provider will offer a standard range of environmental control system (ECS) temperatures from which the user can select, once they pass within the provider's processing control for encapsulation in the payload fairing and launch. The provider will offer clean filtered air in their standard ECS, but the precise level of cleanliness may be something that may be important to you if your mission carries instruments sensitive to dust or hydrocarbon contamination [IES, 1992]. Additional cleaning measures, as well as additional cooling or contamination purges can be provided, but usually not as a standard service. Adding auxiliary cooling measures beyond simple purges and vortex coolers is something to be avoided when practical, due to the potentially high associated costs unless the provider has an existing system constructed for other customers that they can make available.

More-exotic requirements, such as strict control over hydrocarbon or helium exposure, or spore and bacteria cleaning and control for interplanetary lander missions, consumable and safety support for instruments cooled by liquid hydrogen or helium, or processing precautions for instruments with extreme sensitivity to radio-frequency or electromagnetic fields are outside the intended scope of this section, but be assured that each can pose considerable challenge to your provider's capabilities, with commensurate costs incurred. Safety and government launch approval for spacecraft containing significant quantities of radioactive materials occupies its own special category, and should only be undertaken by the most well-funded and practiced of spacecraft and launch vehicle teams.

If you must add capability beyond the provider's standard offering, then try to negotiate that as early as possible. Waiting until the launch service contract is signed and the mission integration is well along before

you identify a new requirement is risky from a cost standpoint, because you're already committed to using their launch service by that time.

26.1.8 Trade Study B—Cost, Schedule, Risk, Availability, Capability, Reliability

Per the process flow diagram in Fig. 26-4, this is the point at which I would recommend that you conduct a second of the two mini-trade studies. Recall that what I term "Trade Study A" focused on aspects of your launch requirements where the details are under your control. The second trade study is different in that the details of launch costs, schedules, capabilities, availability, and reliability are mostly outside your control, other than your decision to select the candidate launch service provider that best fits your requirements and risk tolerance.

By this point, your homework is done and you've contacted the candidate launch service providers to gain answers to the many questions you have, and hear their preliminary recommendations and arguments. Crucial in this series of steps was the cold-blooded assessment of your risk tolerance. You remember that assessment, yes? Put the ice water back in your veins and let's reassess the situation. Unless you're a seasoned veteran of developing and flying spacecraft missions, you will have done well through these three boxes on the process flow diagram if only about half of your assumptions and conclusions were questionable, unaffordable, or completely wrong in hindsight after having contacted the launch service providers.

Eliminate the emotional content (i.e., the "abject despair") from the equation and discipline yourself and your team to look only so far back in the process flow as to see what improvements in the situation you can gain by changes in your cost and schedule budgets, or how you can adapt your risk tolerance. If you haven't done so already, now is the time to make sure that you engage an experienced manager that has been where you are and, most importantly, one who thinks through problems in a different manner than you. Write your evaluations and conclusions down before you move to the next trade study. Without changing any of your mission requirements other than cost, schedules, and risk tolerance, document the options you have to improve the field of launch service provider candidates, make a selection, and implement a successful mission.

26.1.9 Trade Study C—Revisit the Options in the Process Flow

My recommendation is that you only now turn your attention to identifying what technical requirements you can change to make your cost, schedule, and risk posture more achievable, and vice versa. You've managed each area individually, technical requirements and cost/schedule/risk, to the best of your ability. Now each side of the equation has to be adjusted in order to enable the whole to achieve a successful conclusion. The intent of this third trade study is not to "revisit everything," as the process flow figure might unintentionally imply, but rather to accomplish a directed search to identify those requirements that have a hitherto unknown significant impact across the boundaries defined for the prior trade studies. You are essentially seeking those few small adjustments in technical requirements that have a significant, better-understood effect on the cost/schedule/risk elements of launch service provider selection.

26.1.10 Selecting Candidate Vehicles and Assessing Launch Environments

Upon conclusion of the third recommended trade study described above, you should be in a position to select a short list of candidate providers and proceed to the next phase of assessing the influence of their launch environments on your design. I use the term *launch environments* here to encompass the wide range of vibration, shock, acoustics, coupled loads, thermal, and electromagnetic influences to which your spacecraft will be exposed during launch. The significance of their effects on your spacecraft will vary according to your size, choice of structural design, nature of your instruments, and configuration of your solar arrays, to name only a few target areas for your evaluation. The short list of candidate providers should be determined exclusively from the outcome of the process flow steps completed prior to this point, and revisited in Trade Study C. The reason for this is that the process of evaluating the detailed effects of launch environment on your spacecraft design moves you one layer deeper in the design of your mission. To put it another way, the impact of accommodating a 20 Hz sine vibration environment, 141 dB overall sound pressure level, and an 8 g lateral load factor should be a little bit lower on your scale of immediacy than finding a vehicle that can (reliably) deliver your mass to the intended orbit, on time, all within your cost budget and risk tolerance. The best advice here is to save the details of launch environments until the other important drivers of launch (i.e., the first seven boxes on the flow diagram) have been, if not worked out, then at least identified sufficiently to be able to narrow the list of candidates.

In terms of seeking a final list of launch vehicle candidates, the idea is to seek not less than two that can do the job within your cost, schedule, and risk tolerance, and probably not more than four or five. Wanting more than one is logical in that you really ought to have a backup solution if for some reason your first choice doesn't work out. It's also desirable for your prospective launch service providers to know that you're seriously considering flying on their competitor. It may not work out that you have more than one launch service provider solution at this stage of the selection process, but it's worth making it a goal. Remember also that every launch service provider wants to fly the next mission; they're always looking for the next customer. While losing your business to a competitor is stinging on an emotional level, losing your business for the reason that you simply can't afford to fly at all, still means that they lost your business. They can't fly you if it means that they will lose

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26.1.11 Proceed Mission

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money, but they should generally be motivated to negotiate a deal you can accept, even if they're the only vehicle that meets your needs.

Section 26.5 of this chapter addresses launch environments in some detail, so I won't delve into the detailed discussion here. The words "assess the associated launch environments" in the process flow represent a detailed comparison of the launch environment specifications in the user's guides and any updates that the candidate providers might offer you when you contact them in the preceding step in the process flow. What you're seeking is a coarse check of whether or not it's feasible to design and qualify your spacecraft to launch environments and loads that encompass competing providers that meet your needs. If you only have one feasible candidate provider, then you should still make such an evaluation of launch environments before you proceed with the detailed design of your spacecraft.

Consider again, the now-familiar example of the commercial telecommunications spacecraft. Spacecraft manufacturers in this market sector find it little short of mandatory that they design, build, and qualify spacecraft that can be launched variously on a Russian Proton-M, European Ariane 5, US Atlas V, or Chinese Long March, plus at least consider the Indian GSLV and the Japanese H-IIA. This is because their customers all have roughly-similar missions seeking cost-effective and time-critical solutions, requiring them to be flexible in their selection of launch service providers for each mission.

Unique scientific spacecraft occupy a different part of the continuum, wherein they frequently have nonstandard antennas, mirrors, lenses, detectors, and the like that require optimized support structures if they are to stay within their mass budget. Such a spacecraft may have a unique problem if their candidate providers are a Russian Dnepr or a US Pegasus XL. The Russian vehicle is launched from a land-based silo with a surge of gas from an ordnance charge, same way as the original RS20 (NATO designation SS-18) missile from which it is derived. By contrast, the Pegasus XL is dropped from a converted L-1011 jetliner. The structural loads may therefore be a bit different from other vehicles that are launched vertically from a fixed pad within a few meters of ground level, and your instrument structural designers may face challenges as a result. If this represents the variety of launch options available to you, then you need to know early so you can decide, as an example, whether to consume part of your mass margin to beef up your structures and instruments to accommodate a launch on either vehicle.

26.1.11 Proceeding with Detailed Spacecraft and Mission Design

This section is merely an extension of the paragraphs above. It's a separate box on the process flow for the reason that you're now another layer deeper into the detailed design of your mission than in the previous box. Commensurately, you're also deeper into your evaluation of the effects on your design from the launch environments and interfaces of each candidate launch service provider, as you complete your detailed trade studies and design concept definitions to show feasibility of your preliminary spacecraft and mission design.

I offer the advice to maintain two or more candidate providers through the completion of your preliminary design as a notional compromise among possible mission types in the process flow, so please consider it in that light. The commercial telecommunications spacecraft program is going to maintain the ability to fly on multiple candidate vehicles all the way through manufacturing and delivery, for the reasons already discussed. The micro-, nano-, and pico-satellite missions of the world may find it simple enough, because of their small size, to build their spacecraft in a manner so robust as to launch on pretty much any provider that presents an opportunity for flight. However, the complex large space telescope mission with the need for highly-optimized structure built of ultra-light exotic materials skipped the rest of this chapter long ago and has already committed to designing with a single set of launch vehicle environments long before completion of the preliminary design.

Unlike the bounding examples in the previous paragraph, other spacecraft missions may be able to maintain, or at least attempt to maintain compatibility with more than one candidate provider, and even incorporate differences in the candidate provider environments and interfaces in the detailed trade studies. Classically, most of the detailed design trade studies should be complete by the time the *preliminary design** is declared to be complete and shown to be feasible, so this is the rationale by which it makes some sense to proceed into launch vehicle selection after completion of your mission's preliminary design.

26.2 History Prior to 2010

The two decades between 1990 and 2010 experienced a few significant evolutions in the launch industry. The commercialization of the US launch vehicle fleet through a succession of "Commercial Space Launch Acts" passed by the US Congress and signed into law by the President, beginning in 1986 and revised over the ensuing years, was in full swing by 1990. The US Government (principally NASA and the DoD) ceased to control the designs and commercial profit-oriented pressures increased. Many consider that beneficial changes resulted, but nevertheless it did change the way the industry interacted with itself, the US Government, and the spacecraft customers. The rapid evolution of the Atlas II/IIA/IIAS family into a commercial service provider between 1991 and 2004, paving the way for the Atlas III as a step to the development of the Atlas V, is a

* The context of the term *preliminary design* being the same as in the term *Preliminary Design Review* (PDR) in NASA Standard 7123.1 "NASA Systems Engineering Processes and Requirements."



Fig. 26-9. India Space Research Organisation's (ISRO) GSLV. Launching from Satish Dawan Space Center (SDSC), Sriharikota, India. (Photo courtesy of NASA)

case history that bears some study outside the scope of this book. Likewise, the evolution of the European Ariane family of launch vehicles to the current Ariane 5 and significant growth of the Russian Soyuz, Proton, and Zenit vehicles within the international market were all accomplished during this period.

The entry of the Commonwealth of Independent States (Russia) to the international commercial launch service market, first hesitant and unsuccessfully-marketed in the late 1980s, gained strength and success by the mid-1990s, marked by the launches of the first international commercial communications spacecraft on the Proton vehicle in the middle of that decade. The European Ariane 5 vehicle first flew around the same time, and after initial struggles achieved a dominating succession to the highly-successful Ariane 4 family. The Chinese Long March family also entered the international market at the end of the preceding decade, gaining a slow but surprisingly inexorable foothold in the market. The commercial telecommunications spacecraft launch market in 2010 is dominated by these three families of vehicles, the Proton-M, the Ariane 5, and the Long March 3. The Indian GSLV (Fig. 26-9) and Japanese H-II families have become noteworthy, and are gaining ground, but still occupy a much smaller fraction of the current international launch business base.

The previous two paragraphs are focused largely on the international telecommunications market, for the reason that this is where the money and the launch vehicle performance count the most in the commercially-available market. This is still true, commercially, in that the geosynchronous telecommunications market is where the money and performance follow, as simply an artifact

of revenue. Competitive vehicles in this market will push 5,000 kg–8,000 kg to geosynchronous transfer orbit (GTO).

The beginning of the 1990s also saw the success of the far-smaller Pegasus air-launched vehicle in the US, one which arguably spawned a series of competitors, with mixed success, at the lower-performance end of the industry in hopes of attracting smaller "better-faster-cheaper" scientific spacecraft, as well as the believed-to-be-lucrative low altitude commercial lightweight communications spacecraft market that somehow just never appeared. This period, with the Athena, Taurus, and unsuccessful Conestoga joining the Pegasus to make a credible push as successors and exploiters of a market previously only served in the US by the aging Scout vehicle, was important, particularly in accelerating the innovation in directions, successful and otherwise, away from the classical government-directed standards established in the preceding three decades. They "pushed the envelope," to use an overused but still appropriate term. There was a great deal of healthy challenge occurring in the community at the time, and it continues in a similar manner at the time of this writing in 2011.

The preceding paragraphs are significant to you because understanding the history behind certain classes of vehicle will aid in understanding some of the attitudes and mindsets that will be encountered within different performance classes across the industry. Today's small class is bounded on the lower end by Pegasus XL (just over 450 kg to 370 km) and Falcon 1 (just under 400 kg to 370 km) at similar inclinations, both launching from the near-equatorial Reagan Test Site operated by the US Army. Athena IIc (not yet flown, but similar to the Athena series from the 1990s) and Minotaur IV advertise a bounding upper-end performance between 1,100 kg–1,200 kg, albeit to higher inclinations. The theme running through it all is that the small launch vehicle survivors of the early 1990s are the ones that successfully came to grips with the reduction in government oversight, control, funding, and direction that started at the end of the 1980s, together with the commercial market emphasis, without the guidance of a successful heritage. The successful approaches were different in each fleet. Those at the lower end can be said to have clawed their way up with no government support. It's not exactly true, but neither is it terribly far from the truth. They paved a rough road, and their achievement is, therefore, no less significant than those at the highest-performance end of the industry.

The *medium class*, described by the National Academy of Public Administration in 1992 as 4,000 lbm–20,000 lbm (1,814 kg–9,071 kg) to low earth orbit (LEO), also emerged in the early 1990s as a useful niche for a great number of scientific missions. The US Air Force moved in the wake of the Challenger accident in the mid-1980s to update the Delta II launch vehicle to seed the constellation of Global Positioning System (GPS) spacecraft. NASA had a long and successful history of managing the Delta evolution, but acceded to the commercial, then mixed with military, management of

the future fleet development as a part of NASA's mixed fleet strategy in the wake of Challenger and the first of the Commercial Space Launch Acts. The timing was excellent, and the combination was interesting.

NASA's emphasis on the better-faster-cheaper strategy early in the 1990s, that was ultimately met with mixed success, pushed scientific missions originally planned for the Atlas II, Titan III, and the newly-developed Titan IV "down," if you will, to the Delta II-class. Call it 2,300 kg or 5,000 lbm for convenience. The only one there in the US fleet was the Delta II. The outcome was successful in that the "dinky little Delta II," as some noteworthy interplanetary customers deemed it in 1996, became the vehicle of choice for US (and several international) science missions over the ensuing decade. One military GPS mission was lost in 2007 due to a launch vehicle failure, and three NASA spacecraft missions were lost (Contour, Mars Polar Lander, and Mars Climate Orbiter) through no fault of the launch vehicle. Over that period, even as the GPS constellation planned a transition to the Evolved Expendable Launch Vehicles (EELV) fleet funded by the DoD, the science mission successes were otherwise tremendous.

The capability was just too useful to ignore. The commercial customer base waned rapidly as soon as the Iridium constellation was seeded. Geosynchronous-orbit communications spacecraft were becoming larger to serve the global demand within a finite number of orbit locations. Accordingly, the NASA and remaining DoD GPS missions dominated the previously-commercial manifest. The retirement of the Delta II fleet and loss of the medium-class niche prompted Congressional investigations, even as the Falcon 9 and Taurus II are approaching readiness to fill the same niche as this is written.

26.3 Basic Mechanics of Launch

Many, if not most, launch service providers surround the final preparations for launch with great ceremony. The launch vehicle folks are well-practiced, cultivating a cool calm air, almost nonchalant, but not jaded. The spacecraft customers are at times melodramatic as they describe the importance of their missions during the final readiness reviews, always thanking their teams and management for keeping faith during the "long road to get to this day." It is a big deal, and rightly so. The professional (and realistically, some degree of emotional) investment of everyone involved has reached its peak. Launch will be preceded by a precisely-spaced set of readiness review meetings among the various entities involved; spacecraft, range, vehicle, and finally all parties coming together one last time to signal their readiness to proceed into the actual launch procedure itself. The final launch commitment is accumulated through a series of readiness polls on the voice networks during the launch count, and then the last few seconds count down to t-0.

Somewhere within the last five to ten seconds (for most vehicles), the control of a launch passes beyond human operators and is now handled by machines and software. All elements have previously and exhaustively tested to confirm they will function as expected, but there's a subtle transition in that the engine ignition, final health checks, structural separations and umbilical separations are all happening too fast for human intervention. The commitment has been made, and now the machinery works, or it doesn't. Beyond the smoke and fire in Fig. 26-10 that we all enjoy, it's the visceral realization of the enormity of this commitment to flight, and the consequences of a failure, that generate the adrenaline rush among even the most experienced among us. The thrill and the fear never get old.



Fig. 26-10. Orbital's Taurus XL Launching at Night from Vandenberg Air Force Base, California. (Photo courtesy of NASA)

The engines ignite (Fig. 26-11). The ground and flight systems complete final mechanical, electrical, or digital negotiations milliseconds later with an agreement that the systems are healthy and the vehicle is indeed flying. The last of the umbilical booms or lanyards start the process of retracting, and the vehicle begins rising off the pad. The acoustic noise of the engine is reflecting off of the ground and the umbilical tower, hitting the vehicle with what will be one of two driving peak vibro-acoustic environments that can do real damage to hardware if not properly designed and qualified. If not considered and mitigated, any vehicle can literally shake itself apart with the echoes of its own noise.

The big vehicles that use only liquid rocket engines seem to move in painfully slow motion at first. The Atlas V 401 configuration shown in Fig. 26-12 is a good example. There's enough time here to grind your teeth,



Fig. 26-11. A ULA Atlas V Vehicle with Engines Ignited, with Umbilical Separation in Process. (Photo courtesy of NASA/Tom Farrar and Kevin O'Connell)



Fig. 26-12. A ULA Atlas V Slowly Ascends Past the Mast of the Mobile Launch Platform. (Photo courtesy of NASA)

ponder your sins, and wonder briefly whether everything is working properly. The reason for this is that the optimal thrust-to-weight at liftoff for such a vehicle works out to be somewhere between 1.1 and 1.3, with the variation depending on a few second-order effects. What it means is that your most-efficient liquid rocket booster design is going to move slowly off the pad. In an optimal design there's no point in adding any more thrust to the engine than what the vehicle needs in order to reach this initial thrust-to-weight range. On the other hand, vehicles with solid rocket motor first stages, or those like the Atlas V, Ariane 5, GSLV, H-II, and Delta II/IV that use varying numbers of liquid and solid rocket motors, usually have much higher thrust-to-weight ratios at launch, and so appear to get off the pad much faster and depart the vicinity at a good clip.

The vehicle is moving faster now and about to rise past the tallest point of the umbilical tower. One more hazard is retired once the ground-level winds can no longer blow the vehicle into contact with the tower as it passes by (yes, we have to worry about that). The vehicle may have one last collision hazard to clear if the launch site employs protective *catenaries* like those shown in Fig. 26-13, which are a network of wires over the pad that serve the same purpose as lightning rods on a building, but this hazard is also swiftly passed as the vehicle gains speed.



Fig. 26-13. An Atlas V Clearing the Last of the Launch Site Collision Hazards. (Photo courtesy of NASA)

As soon as the last collision hazard has passed, most vehicles will begin their pitch-over maneuver and start heading downrange as quickly as possible (Fig. 26-14). It's a common misconception that orbit is achieved by reaching the "altitude of space," unintentionally promulgated in the early media coverage of spaceflight more than fifty years ago. Achieving orbit is all about speed. The reason why no one orbits the Earth at 10,000 m or even 100,000 m is that the aerodynamic friction would burn them up (however, an orbit in that altitude range is perfectly feasible around the Moon). Rising vertically doesn't do much for you besides getting the ground out of your way and reaching thinner air more quickly. The part about reaching thinner air more quickly can work in a vehicle's favor by trading performance (performance that is lost fighting gravity while flying nearly straight up) for reduced air density when the vehicle starts to put on the speed, thus reducing maximum dynamic pressure and, by extension, reducing the associated vibro-acous-



Fig. 26-14. An Atlas V Beginning to Head Downrange. (Photo courtesy of NASA)

tics from this second peak driving conditions. We refer to the maneuver as a *lofted trajectory*, when we fly away at a higher angle and bend flight path higher than what an optimal trajectory would dictate. If the guidance and performance engineers have their way, however, we will get the vehicle flight path bent over and headed down range just as fast as the stability/controllability, thermal, strength, and vibro-acoustics considerations will allow.



Fig. 26-15. An Atlas V Flying at Supersonic Speed, Nearing Peak Dynamic Pressure. (Photo courtesy of NASA)

The initial pitch-over maneuvering phase is complete within a few seconds. Now the vehicle is approaching Mach 1, and the dynamic pressure is building (Fig. 26-15). *Dynamic pressure,* or *Q,* is the pressure added by passing through any medium with a non-zero density. The resulting pressure is computed by multiplying half of the density by the square of velocity. The only thing about this equation that's important to you is that the dynamic pressure varies as the square of the velocity, so a modest velocity increase will have a much larger effect on the dynamic pressure. As noted previously, a higher maximum dynamic pressure increases the associated vibro-acoustic environment. It also makes the vehicle harder to control, because the disturbing moment (the torque attempting to make the vehicle turn around because it's actually aerodynamically unstable, but don't tell anyone that because it makes people nervous) is a linear function of dynamic pressure. Flight through high dynamic pressure lasts anywhere from thirty seconds to a minute, while the peak dynamic pressure lasts less than ten to twenty seconds. Flight through maximum dynamic pressure is usually preceded by a couple seconds spent crossing from subsonic to supersonic flight as the vehicle passes Mach 1. The flow conditions over the vehicle as the vehicle passes through Mach 1 are difficult to predict, so we settle instead on designing the autopilot to handle large uncertainties in the disturbances. If you haven't picked up on it yet, we're just trying to survive at this point. There are no points for style. Flight through Mach 1 and peak dynamic pressure is the most-stressful period of the flight, if one considers the aggregate effects and minimum margins on controls, structures, and environments for the whole vehicle system. Things go wrong in a hurry at high dynamic pressure. Small errors here can quickly drive anyone's control system beyond the capability to recover, or overload the vehicle structure.

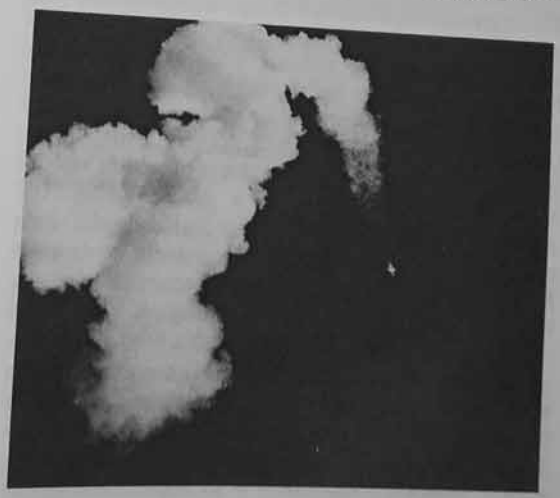


Fig. 26-16. High-Altitude Winds Twisting an Atlas V Contrail as the Vehicle Rapidly Heads Downrange. (Photo courtesy of NASA)

The vehicle flight software will be pre-programmed to fly a zero angle-of-attack profile throughout the period of high dynamic pressure. This is also sometimes referred to as a *gravity turn*. Angle of attack, or alpha, is the angle between the vehicle long axis and the velocity vector relative to the local medium through which you are flying—that's a long-winded way of saying that the

air through which we're flying is moving, too (see Fig. 26-16, the technical term is wind). The fact that we're flying through wind can make things more difficult because even very small angles of attack can create large disturbing moments that require the engine's thrust vector to be pointed to near-maximum capability. That's not a place where we want to be, so some vehicles implement a correction factor loaded into the software on launch day, based on measurements taken by weather balloons or special ground-based Doppler radar [Dhuyvetter and Lam, 1989].

As we climb away from the thicker air, and the dynamic pressure falls off quickly, the vehicle can once again start commanding various pitch profiles to make the trajectory more efficient for reaching the final target orbit. Some vehicles, those with more than two stages, will execute a staging event here, and so, delay the resumption of trajectory optimization maneuvering until after the spent stage has separated and the next stage has ignited the engine and achieved stable powered flight. On certain vehicles, we may also "close the guidance loop" at this point, meaning that the flight software is no longer exclusively following a pre-programmed path, and instead making changes to the trajectory based on the actual flight performance achieved thus far.

The vehicle is really putting on the speed, now that most of the atmosphere has been left behind. There may also be staging events here, depending on the vehicle. The next big event is to jettison the payload fairing, once the density of the now-rarified atmosphere has fallen below a point where the resulting aerodynamic heating will remain below what you were told to expect for your spacecraft—hopefully someone mentioned that along the way, as there's a certain small amount of aerodynamic heating you'll need to be able to accept. They'll usually talk about it in the user's guides. Dropping the payload fairing also drops all pretense of vehicle aerodynamics, so we're now truly a space vehicle. The guidance loop, briefly interrupted for payload fairing separation, is active again, and the vehicle guidance is making minute adjustments to the trajectory to attach the target orbit. Again, we're still putting on all possible speed in an effort to reach a sustainable orbit.

The first orbit may be your final target, if you've asked for a low-altitude orbit, or it may be a parking or transfer orbit that will be followed by another engine firing (or two) to reach your final destination. A *parking orbit* is technically equivalent to an airport holding pattern, in which we would wait to reach a correct orbital geometry from which to continue your flight. We don't often use true parking orbits in expendable vehicles, as they are something of a holdover term from the early days of manned flight where the crew could conduct a quick checkout of the capsule prior to moving on to a higher altitude or a lunar transfer. A *transfer orbit* is an optimal orbit where you are coasting outbound to a point where we will either place you in your final orbit, or apply one more engine firing to achieve yet another optimal transfer to an eventual high-altitude destination that you specify (note that a three-impulse transfer is only more efficient for high altitudes, as opposed to an optimal two-impulse or *Hohmann* or *Hohmann-Vetchinkin Transfer** that we normally try to approximate to the most-practical extent consistent with our vehicle is capability).

One consideration for you while we have you in a transfer orbit is deciding where you will want to be pointed relative to the Sun. Your attitude is obviously not under your own control during this period, nor is your spacecraft likely to be in the final configuration where you can best handle the heat and direct radiation from the Sun. So this will need to be a subject of discussion between you and your launch service provider to plan out what solar exposure you can tolerate. An example would be the sensitive and demanding space telescope that they forgot to design a cover for their aperture. Such a space telescope would obviously never want to find its aperture pointed directly (or even near) the Sun, unless of course the Sun was its intended subject of study. Some spacecraft desire to be in a slow roll as they coast to orbit to even out the heat load over their surface until their own thermal management radiators can be deployed. Still, others have no issues so long as their fixed solar arrays on their bus structures are exposed to sunlight so they can become power-positive as soon as possible.

Finally, we reach your required orbit. We will be grateful to be rid of you, excuse me, to allow you to start your wondrous journey through the cosmos. We mean that. But there are a couple more things we need to do first. The first is to point you in your required direction. Some spacecraft have no issue in being deployed in any attitude, and are insensitive to tumbling. Others are only equipped to hold an attitude efficiently, and have great difficulty accomplishing significant attitude reorientations. Many are sensitive to deploying in an attitude in which they are passively power-positive with their solar cells/arrays, to guard against any of their own potential attitude control problems as they accomplish their turn-on and checkout.

The next order of business in the separation sequence is to send any *wakeup* or separation warning signals (aka *discrete*) to your spacecraft to announce that you are about to be dropped off at your destination. Shortly after that, we will separate you. This is not trivial. I will always recall clearly the day when we delivered two spacecraft to the correct orbit, but then were unable to separate them. It was not a good day, and we like to keep that kind of thing to a minimum. When we separate you, it will either be with springs that impart a significant mechanical separation impulse, or else, if you so request, we might separate you as gently as possible, then back away. In either case, such active measures as are possible and practical will be employed to ensure that our spent upper stage poses no collision or contamination threat to your spacecraft. If

* Named for German mathematician Walter Hohmann, this orbit is also known as the Hohmann-Vetchinkin transfer orbit, to include Russian mathematician Vladimir Vetchinkin. [Source: Websters Online Dictionary]

possible, and for many vehicles it is possible, we will burn the remaining propellants to move the spent stage into a disposal orbit to either await a near-term decay and reentry or else be safely marooned outside the orbits of useful spacecraft. After this is completed, those of us that are vehicle hardware will happily become benign orbital debris, while those of us that are human will wait to hear that your spacecraft is reporting good health and safe attitude, and only then retire to a place of comfortable liquid celebration while you begin your mission.

One final word of concern, "caution" doesn't seem right, the cliché is overused. It is a serious concern for unique spacecraft designs that the transition from being a launch vehicle payload to being an autonomous spacecraft be managed with the utmost attention to the potential for things that can go wrong once your hardware is on its own. The vehicle community is painfully aware that our 95%-on-a-good-day success rate poses the greatest statistical threat to your mission once you get so far as reaching the launch site. No argument there. Everything practicable will be done to deliver you successfully, but it remains a high-risk business. Even so, there have been occasions where the spacecraft was deployed in the correct orbit, in the correct attitude, and was lost to a failure (or just nearly so) in the transition. The message is that spacecraft events like power systems turn-on, array deployment, GN&C handoff, and the like deserve extra attention in order to make certain that each sequence has been thoroughly modeled and tested on the ground in as flight-like a manner as possible. Here is the point at which you cease to be a payload and instead become a true spacecraft, responsible for your own fate thereafter. Please make a point of analyzing and testing every contingency you can possibly imagine while you're in that transition.

26.4 Launch Environments

This section addresses the environments that are peculiar to launch (Fig. 26-17), atmospheric flight, and separation. Table 26-3 outlines the sequence of topics in this section. Elsewhere in this book, there are ample treatments of thermal and space radiation environments, so those will not be addressed here. As previously noted in this chapter, the explanations offered here are intended to be at a fairly-high level; heavy on qualitative examples with minimum use of equations. Each subsection lists useful references that provide further detail and explicit guidance on approaching each environment. Mostly, this section is written to target the things that my colleagues and I wish that all spacecraft customers knew before getting themselves involved with the likes of us.

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26.4.1 Coupled Loads

Unlike the more-straightforward shock and random vibration environments, coupled loads are dependent upon launch vehicle and spacecraft structural interactions as a complete structural system. This creates the potential for unpleasant surprises late in a spacecraft integration cycle when the spacecraft mathematical model is finally verified with test data. Having had exposure to a good many one-of-a-kind scientific and military spacecraft over the years, I can assure that you will (and should) lose sleep over their coupled loads analysis if you're planning to fly a unique spacecraft bus or large, fragile instrument.

Most project managers, and I'll go so far as to say most engineers, don't like to think much about coupled loads. First, it's all about the mathematics and uncertainties. It's not one of those "honest" environments that you can test, except when you go fly a mission. Even when you make flight measurements, you can somehow never quite seem to convince the community to use those to reduce the conservatism applied in their analyses. It's all about obscurely-derived forcing functions, large finite-element-derived matrix models, tedious verifications,



Fig. 26-17. Space Exploration Technologies (SpaceX) Falcon 9 Vehicle Lifting Off from Cape Canaveral Air Force Station, Florida. (Photo courtesy of NASA)

Table 26-3. Launch Environments.

Environment	Subsection
Coupled Loads	26.4.1
Shock	26.4.2
Vibration and Acoustics	26.4.3
Electromagnetic Compatibility	26.4.4
Ascent Venting	26.4.5

lengthy analysis cycles, long reports full of numbers, and vociferous debates over uncertainty factors. You don't want to think about it, the subject makes your head hurt. Coupled loads analysis and the resulting effects on spacecraft structural design is a new case study in technical risk management for every single mission. It requires careful attention from day one of your mission planning cycle until the receipt and review of the final verification loads cycle analysis. Hopefully this has the reader's attention. The challenge is far less intimidating if faced head-on, but can become terribly costly and nearly unmanageable if ignored.

Physical Mechanisms

Physically, the launch vehicle is a large, lightweight structure wrapped around a much greater mass of liquid or flexible solid propellants (yes, solid rocket propellant is quite flexible, equate it to your pencil eraser, it's very similar) balancing on a rocket engine that's continually changing thrust levels and moving from side to side in two axes. Your spacecraft is part of that structural system. Your spacecraft's own structural flexibility directly and indirectly affects the larger launch vehicle because, like your spacecraft, the launch vehicle is only as stiff as it needs to be. They design on a budget, just like you. Designing propellant tanks and thrust structures (and that's the next largest fraction of vehicle mass, behind the mass of the propellants themselves) that are one bit stiffer than the minimum design requirement imposed by the loads and controls analysts is a waste of performance.

It's good practice to think of the vehicle and spacecraft as a collection of point masses coupled together with a complex network of lightly-damped, massless springs of varying stiffness, suspended without being fixed to the earth. That's essentially how it's mathematically modeled and then transformed into complex matrices for conducting the analysis. This coupled spring-mass system is then subject to a variety of disturbances, both self-inflicted and otherwise. Rocket engine ignition and thrust ramp-up is a good example of the former, while aerodynamic *gust* loads are examples of the latter. This mental model may help you understand why some interactions and forcing functions only reveal themselves through flight data, occasionally introducing analysis cases that weren't part of the deal when you signed up to fly on that particular launch vehicle.

All well and good, you might say, and thanks for the physics lesson, but doesn't a launch vehicle user's guide levy load factors and a requirement that customers keep their lowest structural frequencies (many times referred to as *first bending modes* or *primary modes,* refer to Table 26-4 for examples) above a certain frequency? And isn't it true that a coupled loads analysis is only applicable up to 50 or 60 Hz, so we only have to worry about a 25 Hz–35 Hz span of modes? That can't be so bad, can it? The rub is that this doesn't mean the spacecraft with no primary structural flexible modes below 25 Hz is safe from interaction, only that the interaction is unlikely to directly threaten the vehicle by coupling with the attitude rates sensed by the vehicle's inertial navigation system. Plus whatever the spacecraft bus primary modes might be, there may be a number of localized structural features with lower frequencies. There will also be coupling effects that occur with higher frequency launch vehicle structural modes. And the load factors given by the launch vehicle are preliminary estimates based on the types of missions and customers that they've flown before. You may be unique to their experience, as many of my spacecraft customers have found to be the case. If you have a large optical mirror, large antenna, or delicate microgravity isolation system, we can hurt your spacecraft if we're all not very careful.

Table 26-4. Minimum Recommended Spacecraft Fundamental Structural Frequencies.

Vehicle	Axial (Hz)	Lateral (Hz)
Ariane 5	27–31*	7.5–10*
Atlas V	15	8
Delta II (two-stage)	35	12
Dnepr	20	10
Minotaur IV	n/a	25
Proton	25	8.5
Soyuz	35	15
Zenit	20	8

*Minimum varies with spacecraft mass and interface diameter

Along with the frequencies in Table 26-4, you'll also be given a set of *load factors* that are intended to serve as design guidelines until a true coupled loads analysis can be completed. The load factors in Table 26-5 are representative of those provided in most user's guides, although the degree of specificity will vary with launch vehicle provider. These load factors are universally intended to be applied at the spacecraft center of mass, assuming the spacecraft to be a rigid body. Table 26-5 gives you something to use in your designs that is intended to envelope (for most customers) what the more-refined coupled loads cases will give you eventually. The caveat to this design guidance is that the load factors are something of a compromise because it's not realistic for the launch vehicle provider to try to envelope all spacecraft configurations that will eventually be brought to them. If your spacecraft is short, symmetrical, and has relatively-high primary modes (i.e., if you resemble a brick), then the factors given to you will be conservative. If your structure is long and delicate with internal modes at or below the recommended minimums in Table 26-4, then Table 26-5 may not be conservative. Note that the values shown are a combination of steady and quasi-static loads. The values shown are approximately comparable as representing flight through high dynamic pressure.

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Get Stnd

NASA-STD-5002 illustrates the coupled loads process with the flowchart shown in Fig. 26-18. Preliminary structural design of a spacecraft is

Table 26-5. Representative Design Load Factors for Several Fleets.

Vehicle	Axial (g)	Lateral (g)
*Delta II (liftoff and high dynamic pressure)**	2.8 ±0.2	±3.0
Dnepr (maximum dynamic pressure)	3.0 ±0.5	0.5 ±0.5
Rockot (maximum dynamic pressure)	2.7 ±0.4	±0.9
Minotaur I (supersonic)	3.8 ±0.5	0.4 ±1.1
Pegasus (aerodynamic pull-up)	3.7 ±1.0	−2.3 ±1.0**

*Valid for payload mass range of 1,134–2,268 kg.
**Longitudinal axis only. Lateral load factor is ±0.9g

accomplished using, among other inputs, recommended load factors for the launch vehicle on which they'll be flying. Then the designer of the spacecraft (or payload in the figure) develops finite-element-based mathematical dynamic models for use in conducting an analysis of the actual loads expected for that specific vehicle/spacecraft combination. The model is submitted to the launch vehicle team, and the launch vehicle team performs the analysis and provides results to the spacecraft. This is a "preliminary coupled loads analysis," often times referred to as a *PCLA*. At the time of a PCLA, a spacecraft may be in any of a variety of design phases. It's possible that the design is still evolving, or the spacecraft may be one of those that were built years ago and still sitting around waiting for a ride. Let's assume the former condition, where the spacecraft may be approaching CDR or still early in the build process. The first coupled loads analysis is intended to provide a check point to a spacecraft while design changes might still be made. It's the first look that a spacecraft will have into the forces, stresses, accelerations, and deflections that are recovered with real vehicle and spacecraft models under realistic forcing functions. The spacecraft team will assess structural margins against the predicted loads and may find it necessary to alter the design to adjust modal frequencies or increase strength.

Once the design is finalized and no further changes are anticipated, then what's often referred to as a "final design coupled loads analysis" (*FDCLA* or *FCLA*) will be performed by the launch vehicle team using the updated model submitted by the spacecraft customer. The loads obtained by this loads cycle are typically used to support the spacecraft CDR.

NASA-STD-5002 and Preferred Reliability Practice PD-AP-1317, referenced earlier, each offer some good high-level language describing the requirements for verifying the accuracy of a model. This is most often done with the aid of modal frequency measurements acquired during spacecraft vibration or preferably through dedicated modal surveys. This process is where nonlinear structural behavior is investigated, predicted and test-generated structural frequencies are compared and the model adjusted (5% or better is the goal for significant modes), modal damping values are confirmed (or adjusted as necessary), and orthogonality checks are completed. Once the model is said to be correlated, it is ready for submission and use in what's usually termed the "verification coupled loads analysis" or *VCLA*. The VCLA is intended to be the final analysis cycle, from which final strength margin numbers and predicted deflections (for use in clearance analyses) will be generated.

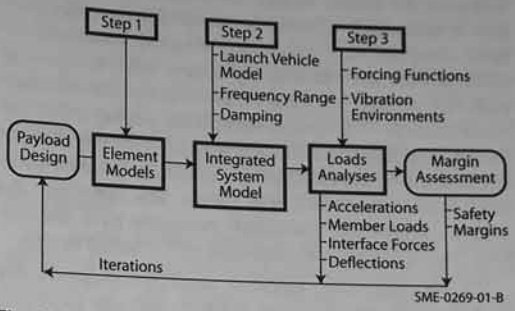


Fig. 26-18. The Load Cycle Analysis Process. (From NASA-STD-5002, "Load Analysis of Spacecraft and Payloads" NASA Technical Standard, June 1996.)

Notice the word, iterations in the lower left hand corner of Fig. 26-18. As described above, the process of design, loads analysis, and safety margin assessment is very much an iterative process unless steps are taken to reduce the likelihood of finding a negative safety margin.

It is understood that a test-correlated model may have to wait until spacecraft structure is assembled and subject to modal testing, which may occur relatively late in the mission integration cycle. Here's another detail that requires some clarification. The finite element modeling doesn't just automatically spit out a complete, ready to use, mathematical model of the spacecraft. There's some art to adjusting the model so that it appropriately reflects the spacecraft for the purpose of the analysis. To mitigate the risk of generating surprises late in the mission integration cycle at VCLA, we use model uncertainty factors in all load cycles. These factors typically decrease with design maturity and by gaining confidence in the mathematical models via testing. Recommended values for the model uncertainty factors for different load cycles are:

- For PCLA: 1.4–1.5 to account for future design changes and using mathematical models that are not test verified
- For FCLA: 1.2–1.25 to account for using mathematical models that are not test verified
- For VCLA: factor decreased to 1.1 with test-verified models

It should also be noted that the model uncertainty factor can never be a substitute for model verification.

Mitigations

There are no mitigations to reduce the length or iterative nature of the analysis process. Having a spacecraft

that's already built and tested can be an advantage, but only if the proper dynamic model correlations have already been performed. The drawback for this situation is that, if a negative margin is identified, then either a redesign or some manner of motion control system becomes necessary.

One risk management option is to seek out spacecraft bus manufacturers with designs that have previously flown on the launch vehicle of interest. The idea being that a spacecraft primary structure having previously been analyzed in a completed coupled loads analysis cycle affords less uncertainty than a unique, one-of-a-kind bus, or one that hasn't flown on the vehicle under consideration. The unique instrument package that you hang on that bus will still carry some uncertainty, but you can be fairly-to-reasonably assured that the primary structure of your spacecraft will harbor few surprises. For unique instruments, spacecraft providers have successfully used a *mass-acceleration* curve for initial design.

Finally, the same vibration control companies that market shock and vibration isolation services have also demonstrated successes in mitigating negative strength margins that arose from late-breaking coupled loads analysis findings.

Summary of Things You Need to Know About Coupled Loads Analysis

- Coupled loads are the product of spacecraft-vehicle flexible body interactions in the presence of disturbing functions arising from vehicle system operation and external disturbances. Frequencies of interest are generally below 60 Hz, but not always
- There are a variety of methods by which a forcing function is created and modeled, generally in an attempt to conservatively model the known physics associated with an event, but estimating magnitudes to bound three sigma extent without precise knowledge
- Coupled load forcing functions are difficult if not impossible to predict and model from ground tests alone. Many loading events do not become apparent until flight
- Launch vehicle coupled loads forcing functions are subject to change as new events are discovered and uncertainty factors are updated. You should regularly ask if any updates are being planned during your 3–5 year integration cycle
- Coupled loads analysis is almost always an iterative process, especially for one-of-a-kind spacecraft
- The quality and accuracy of the spacecraft structural dynamic model is of critical importance to the credibility of the coupled loads analysis results. Simple model checks for all load cycles, and test verification of the model prior to VCLA are, therefore, required
- As with shock and vibration, isolation measures are available, and have been effectively used at both the whole-spacecraft and component/instrument level

26.4.2 Shock

Shock is the first of two environments to which most people can easily relate. The other is vibration. Both seem honest and straightforward in their causes and physical manifestations. We understand what shock and vibration feel like. We know them, we "get" them. There's seemingly nothing arcane here. "Shock" is your old friend from college, you know, the one (…the one your spouse hates). Furniture might get broken, a mailbox run over, and something always gets spilled on the tablecloth, but we're usually glad to see Shock. When expected, his presence signals good things; things for which we've been waiting.

It used to be a recurring feature on launch day, back when we used old thermal paper strip charts to plot telemetry signals in real time, where engineers of many disciplines would gather in the telemetry lab around the strips showing the accelerometer traces. We could recognize events by the timing and the shape of the sensed shock as the hot needle flickered across the paper. "Ah…" we'd say to each other, eyes lifting knowledgeably to each other just above the top edge our eyeglasses, "there's the shock…signature looks good…good spacecraft separation." Not quite wearing white lab coats and thick accents, but maybe imagining something along those lines. Afterward, a good spacecraft separation from the launch vehicle would be confirmed by binary signals from breakwires on the in-flight disconnects and telemetry from an awake, alive spacecraft, at which time we'd all celebrate by standing and applauding a successful mission. But we knew the outcome seconds or minutes beforehand. We'd seen the shock, and the signal looked good.

Physical Mechanisms

Physically, the mechanisms that induce shock environments on a normally-operating launch vehicle are pyrotechnically-initiated, but not necessarily dominated by the operation of the pyrotechnic ignition itself. Many spacecraft are joined to their launch vehicle using a *Marman-style clamp* band. A Marman clamp uses a tension strap to apply inward radial forces on a set of two *V*-segments, thus wedging two cylindrical structural flanges together [NASA, 1998]. For spaceflight applications, the clamp is often released through the use of a pyrotechnically-driven bolt cutter at one or more locations, abruptly detensioning the band and allowing compressed push-off springs to force the spacecraft away from the launch vehicle stage. In this case, it's the stored energy in the metal band that dominates the shock environment, not the pyrotechnic cutting device. Other separation systems like frangible joints or linear shaped charges use a greater quantity of explosives and result in a true *pyrotechnic* dominated shock environment. In any case, the

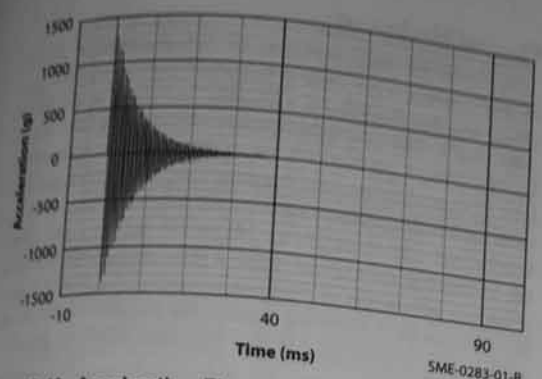


Fig. 26-19. Acceleration Time History of a Typical Shock. (Adapted from Test Operations Procedure 5-2-521 "Pyrotechnic Shock Test Procedures," US Army White Sands Missile Range, 2007.)

common elements of short duration and broad frequency content are maintained.

Figure 26-19 illustrates a representative time history of a pyroshock. Note the high accelerations and rapid decay typical of the environment. The decay is largely complete 20 m/s after initiation.

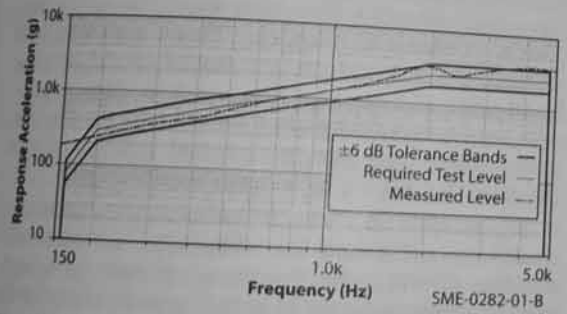


Fig. 26-20. Shock Response Spectra (SRS) of a Typical Single Shock Event. (Adapted from Test Operations Procedure 5-2-521 "Pyrotechnic Shock Test Procedures," US Army White Sands Missile Range, 2007.)

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Figure 26-20 better illustrates the broad-spectrum frequency content of the shock recorded in Fig. 26-19. Figure 26-20 is a frequency-domain expression of what's called a *shock response spectrum* or *SRS*. An SRS is the peak response to the acceleration time history of a single degree of freedom ideal mass on a uniformly-damped spring, where the natural frequency of the system is swept through each of the frequencies of interest. By providing the information in a time-invariant, frequency domain, an SRS gives insight to the frequencies at which the energy is greatest. Something you should keep in mind is that there are multiple methods by which an SRS can be computed, so the conservatism of the levels shown (and the levels to which you must design and test hardware) will vary depending on the SRS method used. There's no one way to compute an SRS, so an SRS like Fig. 26-20 should not be taken as fact without understanding the method by which it was calculated. An SRS is an analytical interpretation of measurements, not a measurement in its own right.

One convenient property of shock is that it attenuates rapidly over distance and does not transmit through structural joints as well as other dynamic environments. Another is that, due to the nature of the physical mechanisms that create the environment, shock is easier than some of the other dynamic environments to characterize on the ground in a laboratory test environment. Again using the Marman clamp band as an example, clamp band separation tests are a routine feature of spacecraft structural interface fit checks, and these test environments are usually recorded by a number of non-flight structural accelerometers at key locations on the spacecraft.

Mitigations

As noted in the previous section, shock can be attenuated by distance and structural joints. It can also be addressed at the physical source, whether through installing the clamp band at a reduced tension, or by changing the nature of the band release system such that the strain release is not as explosive as when released by a pyrotechnic bolt cutter. The Russian Proton-M [ILS, 2009] and US Atlas V [ULA, 2010a] vehicles both offer separation shock mitigation through use of mechanical band release systems, rather than pyrotechnic bolt-cutters. Likewise, the Planetary Systems Corporation [PSC, 2010] offers their Lightband "space vehicle separation system" as mitigation for shock-sensitive spacecraft customers.

If necessary, mechanical isolation systems can also be added to add damping and change the frequency content of the environment before it is transmitted to sensitive hardware. At the spacecraft system level, it is possible to introduce shock attenuation methods directly in the structural path to the spacecraft. This has been done successfully on several missions over a wide range of spacecraft designs and masses over the last two decades. The Pegasus XL [OSC, 2010a] launch vehicle offers a load isolation system capability in their user's guide. Also, CSA Engineering, now a subsidiary of the Moog Space and Defense Group [CSA, 2010] has provided a number of vibration and shock control mitigations to different spacecraft customers.

At a box or instrument level, it is also possible to apply some of the same techniques to add damping and change frequency content in the secondary structure to which a specific box or instrument will be mounted. Elastomeric, wire rope, or combination isolators are available commercially in a variety of shapes, characteristics, and sizes. This will be obvious to most spacecraft and instrument builders, so I mention it only for completeness.

The important thing to remember is that, though we use the term *isolator*, what the isolator is mostly doing is changing the waveform of the shock or vibration environment before it reaches sensitive equipment. A narrow

peak or broad range of high-energy input will be effectively attenuated, but the overall energy transmitted through the isolator will only be slightly reduced, even using the most-effective damping measures.

Summary of Things You Need to Know About Shock

- Shock is a sharp, intense transient acceleration with broad frequency content and a very short (generally less than 20 m/s) duration
- There are multiple methods of computing a frequency-domain shock response spectrum (SRS). Make sure you know which method is being used before you accept an SRS at face value
- Shock sources are generally specific localized equipment functions (like severing the connection to a spent stage with a linear shaped charge), and straightforward to measure in ground tests
- Launch vehicle shock environments are generally well-understood and not subject to frequent changes
- Shock isolation measures are available, and have been effectively used at both the whole-spacecraft and component/instrument level

26.4.3 Vibration and Acoustics

Big engines shake things. They also make loud noises. Rushing airflow over the skin of a launch vehicle at high dynamic pressure also makes for loud noises. The vehicle structure, and by extension your spacecraft, respond to these disturbances by vibrating with their own unique characteristic response. This section discusses the basics of three main types of vibro-acoustic environments your spacecraft will encounter. These are random vibration, sine vibration, and acoustically-induced vibration.

Vibration and acoustics appear on the surface to be honest and workmanlike environments, more-similar to shock than to the arcane mysteries of coupled loads. Those of us that drink beer and drive trucks will feel very comfortable talking about noise and vibration as things that you just have to accept, preferably with ample mass budget devoted to gaining large strength margins. Often you'll be able to make it through a mission with this mental construct intact, in that the vehicle provider will give you an environment specification to which you will design and test successfully. All will be well. Other times it won't be so simple, and a greater investment in understanding this inexact science will be useful to you when you have to choose among difficult and unattractive options.

Just as a note, *acoustics* are sometimes broken out separately for analysis within the launch vehicle community as *aero-acoustics*. In your case, given that your spacecraft is shielded from the direct airflow by the payload fairing, you will not be encountering any transient or unstable flow conditions during ascent. It is therefore appropriate to simplify the treatment and spare you all but the most elementary descriptions of fluid flow dynamics over the skin of the vehicle.

Physical Mechanisms

Converting large amounts of stored chemical energy to mechanical work over a sustained period of time generally creates noise and vibration, whether it occurs in your car's internal combustion engine, an old steam locomotive, a jet aircraft, or even a nuclear submarine (the first nuclear-powered submarine, the USS Nautilus, was said to be as loud as a diesel locomotive... they've worked on that since then). Marathon runners and owners of electric cars could make a good argument that this assertion falls short of being universally true, but it's useful enough as a working model for launch vehicles, as well as any motor vehicle that I'm ever likely to drive.

Liquid rocket engines are particularly good examples of this, as most launch vehicle engines convert propellants to heat and kinetic energy at much higher rates than the other examples cited earlier in the previous paragraph. Most times these engines are driven by high-horsepower turbo-pumps which add their own significant vibration signatures. Large solid rocket motors lack the turbo-pumps of their liquid counterparts, but consume propellants at even higher mass flow rates than comparable liquid engines. The effect is a set of sustained high-frequency (e.g., 50 Hz to 200 Hz and beyond) mechanical displacements that are borne along the vehicle structure from their source to your spacecraft's mechanical interface. These high-frequency mechanical displacements are mostly spread across a broad range of frequencies (i.e., *random vibration*), but with some relatively-coherent peaks where the displacements remain consistently sustained within a narrow frequency band (i.e., *periodic* or *sine vibration*).

Figure 26-21 illustrates a few noteworthy aspects of specifying (and making use of) a random vibration environment. As will be discussed in an upcoming paragraph, there is wide latitude within the industry practices for what is kept under the heading of random vibration and what is kept under acoustics. The two Minotaur profiles exhibit characteristics typical of ground-launched vehicles that use a single random vibration specification to envelope liftoff, flight through high dynamic pressure, and structure-borne random vibrations during steady-state powered flight. By contrast, the Dnepr User's Guide splits the liftoff and steady-state powered flight environments into two specifications. Doing so offers the occasional advantage to a spacecraft customer with a sensitivity to qualifying their spacecraft to enveloping levels for periods encompassing the entire flight through high dynamic pressure (typically 30–60 sec is used with such envelopes). The decision to differentiate envelopes in the manner used by the Dnepr's provider is more dependent upon such factors as heritage specification practices and the number of spacecraft customers that have requested relief from an overall envelope. It is not dependent on the manner in which the different vehicles fly to orbit. For most spacecraft customers, the environments specified in Fig. 26-21 are not exceedingly harsh in any way.

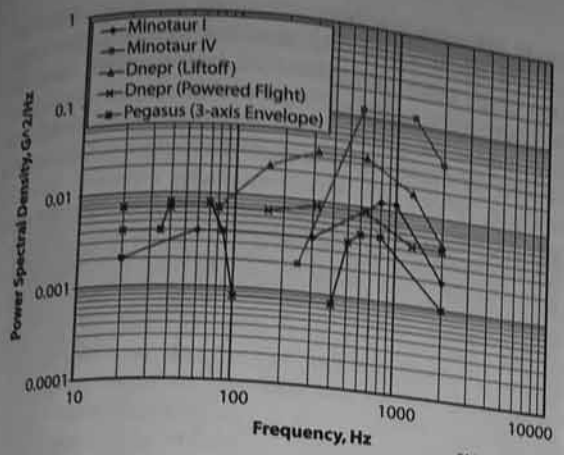


Fig. 26-21. Example Vehicle Random Vibration Environments. [OSC, 2006a; OSC, 2006b; ISC Kosmotras, 2001; ISC Kosmotras, 2001; OSC, 2010a]

Analytical and Test Techniques

Acoustics and vibration environments contained in vehicle user's guides are derived from flight measurements because many of the contributing sources are impractical to measure in ground tests. This is the reverse from shock, where shock is best characterized on the ground, and difficult if not impractical to adequately characterize in flight. In a new launch vehicle with no prior flight measurements, we'll do our best to combine those ground test data sources we can accumulate, compare it to flight data taken from similar vehicles and flight conditions, and add large uncertainty factors. Statistical energy analysis (SEA) tools have also come into wide usage for predicting flight acoustic environments in the absence of applicable test data. The environments provided to you for an existing launch vehicle will have been derived from a limited number (possibly as few as one or two) of flights with special instrumentation packages, so there will still be uncertainty factors embedded in the environments provided to you in order to address flight-to-flight variability.

The simplest way to communicate the acoustic and vibration environments is to develop an enveloping specification that covers all events from engine ignition to the completion of flight through high dynamic pressure, and offer that with expected flight duration of about one minute. The acoustic sound pressure level (given in decibels (dB) against a reference pressure) or sine vibration acceleration (usually given in multiples of one Earth reference gravity (g) or random vibration spectral density (g^2/Hz) at each frequency will be chosen to reflect the maximum of the predicted values from a combination of events such as engine start, liftoff, Mach 1, or maximum dynamic pressure. The predicted intensities will be given with a statistical reliability and confidence consistent with the provider's own technical policies; for example, "95% reliability with a 50% confidence," and "99% reliability with a 90% confidence" are common descriptors indicating the level of uncertainty in the values provided. What's really happening here is that peak levels driven by short duration events (<10 sec) are being used to define an envelope that will be given to you as occurring over a much longer duration (~30–60 sec) than is physically correct. This may sound unjustly conservative to those unfamiliar with the practice, but simplifications like these came into common usage within the launch community decades ago out of convenience when it was found that the customers were largely insensitive to the added conservatism. The message here is to be aware of the conservatisms inherent in the commonly-practiced simplifications within the industry, just in case you're one of the few spacecraft customers that can't handle qualifying for a high vibration event that's assumed to last for one minute when in fact it only lasts for 5 or 10 seconds. If you find yourself in trouble, start asking these kinds of questions. You might have to pay for additional analysis by the launch provider in order to split up the environments and durations into more-realistic chunks, but that could be substantially cheaper than redesigning part of your spacecraft.

Summary of Things You Need to Know About Vibration and Acoustics

- The acoustic environments to which your spacecraft will be exposed are principally driven by overpressure and reverberations at launch, followed by flow noise over the surface of the payload fairing (and the reverberations within) during flight through high dynamic pressure
- As with shock, there are multiple methods by which measured environment data is processed to develop an acoustic or vibration intensity prediction. Even a summary-level understanding of the differences in predictive techniques, and the embedded simplifications, may save you from an unintended overtest condition
- For your spacecraft, maximum acoustic and random vibration environments will coincide with liftoff and flight through high dynamic pressure (roughly from transonic to maximum dynamic pressure)
- Sine vibration environments are generally associated with a physical coupling between vehicle structural modes and pressure oscillations within the propulsion system. As such, they are not limited to a coincident occurrence with liftoff and high dynamic pressure flight
- The launch industry simplifies their representation of random and sine vibration environments as a matter of good economic practice, in that it doesn't make sense to perform in-depth analyses of certain vibration environments (mostly sine events) if 99% of their customers are insensitive to testing at the simplified recommended levels. You may be among the 1%, so don't hesitate to ask your launch

provider for help if you identify an environment for which it will be difficult or risky to demonstrate qualification

- Motion control solutions are available to aid in attenuating sine vibration events, and even random vibration environments borne through the launch vehicle structure. However, these will not be effective to attenuate your response to the acoustic environment within the payload fairing

26.4.4 Electromagnetic Compatibility (EMC)

Electromagnetic interference (EMI) and electromagnetic compatibility (EMC) are accorded the same arcane mystery as coupled loads by most folks. Even some experienced launch vehicle engineers and managers are prone to unconsciously view this area as somewhat sneaky or dishonest. The math can be a bit difficult to follow, and the visceral reaction is that the environment can't be seen or felt, as folks might forget that light and radiant heat are, themselves, electromagnetic emissions. The idea of a radar beam sweeping over the vehicle and spacecraft is fairly easy to grasp. Yes, it's radar. Yes, it's potentially dangerous to the spacecraft and vehicle. But the physical understanding of why this is the case, what the mechanisms are for causing real damage, is not intuitive for most of us. Jump forward to explaining why a one-watt spacecraft RF telemetry transmitter is potentially dangerous to both its own spacecraft and the launch vehicle during flight, and you begin to strain the bounds of credulity in much the same way as being told that your wireless communication device poses a danger to an airliner and all those aboard (at least in the US and EU in 2011). The wireless phone on an airliner is a great example of the challenge in confronting this analytical discipline, because the whole issue of wireless phones being dangerous to airliners is seemingly unjustified, given the number of wireless phones that inadvertently fly in the "on" condition (ahem, including mine), unknown to their owners that thought they had turned them off properly when instructed. "Obviously this is bunch of garbage, no airplanes are falling from the sky due to wireless phones" Ever hear that comment or something like it? Granted that it might sometimes come across that way, but the precautions, analyses, and tests are there for good reasons. Consider also that the nature and power of the emitters are evolving, as are the density, sensitivity, and materials in critical circuits. The solution for those of us that do not work directly in this specialized discipline is to understand the physical mechanisms a little better than a simple gut reaction will allow.

Be aware that the sources and effects of electromagnetic interference are expected to continue to change more rapidly than the rate at which most of the authoritative guidance on the subject can be revised. The same colleagues that recommended the older classic military standards as good guidance were also matter-of-fact in their caution that even the most-current updates and newly released guidance will likely not keep pace with new RF applications and electronic circuit technology. This is an important difference between EMC and the other environments described in this chapter. It is a useful caution to observe that the density of integrated circuits in high-performance electronics continues to increase, while new electromagnetic emission sources (e.g., wireless personal voice and data connections) expand their range into new and higher frequency ranges [Brewer and Trout, 2006]. The important message here is that technological advances in avionics and RF applications make EMC a more-dynamic area for guidance due to the potential for rapid change to previously-established ground rules and assumptions. Good guidance is available to address the fundamental physics, but increased attention should be paid to understanding the areas of conflict between accepted standards, as well as seeking the most-current descriptions of the expected RF environment to which your spacecraft will be exposed during processing and launch.

Physical Mechanisms

Typically, the term EMI refers to unintentional disturbances, either radiated or conducted, and is generally addressed at the box level. EMC is generally understood in the context of electronics operating together in a system. The overall discipline is normally referred to as EMC. A third term, electromagnetic effects (EME) is sometimes used to include other electromagnetic events in the discipline such as lightning, or charging but this section will adhere to the simplification of grouping EME within the larger EMI. Also keep in mind that EMI does not solely refer to electromagnetic radiation through free space, and likewise EMC does not only encompass electromagnetic noise/signals conducted through metal. The practical electrical engineers among us will understand the old admonishment given to junior folks that "a plug doesn't necessarily mean it has pins, and a receptacle doesn't mean it has sockets."

The common physical mechanism of greatest interest to launch is temporary exposure to electric fields or conducted interference sufficient to cause latent damage (or outright failure) of your most-sensitive circuits. Damage is the thing we worry about. Simple interference is only a transient concern, given that you will be gladly rid of your launch vehicle some minutes or hours after launch. You should assume an inability to transmit or receive RF communication with your spacecraft during launch unless you make expensive provisions with your provider to do so.

Electrostatic discharge (ESD) won't be anything new to you as a spacecraft customer. Opportunities are probably far greater for developing static charges in normal processing between electrically-isolated articles until a sudden discharge (spark) occurs. These normal sea-level ambient ESD hazards are unchanged once integrated with a launch vehicle. Depending on your target orbit, some missions may face the hazard of flight through bands of energetic protons trapped in the Earth's magnetic field that might cause detrimental charging over

exposed and improperly-grounded surfaces, but this is not a significant hazard to missions in low Earth orbit. The best and only real ESD mitigation here is the same as you would practice during processing, specifically to ensure that all elements are adequately grounded at all times, with no opportunity for charge accumulation.



Fig. 26-22. 3 Megawatt Debris Detection and Tracking Radar at KSC. (Photo courtesy of NASA)

Exposure to sweeps from high-power radar like the debris detection site in Fig. 26-22 will likely be a new hazard at the launch site, compared to what your spacecraft would have encountered during manufacturing, shipping, and processing. The momentary RF electric fields thus induced can exceed 200 volts per meter, sometimes several multiples of that figure, depending on launch site. Buildings, shipping containers, and payload fairings offer some attenuation, but not nearly as much as you might think. It is intuitive to think of a metal payload fairing as being a Faraday Cage*, thus impenetrable to electric fields, but strictly this is only true for static electric fields. Oscillating fields are more devious and inventive. Remember that radar is an RF emission, with wavelengths that are inversely-proportional to frequency. Remember also that buildings, shipping containers, and payload fairings all have openings to pass people, equipment, cables, airflow and the short wavelengths of radars in use at most ranges do a fine job of penetrating what your undergraduate basic electricity and magnetism class recollection would tell you is an impenetrable shield.

* Named for English physicist Michael Faraday.

There is another hazard associated with radar that is rarely mentioned, though, specifically that the RF environment described to you by your launch vehicle provider is unlikely to encompass all of the radar sources to which you may be exposed. Your provider will certainly give you their best description of all the RF emitters, radar and otherwise, of which they are aware. This isn't their fault; they won't hide any known emitters from you. The problem is that they might not know about all of the emitters, mostly radar, which exist on the launch range. Consider that many launch ranges around the world serve dual civilian and military roles, and most are located on coastline. The radar emitters that are dedicated to the safety functions of maintaining positive launch vehicle tracking and local airspace clearance will be well-known by your launch provider. No problem there. The range organizations are typically very cooperative in avoiding sweeps by the radars they control when they are apprised of a particular sensitivity on your part. The term used in the US is *sector blanking*. Your launch range wants to see you succeed, too.

The basic problem is that the launch range safety organization may not control all of the radar emitters on or around the range. High-powered military radars aren't well-advertised and upgraded power and frequency capabilities are even less likely to be advertised than is their existence. This makes sense, as knowledge of military capabilities are issues of national security; anyone's national security. Many nations have missiles, and every nation has enemies. Unlike the launch range, your spacecraft mission success is well down on a military air and space defense priority list, though they will take what steps that they can to reduce your hazard if they know about your mission sensitivity. They may not know about your unique sensitivity anymore than you being aware that they exist. Otherwise, your launch vehicle makes a wonderful target for tracking practice by those charged with protecting against less-friendly vehicles. The only mitigation for an RF-sensitive spacecraft is to make that sensitivity known far and wide in advance. Ask your provider to arrange a meeting with the range folks long before (years before) your spacecraft's arrival at the range in order to help get the word out to all potential military applications that might exist in proximity.

Beyond military radars, the likelihood that you will be launched from a coastal range offers another hazard, less powerful but realistically impossible to control. Ocean-going ships and even coastal fishing boats carry their own radars. These can't be controlled by the launch range, since any passing ship of near-infinite international registry can pass freely off the coast in international waters. Ranges can and do conduct site RF surveys, however, their information could easily be several years old. Likewise the guidance in published standards, such as that shown in Table 26-6, may be outdated soon after appearing in print. If your spacecraft is particularly RF-sensitive, then you might consider requesting that an updated survey be performed by your launch service provider as an additional (paid) service in

Table 26-6. "Electromagnetic Compatibility Requirements for Space Equipment and Systems." Adapted from Table 6.27e2-2 in SMC Standard SMC-S-008, (US Air Force Space Command, 13 June 2008.)

Frequency	Ascent RFI Limits (V/m)	Source Notes
10 kHz–100 MHz	20	MIL-STD-464A, Table 1C, External EME for Space and Launch Vehicle Systems, and MIL-STD-461F, Table VII, RS103 Limits (space)
100 MHz–1 GHz	100	MIL-STD-464A, Table 1C
1–4 GHz	200	MIL-STD-464A, Table 1C
4–11 GHz	1000	According to the RF Power Impingement Analysis, 1997 and 1998 composite analyses for Titan Core Vehicle from 0 to 80 nmi (an internal Aerospace Corp. briefing written by D. Oltrogge and A. C. Bustillos), there is a source capable of generating approximately 1000V/m, with approximately 11 RFI threats between 100 and 500V/m in the frequency range from 4–11GHz. In addition, Aerospace Report No. TOR-2005(1663)-3790 shows four locations at CCAFS with field strengths in the 4–11 GHz range from 1,000–43,000 V/m, plus about 35 occurrences of field strengths from 100 to 500 V/m. 500 V/m is a compromise which captures most of the identified RFI threats in the 4–11GHz range
11–100 GHz	20	MIL-STD-464A, Table 1C, and MIL-STD-461F, Table VII, with additional SMC-S-008 guidance on extension above 40 GHz

order to develop a most-current understanding of the ambient RF environment. Nevertheless, keep in mind that such surveys are only a useful snapshot, not a guarantee against the worst case of having the odd naval vessel cruise by in international waters while conducting some convenient air-defense exercises at your expense.

Analytical and Test Techniques

The published standards recommended earlier in this section provide good guidance for design and test. The environment will already have been analyzed in general, as reflected in the published standards and handbooks, and particular, as reflected in the individual launch vehicle user's guides. However, the differences in analytical and test guidance published by such agencies as the US Air Force, AIAA, and ESA are reflective of the continuing evolution in both the discipline and the environment being studied.

There isn't much in the way of mission-specific analysis that will be needed in this area, other than ability to show compliance with RF emission and circuit isolation requirements. If you are one of those spacecraft with instruments that carry an inherent and extreme sensitivity to RF fields, then there are numerical simulation techniques (the same as those used to predict radar cross-sections for military aircraft) that were unavailable ten years ago. These can be brought to bear on the problem of sharpening the predictions for field levels that your spacecraft may encounter given a sweep from a hypothetical naval vessel taking advantage of your launch to conduct an air-defense drill. More importantly, they can help quantify the benefit of any mitigating shielding that you might employ as a mission-unique modification to the vehicle. But be aware that such analyses and mitigating measures are expensive and require considerable cooperation from your launch vehicle provider.

Table 26-6, Fig. 26-22◀, Eq. 26-0

Summary of Things You Need to Know About EMC

- The electromagnetic environments to which your spacecraft will be exposed are principally driven by exposure to tracking radars and launch vehicle RF transmitters, with some potential for charging due to flight through regions of energetic protons on your way to final orbit. And always, there is atmospheric lightning to consider as a source broadband emissions that are potentially-fatal to sensitive instruments
- Vehicle RF emitters are well-characterized by the emission spectra included in pretty much any vehicle user's guide. However, though launch ranges make good efforts to characterize their local RF environment (primarily concerning high-power radar sources), they will not necessarily be able to control all high-power local emitters, nor may they be aware that some (like maritime navigational radars or naval fire control radars) are even in the vicinity from one day to the next
- Published guidance continues to be updated as the discipline is studied further, and while the nature and power of electromagnetic environment and applications evolves on a continual basis
- System-level electromagnetic compatibility is normally addressed through integrated avionics testing, combined with rigorous attention to bonding and isolation requirements for those interfaces (i.e., with the launch vehicle) for which it is not practical to accomplish full system integration until arrival at the launch site for flight

Again, while there is nothing particularly new that the launch vehicle itself brings to the EMC issues confronting any spacecraft, the magnitude of the collateral radiated environments (e.g., radar and lightning) can be

daunting. No less daunting is the prospect that once your own spacecraft team has settled on the set of published EMC guidance that you intend to follow in your development, you may find that the launch vehicle provider bases their recommendations on a completely different and conflicting standard. Always remind yourself that such conflicts are not based in lack of expertise, but rather are generated by differences in experience, sensitivities, and the timing of the published guidance. This is a complex and healthy field of study, but cookbook solutions may be limited or dated. The best approach is to understand the physics of the problem and make your own determinations based on the best-available guidance. Even then, you may spend a lot of time looking over your shoulder wondering what it is that you forgot.

26.4.5 Ascent Venting

Choosing to end this section on environments with a bit of good news, your analysis and mitigating measures to accommodate the launch vehicle provider's ascent venting predictions should be straightforward. The physics are uncomplicated, and so is most of the available guidance. While computational fluid dynamics simulations are always refining the state of the predictive art, the design guidance in use today was developed without dependence on those techniques, and is not generally onerous in the implementation. However, you should plan on paying special attention to vent path routing and assuring that your as-built configuration matches the configuration that you analyzed.

Physical Mechanisms

The basis for the ascent venting environment is straightforward. We'll be carrying you from sea level to essentially zero sensed-atmosphere in 80 to 120 seconds, depending on the vehicle and the target orbit. Figure 26-23 shows an unrealistically-conservative dispersion envelope developed from a composite of profiles encompassing several different ground-launched vehicles. It is only intended to be roughly illustrative of what your launch vehicle provider will specify to you.

The curves start at ~101.4 kPa, which was the ambient atmospheric pressure at my hotel outside Vandenberg Air Force Base this morning. The end point, obviously, is essentially zero pressure, just prior to jettison of the payload fairing. For the vehicle, designing the venting implementation is actually quite challenging, owing to the dynamic interaction of free-stream flow over the vehicle skin and vent orifices during flight. What you will be given is not the actual outside atmospheric pressure profile, but the ambient pressure within the payload fairing.

Your design problem will not involve any supersonic flow. You will merely have to accommodate variations in vent rates over a much smaller range of uncertainties than is shown in Fig. 26-23. This simplifies your analysis problem.

One area that will require your increased attention, however, is assuring that all the vent areas assumed in your analysis actually exist on your finished spacecraft.

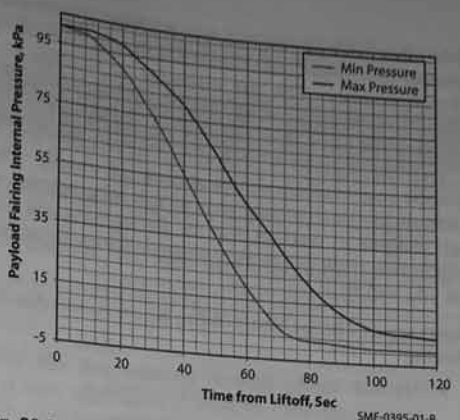


Fig. 26-23. Composite Envelope of Payload Fairing Internal Pressures.

Your venting analyst looks at the released structural drawing of the bus or the instrument that shows all of the openings through which air can vent. Unless your analyst is very experienced, they may not be looking at the thermal blanket drawings that show which structural openings are covered up. Likewise, your analyst knows that the blankets are built with so many holes per square meter, per your specification, but may possibly be unaware that your technicians have been enthusiastically taping over them for the last few weeks during installation. The obvious imperative of making sure the blankets stay in place and patching repairs (blanket damage repair never seems to end) can easily take unintentional precedence over the more subtle requirement that the blankets vent in a predictable manner.

The second area that merits increased attention, pointed out by one of the editors of this book based on our shared experiences, is the direction of the vent path. It is unwise to vent blankets, for example, in proximity to debris-critical instruments without taking steps to mitigate manufacturing debris. Your spacecraft may have been built in a clean laboratory; however your blankets were not. Neither were many of the components you installed during assembly. To be sure, they were built in what's typically called *shop clean* environments, but not strictly-controlled clean rooms. What this means to you is that the act of venting during ascent will somewhat resemble the act of running a vacuum cleaner over your spacecraft. If the vent path runs past any good traps, like a detector array or optical surface, then that could be where some of the remaining debris ends up. One mitigation for this is careful attention to routing your internal and external vent paths. Another is to apply special debris traps at key locations in the vent path such that any residual debris is confined to known areas where is can do no further harm.

Analytical and Test Techniques

Ascent venting effects should be handled as part of your strength analysis, since the driving concern is the

stress that will be created by differential pressures between your spacecraft volume and the payload fairing internal pressure. The gas discharge and differential pressure equations are not complex, and are amply treated by the NASA references already cited.

You will be asked by your launch service provider to describe your *ventable volume* (this is just exactly what it sounds like) and venting area, as this becomes part of their analysis to confirm that the fairing structure and vehicle vent areas will remain adequate for the trajectory that you will fly.

Summary of Things You Need to Know About Ascent Venting

- Payload fairing internal pressure will fall from roughly sea-level ambient to essentially zero in a period less than two minutes. Your venting problem analysis will involve only simple, subsonic flow over a venting profile that will be specified by the launch vehicle provider, usually in their user's guide
- You'll use the fairing pressure profile and your own ventable volume to size your vent area in order to keep your maximum differential pressure within bounds that won't overload your structure
- You will be required to specify your ventable volume and vent rates to the launch vehicle provider for use in their analyses
- The structural effects of differential pressures due to ascent venting will have to be addressed in your spacecraft strength analysis
- Pay special attention to consistency between what the configuration you analyzed and the configuration that you actually fly
- Observe caution for residual manufacturing debris becoming trapped at unintended locations after being swept out by the vent flow. Keep the vacuum cleaner analogy in mind and try not to let your instrument serve as the dirt bag

26.5 Available Launch Vehicles

Get More

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This last section will only attempt to provide qualitative observations, avenues for gaining additional information, and a rough overview of the fleets, such as the Soyuz in Fig. 26-24, that are in actual service worldwide today. The book website is updated from time to time with more current vehicle information. In addition, vehicle providers have embraced the internet as a means of communicating performance, contact, and availability information to a wider audience than was possible before, so that kind of contact information will be included here, as available. There are two caveats to this section, however. The first being that a detailed treatment of world fleet performance, features, and history would fill a book in its own right, e.g., Isakowitz, Hopkins, and Hopkins Jr. [2004], which provides an authoritative and widely-used survey of world vehicles. An extension of this caveat is yet another reminder to take published vehicle performance data as only a starting point in your investigations, even if it comes from the vehicle providers themselves. Again, what's published in any forum or media is a best-faith estimate of the vehicle performance offered at that time, but vehicles, even those with venerable histories, usually see a series of changes and upgrades as parts become obsolete, or manufacturing techniques change, or even as they build in measures for greater reliability. All of this affects the performance available for your use. Refer to Table 26-7 for a summary of vehicles providing launch services today, along with their advertised LEO and GTO performance capabilities.



Fig. 26-24. TsSKB-Progress' Soyuz Vehicle during Rollout at Baikonur Cosmodrome. (Photo courtesy of NASA/Carla Cioffi.)

The other caveat is that I personally tend to apply a ruthlessly-pragmatic filter on vehicle availability and capability, and that is "Is it flying to orbit today?" There are a number of vehicles in various stages of development at any one moment in time, and most carry a high degree of credibility in their development plans. Governments and other investors tend not to throw their money into such ventures unless the credibility is there. Even so, many never fly. This is as true in 2010 as it was in any year of the previous three decades and prior, and this observation respects neither nationality nor vehicle performance class. Obviously, new fleets are developed and successfully placed into service; however, the number of fleets under development with this intent consistently exceeds the number that actually reaches their goal. If it hasn't yet flown, then there's considerable risk that it will never enter service at all. I'm a great fan of "daring mighty things," but you should be pragmatic about finding a ride to orbit that will be there when you need it.

Table 26-7.Summary of Available Launch Vehicles. The book website contains a more extensive table with a more complete description of each vehicle, web references for each vehicle, and is also updated as vehicles change.

Nation	Vehicle	Orbits Serviced	Performance (mT) LEO	Performance (mT) GTO	Active Launch Sites*	Comments
USA	Atlas V	LEO, GTO, GSO, Escape	9.4–18.5	4.8–8.9	CCAFS, VAFB	Two payload fairing sizes, one to five strap-on solid motors
	Delta II	LEO, GTO, Escape	2.5–5.5	0.8–2.0	CCAFS, VAFB	Three, four or nine strap-on solid motors
	Delta IV	LEO, GTO, Escape	9.2–22.6	4.3–13	CCAFS, VAFB	Two upper stage sizes, two or four strap-on solid motors, or three-core "Heavy"
	Falcon 1	LEO	1	n/a	RTS	
	Falcon 9	LEO, GTO	10.4	4.5	CCAFS	
	Minotaur I	LEO	0.6	n/a	WFF, VAFB	Uses decommissioned ICBM motors for the first two stages
	Minotaur IV	LEO	1.7	n/a	WFF, VAFB	Uses decommissioned ICBM motors for the first three stages
	Pegasus XL	LEO	0.45	n/a	Air-mobile	Prior launches from CCAFS, RTS, VAFB, WFF, and the Canary Islands
	Taurus	LEO	1.4	n/a	VAFB	Vehicle and launch equipment designed to be road-, rail-, and sea-portable to austere launch sites
China	Long March 2C/D	LEO, GTO	3.8	1.2	JSLC, TSLC, XSLC	
	Long March 3A/B/C	GTO	n/a	2–5.5	XSLC	Primarily marketed for GTO missions
	Long March 4	LEO	4.2	n/a	JSLC, TSLC	Mostly sun-synchronous use
European Union	Ariane 5	LEO, GTO	20	10	CSG	Two upper stage choices, ECA (cryogenic) and ES (storable) for LEO and GTO missions, respectively
India	PSLV	LEO, GTO	1.6	1	SDSC	The LEO performance figure quoted by ISRO is for sun-synchronous
	GSLV	GTO	n/a	2.5	SDSC	No LEO performance quoted by ISRO
Israel	Shavit	LEO	0.3*	n/a	Palamchim AFB	Launches to date have been westward, however a polar capability is being offered and the launch system design is intended to operate from austere sites
Japan	H-IIA	LEO, GTO	n/a	4–6	TSC	Two to four strap-on solid rocket motors. This vehicle is primarily marketed for GTO missions
	H-IIB	LEO	16.5	8	TSC	Used for International Space Station servicing missions to date
Russia	Dnepr	LEO	3.7	n/a	Baikonur, Yasny	Converted RS-20 (NATO SS-18) ICBM, with commercial services marketed by ISC Kosmotras
	Proton	LEO, GTO, Escape	23	6.9	Baikonur	Commercial services marketed by International Launch Services (ILS)
	Rockot	LEO	2	n/a	Plesetsk	Converted UR100N (NATO SS-19) ICBM, with commercial services marketed by Eurockot, GmBH
	Soyuz	LEO, GTO, Escape	5.0–8.2	1.5–3.0	Baikonur, Plesetsk, CSG	Commercial services marketed by Starsem (Baikonur) and Arianespace (CSG)
Ukraine	Zenit	LEO, GTO	n/a	n/a	Baikonur, Sea-mobile	Commercial services marketed by Sea Launch, LLC

* **Site Names/Acronyms**
Baikonur—Baikonur Cosmodrome, Kazakhstan
CCAFS—Cape Canaveral Air Force Station, Florida
CSG—Guiana Space Center, Guiana, France
JSLC—Jiuquan Satellite Launch Center, China
RTS—Reagan Test Site, Kwajalein Atoll, Marshall Islands
SDSC—Satish Dhawan Space Center, Sriharikota, India
TSC—Tanegashima Space Center, Japan
TSLC—Taiyuan Satellite Launch Center, China
VAFB—Vandenberg Air Force Base, California, United States
XSLC—Xichang Satellite Launch Center, China
WFF—NASA Wallops Flight Facility, Virginia, United States
Yasny—Yasny Launch Base, Orenburg Region, Russia

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27 Launch Operations

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27.1 Worldwide Launch Sites and Launch Restrictions
27.2 Launch Site Preparations
Creating a Launch Site Operations Budget; Launch Site Operations Planning; Launch Site Documentation; Ground Support Equipment and Flight Hardware Specifications; Detailed Technical Drawings; Schedules
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The launch site is the last stop before the spacecraft is placed into orbit. Ensuring successful planning, processing, and operations is essential to achieving mission success. Once the launch vehicle and launch site have been selected, you must begin working with each organization as soon as they are available. By familiarizing yourself with the launch site processes, you will be able to successfully develop a schedule and budget for the launch site operations (Sec. 27.2.1). For example, processing a "ship and shoot" mission may be as little as a week or two. A larger mission could take a few months. Of course, this is only true if the spacecraft or vehicle teams do not encounter any issues during the processing. After understanding what the launch site operations, restrictions, and limitations are, we must consider how to potentially save time and reduce launch site costs by implementing changes in the spacecraft or Ground Support Equipment (GSE) design.

27.1 Worldwide Launch Sites and Launch Restrictions

Many factors should be considered when determining when and where a spacecraft will be launched. You have the mission requirements and objectives, the orbit required to meet the mission objectives, size and weight of the spacecraft, and the time/day of the week/month to maximize fuel and sunlight, just to name a few. Figure 27-1 and Table 27-1 show worldwide launch sites and inclinations they can reach.

Not only do you have to consider the spacecraft requirements and objectives, but the launch site and launch vehicle capabilities and availability as well. Is there a launch vehicle available at the launch site that has the capability to support the spacecraft size and weight to achieve the required orbit? Chapter 26 defines in detail the launch systems and the launch system selection process.

Each launch site does its best to accommodate the requirements of the spacecraft team; however, there will be restrictions and constraints that must be considered in the selection process. Can your spacecraft design and Ground Support Equipment (GSE) meet these restrictions or constraints? Can you comply by altering the system design of the spacecraft or GSE prior to fabrication? For example, if the launch site is in an earthquake prone area, does the design of the GSE meet the appropriate seismic requirements so it will not topple over if such an event should occur? While the spacecraft is on a work stand, does it meet the same seismic requirements or will leg extensions or a base plate be required?

Launch Site Range and Tracking Systems

To monitor the trajectory of the launch vehicle from launch through the End-of-Flight (EOF), there are various range assets and tracking systems scheduled to support the launch. Range and tracking systems provide the data-handling interface with time, range, azimuth, elevation, Doppler frequency, and radar status information of the vehicle being tracked.

Launch site range and tracking systems can have a substantial effect on the spacecraft. Even though the

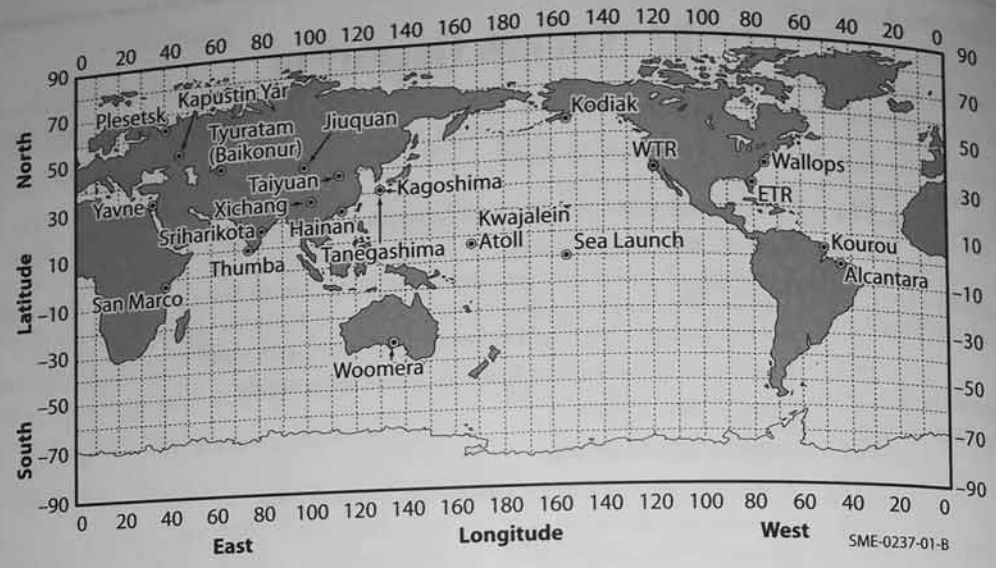


Fig. 27-1. Locations of Launch Sites Around the World. (See Isakowitz, et al. [2004] and Strom [2006].)

Table 27-1. Launch Site Locations and Possible Inclinations. (See Isakowitz, et al. [2004] and Strom [2006].)

Launch Site	Map Designation	Country	Latitude	Longitude	Possible Inclinations
Woomera Launch Site	Woomera	Australia	31.1° S	136.8° E	82–84°
Alcântara Launch Center	Alcântara	Brazil	2.3° S	44.4° W	2–100°
Hainan Space Satellite Launch Center	Hainan	China	19° N	110° E	
Jiuquan Satellite Launch Center	Jiuquan	China	40.6° N	99.9° E	57–70°
Taiyuan Satellite Launch Center (TSLC)	Taiyuan	China	37.5° N	112.6° E	99°
Xichang Satellite Launch Center (XSLC)	Xichang	China	28.25° N	102.0° E	28.3–36°
Kourou Launch Center	Kourou	French Guiana	5.2° N	52.8° W	5.2–100.5°
Thumba Equatorial Rocket Launching Station (TERLS)	Thumba	India	8° N	76° E	
Satish Dhawan Space Centre (SDSC)	Sriharikota	India	13.9° N	80.4° E	
San Marco Launch Platform	San Marco	Indian Ocean	2.9° S	40.3° E	2.9–5°
Palmachim/Yavne	Yavne	Israel	31.5° N	34.5° E	
Tanegashima Space Center (TSC)	Tanegashima	Japan	30.4° N	131.0° E	
Uchinoura Space Center (USC)	Kagoshima	Japan	31.2° N	131.1° E	up to 57°
Kwajalein Missile Range	Kwajalein Atoll	Marshall Islands	9° N	167° E	Any
Sea Launch	Sea Launch	Pacific Ocean at the Equator	0° N	154° W	Any
Baikonur Cosmodrome	Tyuratam	Soviet Union	45.6° N	63.4° E	63–83°
Kapustin Yar	Kapustin	Soviet Union	48.4° N	45.8° E	63–83°
Plesetsk	Plesetsk	Soviet Union	62.8° N	40.1° E	63–83°
Cape Canaveral Air Force Station (CCAFS)	ETR	United States	28.5° N	81.0° W	28.5–59°
Kennedy Space Center (KSC)	ETR	United States	28.5° N	81.0° W	28.5–59°
Kodiak Launch Complex (KLC)	Kodiak	United States	57° N	152° W	
Vandenberg Air Force Base (VAFB)	WTR	United States	34.4° N	120.35° W	56–104°
Wallops Flight Facility (WFF)	Wallops	United States	37.8° N	75.5° W	38–60°

spacecraft is encapsulated inside of a fairing, it can still be exposed to large electromagnetic fields generated by the tracking systems. The spacecraft should be designed and tested to handle the expected worst-case exposure.

The most widely used system for tracking launch vehicles is C-Band radar, also known as "skin" tracking. *Skin tracking* of an object uses radar without a beacon or other signal device on board the object being tracked. Most of the systems are land based, but transportable units are deployed to fill in where land based systems are not available. Many launch vehicles have a C-Band transponder installed to increase the tracking range.

One-way and two-way Doppler tracking are the most common methods of tracking the position of satellites in space. *One-way Doppler tracking* measures the Doppler shift of a radio signal sent from the spacecraft to a tracking station on the Earth. This signal may come from an on-board satellite oscillator or beacon. *Two-way Doppler tracking* measures the Doppler shift of a radio signal sent from the ground station, received by the satellite, and then re-transmitted back to that ground station.

Every vehicle launched in the US has a Flight Termination System (FTS), including the Space Shuttle. The FTS system is a dynamite charge (ordinance package), about the size of a small pipe that runs from top to bottom of each of the launch vehicle stages. During the launch ascent, the Range Safety Officer (RSO) at the launch site monitors the trajectory of the launch vehicle via a real-time display, which shows a pictorial view of the boundaries of the flight trajectory and is updated in real-time with the position of the launch vehicle. If the launch vehicle veers off course and crosses one of these boundaries, the RSO presses two switches that send a coded radio signal to the FTS. This ignites a detonating cord assembly lead to the linear shape charges, breaking the launch vehicle apart. This action effectively zeros the thrust, stopping the acceleration, and rapidly burns the remaining fuel, before it becomes a danger.

27.2 Launch Site Preparations*

Let's assume at this point, the launch vehicle, the launch site, and launch date have all been decided. Now we will take a look at the launch site processing of unmanned missions from pre-arrival at the launch site through the beginning of the launch and early orbit operations.

Launch site operations are often "procrastinated" or not deemed a "high priority" on the list of things to do. Getting the launch site requirements defined, documented, approved, and implemented, however, can take years of work (as discussed in Sec. 6.1). As will be discussed in Sec. 27.7, preparing the spacecraft for the transport to the launch site, arrival, processing, and finally launch requires conscientious attention to detail. If you want a smooth transition from your spacecraft development facility, the key to a successful processing flow is to plan, plan, and plan. The time and effort spent on the upfront planning will not only save you time, but money as well.

As you will see in the sections that follow, there are many tasks and details that will need to be completed for smooth and successful operations at the launch site. With all of the expenses that can be incurred during the launch site processing, this could become your worst budget nightmare if not handled properly.

27.2.1 Creating a Launch Site Operations Budget

One of the tasks of a Project or Program Mission Manager is to define and manage a budget. This includes the many tasks it will take to accomplish the mission and the cost to do so. A typical budget calls for setting up monthly spending limits over a period of time, and then tracking whether the spending stayed within those limits. Budgeting for launch site operations is no different.

Budgeting for the launch site operations can be successfully accomplished following a simple 4-step process.

1. **"What"** needs to be done? Define all of the tasks that will need to be accomplished from arrival at the launch site to the final transportation of the GSE home.
2. **"How long"** is it going to take? Determine the amount of time it will take for each task, taking into account whether the work at the launch site will be:
   - *Quick "ship and shoot" process* with very little testing and little or no mechanical work on the spacecraft?
   - *"Limited" process* with some testing and mechanical work on the spacecraft?
   - *"Full blown"* process with extensive testing and mechanical work that has to be performed on the spacecraft?

   Each task that is performed at the launch site will increase the time required at the launch site, possibly increase the number of required personnel, facilities, resources, and undoubtedly increase cost.
3. **"When"** will it get started? Define when the specific tasks will need to be done.

   A detailed daily processing schedule is produced from steps 1, 2, and 3.
4. **"Who"** is going to do it? Define the personnel it will take, broken down by task (e.g., attitude control system, thermal, electrical, and mechanical), to get the work completed.

Cost is determined from all 4 of these steps.

Depending on the complexity of the operations, the Mission Manager will hold weekly or monthly meetings

*Applicable standards: ISO 17400 (Space Systems). See website for reference.

with each of the system leads to discuss their particular budget and schedule. It is imperative that the system leads report any anomalies or deficiencies in a timely manner to maintain budget and schedule. If the tasks exceed the available time to complete, more time or personnel will have to be added, thus affecting the budget.

You should have reserve funds to account for unexpected events. Reserve funds are allocated to unexpected or unavoidable events, such as broken parts or delays in schedules. From spacecraft delivery to the launch site through launch, there should be at least 1-week of funded schedule reserve per month of launch site operations. The launch site is a new environment for the spacecraft and the team, so there are bound to be unexpected problems. For new or changing requirements, additional funds should be requested from management. Delays that are not the fault of the spacecraft team, such as a launch vehicle schedule slip, should be treated as a "new" requirement.

Because the launch is a major, often public, milestone for a spacecraft team, many organizations include promotional or recognition activities in their budget. Items such as pins, stickers, and posters are distributed prior to and on the day of launch to promote the mission. Other expenses include information packages about the mission and transportation to the launch viewing site for the "Very Important People (VIPs)." Be sure to work with all facets of your organization to understand what costs and requirements are being charged for your launch activities.

27.2.2 Launch Site Operations Planning

Depending on the complexity of the operations that are required, it is not unreasonable to start the launch site planning process 2 to 3 years prior to launch. A word of caution, do NOT wait until the last minute to start this planning. Even a less complex mission will take a minimum of 12 to 18 months (See Sec. 3.4), just to get the required documentation completed, submitted, and approved on time.

If the spacecraft has a propulsion system, once it is fueled, personnel safety will be of great concern. Additional precautions will be required by range safety to ensure operations are successful without incidents. A special processing facility equipped to handle spills or leakage of your particular propellant is most likely required. If there are any hazardous components on the spacecraft (e.g., pyrotechnic systems) or hazardous GSE needed to calibrate/test the spacecraft (e.g., a radioactive source), you must plan to spend more time and money for special processing. What may seem benign to you may not be to launch site safety!

No matter what the size of your mission, a little thought about operations at the launch site very early on in the process will help bound your costs later. Figure 27-2 is a typical launch operations timeline (including working group meetings and interface, and requirements documentation schedules).

Spacecraft Launch Site Interface

The coordination of launch site activities typically requires the full attention of an individual, such as a Launch Site Operations Manager (LSOM). This position requires someone who has a great attention to detail and good communication skills. Many times, teams will try to "stretch the budget" by assigning this task either to the Integration and Test (I&T) Manager or other members of the spacecraft team as a "part-time" responsibility.

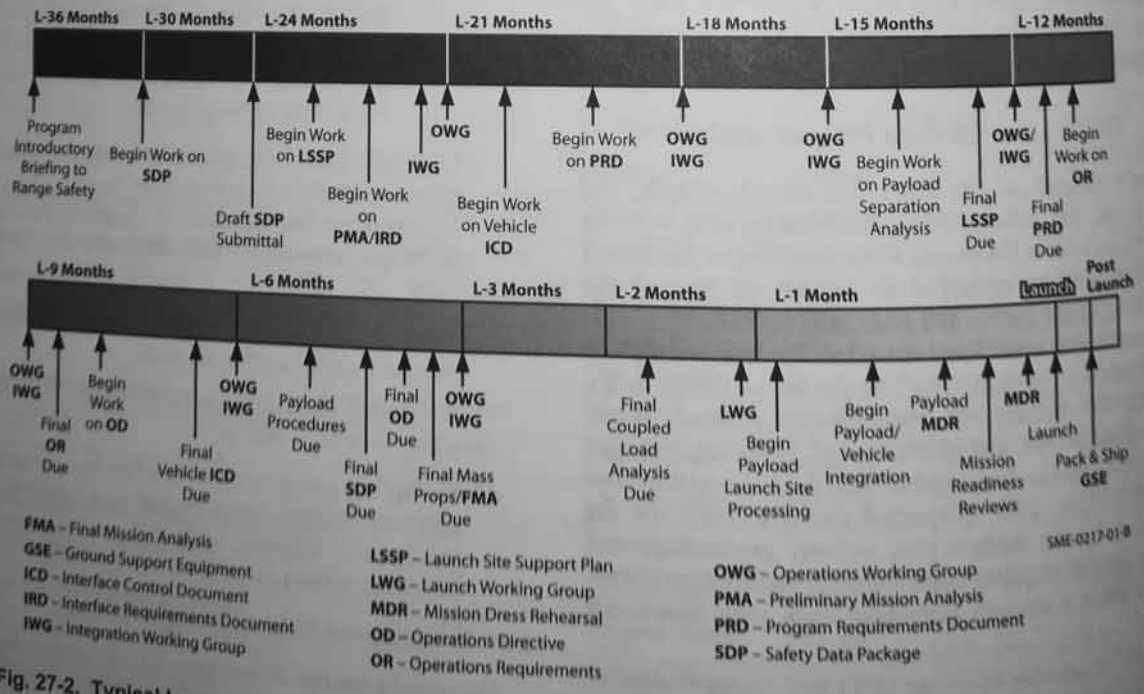


Fig. 27-2. Typical Launch Operations Timeline.

Table 27-1◀, Fig. 27-2, Eq. 27-0

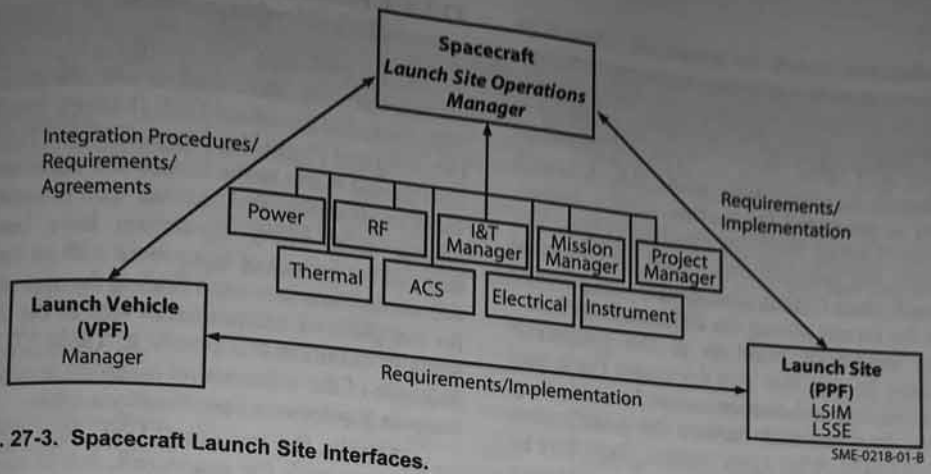


Fig. 27-3. Spacecraft Launch Site Interfaces.

Without a dedicated lead, important requirements are typically missed, causing delays in processing or unneeded disruptions and frustration.

The LSOM is the spacecraft representative and there to protect the project's interests. Knowing who to talk to, what to say, and what to ask for is crucial in saving time, money, and resources. Figure 27-3 is an example of how the LSOM interfaces with the launch site and launch vehicle teams. The LSOM ensures that day-to-day issues are resolved to the satisfaction of all parties (the project, vehicle, and launch site). In addition, the LSOM continuously plans for the upcoming project activities to ensure that little or no time is lost. Pre-coordination and scheduling of consumable resources ensures the smooth continuation of spacecraft processing and operations.

Working Groups

Over the course of a program, the launch team holds various meetings to define and work on the launch vehicle and spacecraft requirements. A few of the most common meetings related to launch activities are the Integration Working Group (IWG), the Operations Working Group (OWG), and the Launch Working Group (LWG). Depending on the launch vehicle and the launch site, the names of the working groups (meetings) may vary, but the objectives are basically the same.

Integration Working Group

IWGs are held every 3–4 months between the spacecraft and launch vehicle personnel and primarily concentrate on the activities that occur between the spacecraft and the launch vehicle (both integrated and stand-alone) before and after arrival at the launch site. This includes the what, when, and where of procedures such as electrical and mechanical integration, as well as monitored data points and testing. Launch site personnel typically support these meetings, even if for information purposes only.

Each vehicle team has a set process it uses when assembling and testing the launch vehicle. This process is typically modified for each mission to accommodate both vehicle and spacecraft integration and testing. During the IWGs, negotiations take place to ensure the requirements of both the spacecraft and the vehicle are met. Detailed drawings, as part of the Interface Control Documents (ICDs) are developed to illustrate the various vehicle/spacecraft interfaces required to ensure a successful launch and separation.

Operations Working Group

OWGs are held every 3–4 months (typically at the same time as the IWG to combine resources) with participants from the spacecraft, launch vehicle and launch site personnel. The OWGs begin early, from 2–3 years prior to launch, and continue up until the spacecraft arrives at the launch site. OWGs primarily concentrate on the spacecraft requirements and activities that will occur while processing at the launch site. This includes what is expected for the spacecraft arrival, the processing facility to be used, mechanical support (if needed), personnel requirements, cleanliness, communications, and safety, just to name a few. OWGs typically occur at the launch site, which allows the technical personnel, as well as management, to participate. Many important documents are developed from the OWGs, such as the Launch Site Support Plan (LSSP), Mission Constraints Document (also known as the Launch Commit Criteria), and the Launch Checklist (or launch script).

Launch Working Group

The LWG is the last meeting held prior to the spacecraft arriving at the launch site (approximately 30 days prior). The purpose of the LWG is to ensure that the launch site is ready for the arrival of the spacecraft. Each supporting entity presents their current status to the launch site upper management. Any outstanding issues are addressed and closed (or near closure) by the end of the meeting. If there are any major open issues, shipment of the spacecraft to the launch site will most likely be delayed.

27.2.3 Launch Site Documentation

Specific documentation is developed to describe in detail all of the support required from the launch site to

process and ultimately launch the spacecraft. These documents will cover communications, transportation, processing facilities, consumables, safety, mechanical and electrical procedures, and more as described in Sec. 6.4. The Launch Service Contractor (LSC) typically generates commercial launch documentation, while NASA launch documentation is generated by another NASA organization called the Launch Services Program (LSP). The military has their own unique format called the Universal Documentation System (UDS). This system was adopted to standardize the format among the different branches.

During the spacecraft build-up at the spacecraft assembly facility, be sure that you document the actual temperature, humidity, and contamination limits as they are defined and implemented. Include the justification for each of these limits. After a few months, there may be personnel changes and the current personnel may or may not know why these limits were chosen. By the time you reach the launch site, all the justification knowledge could have been lost if not properly documented. If a limitation is exceeded, it is very important to understand what components/assemblies are affected by the outcome of limit condition. Without the proper documentation, crucial time can be wasted studying whether or not the exceedance damaged something, whether or not the limits are valid, and can or should be changed. These environmental requirements get transferred into the launch site documentation (LSSP and ICD) for use in the Payload Processing Facility (PPF), during transport, and at the launch pad.

Produce and maintain a spreadsheet of the spacecraft procedures that will be used at the launch site. Develop a schedule for the release and review of each procedure. For those procedures that are identified as hazardous, clearly mark this on the front cover and highlight it in the spreadsheet. Define and list each agency that will perform a review, plus when and the duration they will have to provide comments back to you. Include agencies such as the spacecraft manufacturer, launch vehicle, launch site and range safety, as applicable. Allow enough time for the document to complete the review cycle (which could be as much as a month for each reviewer). Managers typically underestimate how long it will take to perform this review cycle (typically three times the amount), which could impact the operation.

Important: You will not be allowed to perform your operations at the launch site until the appropriate spacecraft procedures have been reviewed and approved by the launch site personnel. Failure to comply typically results in an immediate shutdown.

Depending on the launch site or vehicle, the names of the documents may vary, but the information in them will essentially be the same. Clear and concise documentation is not only required, but also necessary to ensure a smooth process at the launch site. Table 27-2 lists the typical US launch site documentation required (which may or may not apply to a particular mission) and the estimated time frame for first submission and completion (final version).

Table 27-1◀, Fig. 27-3◀, Eq. 27-0

27.2.4 Ground Support Equipment and Flight Hardware Specifications

Accurate Ground Support Equipment (GSE) specifications are imperative for a smooth processing flow. This includes Electrical GSE (EGSE), Mechanical GSE (MGSE) and Flight Hardware. GSE that requires calibration should come to the launch site calibrated and with a valid certification that extends well beyond the planned launch date. Many operations have been halted or delayed for a piece of equipment with an out-of-date certification. Time is wasted finding the appropriate support for that piece of equipment, as well as the time it takes to actually calibrate and certify it. Table 27-3 provides an example of the information included in a typical Ground Support Equipment Specifications table.

Generally, the spacecraft GSE does not reside in the clean room with the spacecraft, but is co-located in an adjoining room in the Payload Processing Facility (PPF) and also at the launch pad. This could be for a number of reasons, including cleanliness, safety, and limitations on physical space. If the GSE must be in close proximity to the spacecraft (e.g., in the clean room), most PPF management require that the GSE be explosion proof.

Interfacing the GSE with the spacecraft is accomplished via two methods, hard-line, or Radio Frequency (RF). The hard-line method is one or more cables (known as a spacecraft umbilical) connecting the GSE to a test or interface panel on the spacecraft. When designing this interface, carefully plan what you would like to be able to do with the spacecraft (e.g., send commands to the spacecraft or instruments, receive telemetry, power the spacecraft, or charge the battery), as this will determine how many cables are required for the umbilical.

There are two ways to interface the GSE with the spacecraft via an RF, open and closed loop. Open loop is via a signal transmitted from the GSE (RF rack) to the spacecraft antennas through the air. Closed loop is done by attaching a "RF hat" to the spacecraft antennas, attaching a cable to the hats, and connecting that cable to the GSE. Closed loop is very similar to the hard-line method, except the signal from the GSE is received via the spacecraft antenna instead of the test or interface panel. When determining which RF method is best for you, work closely with the launch site and vehicle personnel to identify any concerns.

When determining what the maximum length of the GSE cables should be, include their use at the launch site. Take into consideration the distance each rack will be from each other, distance from the spacecraft in the PPF to the GSE, and the interfaces (connections) at the pad. This will allow for the reuse of these cables at the launch site and not require a "rebuild" or an extension. Of course keep the total distance in perspective, as line loss is another factor to consider. In some cases, adaptors or converters may be used to connect with the different interfaces (e.g., coaxial to fiber).

In the special case of spacecraft with biological payloads, additional care must be taken to completely specify

Table 27-2. Launch Site Documentation.

Item	Submittal Time		Description
	Draft/Preliminary	Final	
Submit Systems Safety Program Plan (SSPP)		Launch minus 24 to 48 months	Defines what will be done by the payload organization to meet safety requirements for the Approving Authority.
Safety Data Package	Arrival minus 24 months (or 30 days prior to PDR) Arrival minus 18 months (or 30 days prior to CDR)	5 days prior to the Ground Operations Review (GOR)	Detailed description of hazardous and safety critical ground support and flight hardware equipment, systems, and materials and their interfaces that will be used by the spacecraft team while processing at the launch site.
Non-Ionizing Radiation Use Authorization Request (NIR)		Arrival minus 6 to 12 months	Provide reasonable and adequate guidance for the safe use of NIR sources at the launch site.
Launch Site Operations Plan (LSOP)/Ground Ops Plan (GOP)		Arrival minus 12 months	Defines the day-to-day spacecraft activities that will occur while processing at the launch site.
Program Requirements Document (PRD)		Arrival minus 12 months	Defines the requirements levied on the supporting launch site (also known as the "Range").
Submit Information for Environmental Impact Analysis (AF Form 813)		Launch minus 12 months	Form used for documenting the need to conduct an environmental analysis or the application of certain Categorical Exclusions (CATEXs).
Submit List of GSE Plastics, Films, and Adhesives Used (samples as required)		Launch minus 8 to 12 months	List of all the plastics, films, foams, and adhesives that will be brought to the launch site.
Launch Site Support Plan (LSSP)	Arrival minus 16 months Arrival minus 12 months	Arrival minus 7 months	For NASA launches, an agreement between the spacecraft project and the Launch Service Provider (LSP) as to the products and services necessary (i.e., requirements) for launch site processing from the time of the spacecraft arrival to the time of departure.
Submit Radio Frequency Authorization/Coordination Forms		Arrival minus 4 months	Process for obtaining temporary radio frequency authorizations for testing.
Payload Procedures		Arrival minus 90 days or NLT first use minus 55 days	Spacecraft procedures that may or will be used at the launch site during processing. Submitted to the Range Safety for approval prior to use.
Personnel Deployment Plan (PDP)		Arrival minus 90 days	Defines when supporting personnel are required to be at the launch site to support a particular event during the processing of the spacecraft (Table 27-4).
Contingency Plans (e.g., Off-load at Pad)		Arrival minus 90 days	Defines the procedures to follow should a spacecraft anomaly occur (such as a hydrazine leak).
Material Safety Data Sheets (MSDS)		Arrival minus 60 days	Sheet that provides the information needed to allow the safe handling of hazardous substances used at the workplace.
Final Arrival/Transportation Plans, Schedules and Arrival Badging Requests		Arrival minus 45 days	Defines the details to support the arrival, entry and placement of the spacecraft, equipment, and personnel onto the launch site.
Tool Control Plan (if applicable)		Arrival minus 45 days	Any special tool requirements that are to be used, on or in the vicinity of the spacecraft.
Launch Commit Criteria (LCC)		Launch minus 90 days	Defines the "Go/No-Go" criteria for both the spacecraft and vehicle.
Launch Checklist (Script)		Launch minus 90 days	A "step-by-step" list (Fig. 27-9) used by all supporting personnel on console during the countdown.

NLT = No later than.

Table 27-3. Ground Support Equipment Specifications.

ITEM	Function	Size (H × W × D)	Weight (lbs)	Power (V/A/Phase)	Connector Type	Cable Length	Use Locations PPF	Use Locations PAD	Cert Exp.
			EGSE						
RF Test Rack	Command and TLM Interface via RF Test Equipment for Transponder	72" × 20" × 32"	~500	120V/30A/ 1 Phase	NEMA L5-30P	6'	X	X	N/A
Test Rack	Testing	72" × 20" × 32"	~200	208V/30A/ 3 Phase (5-wire center ground)	NEMA L6-30P	6'	X	X	N/A
			MGSE						
S/C Integration Stand	Used to perform routine S/C operations and testing	4" × 6" × 6"	400	N/A	N/A	N/A	X	N/A	12/XX
Solar Panel Protection Cover	Used when lifting the S/C to protect the solar array panel	4' dia.	20	N/A	N/A	N/A	X	N/A	N/A

the life support function of the GSE, and the critical hand-offs between the GSE, launch vehicle, and on-board life support provisions. Whenever possible, the GSE and payload systems should be designed to support launch scrubs without requiring direct payload intervention.

When designing the GSE, take into consideration ALL of the environments in which the GSE will be used. For example, if the spacecraft manufacture is on the East Coast of the US, a standard rack will generally meet the requirements for most East Coast launch facilities. However, if the launch is to occur on the West Coast of the US, GSE seismic requirements must be met in order to use those racks in the West Coast facilities. This could include the addition of brackets to attach to a ceiling rail, the tying of several racks together, or other structural modifications. If the rack will be used in conjunction with a carrier aircraft, the FAA requires a specific grade metal and bolts be used before the rack can be certified for use in flight. Upfront planning by the Vehicle Production Facility (VPF) manager will help to alleviate modifications to the GSE later in the processing schedule.

27.2.5 Detailed Technical Drawings

It is often said that a picture is worth a thousand words. This statement has never been truer than when it comes to all of the intricacies required to launch a spacecraft. For example, when troubleshooting an anomaly with a custom communication link that goes from the spacecraft; through the launch vehicle; through the launch providers' equipment; to the spacecraft vendor's equipment; to the NASA interfaces; across Air Force property; to commercial communications lines; then back to the spacecraft vendor's facility, a detailed drawing is essential in helping to isolate where the anomaly's root cause could be located.

Facility

Detailed facility drawings for each area to be occupied should be done to scale. Details in these drawings will include power requirements (e.g., plug types, voltage, polarity, and number of wires), space requirements for GSE, clean rooms, storage, and office areas. Typically the building facility manager can provide drawings for the area to be occupied. If not, take a measuring tape and generate the drawing yourself. Use the information collected in the GSE Requirements chart to complete the facility drawing. Be sure to pay attention to the width AND height of the doorways to ensure the flight hardware, test rigs, and other hardware can fit through. Figure 27-4 is an example of a completed facility drawing.

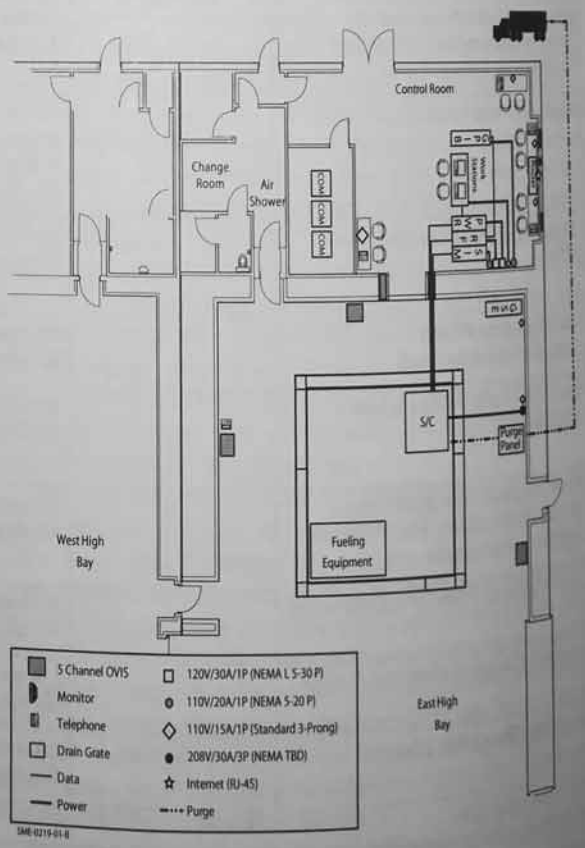


Fig. 27-4. Completed Facility Drawing.

Communication

Detailed technical communication drawings are also very important. Included in these drawings are on-site and off-site communications, as applicable. High-level drawings are typically used for depicting how voice and data will get from point to point. Low-level drawings include not only the high-level information, but also each individual piece of equipment that is required for the end-to-end flow. Low-level drawings can be invaluable when trying to isolate an anomaly. Figure 27-5 is an example of a high-level communications drawing.

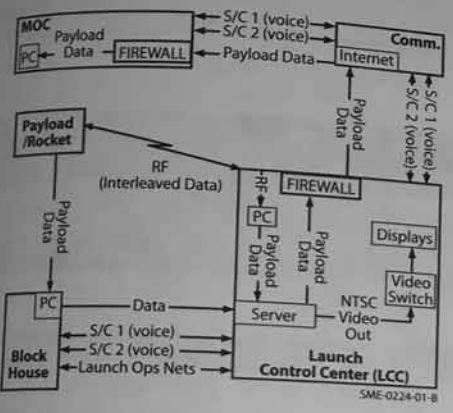


Fig. 27-5. High-Level Communications Drawing.

27.2.6 Schedules

In order to meet the contracted launch date, it is imperative that both the launch vehicle and the spacecraft teams independently develop detailed processing schedules. Each schedule should remain flexible with enough built-in slack time to accommodate issues that may arise. These two independent schedules are then integrated into one master schedule, which will call attention to any potential conflicts in activities. Once the conflicts are identified, who will do what and when, is negotiated between the parties. Off-shift hours or additional days may need to be added to the schedule to accommodate joint or overlapping processing activities. Once the integrated schedule processing flow is complete, you can calculate what date the spacecraft must arrive at the launch site to make the contracted launch date.

Launch Vehicle Schedule

Typically, the launch vehicle team already has a clearly defined schedule of what it takes to assemble, test, and process its vehicle in preparation for a launch. The launch vehicle schedule can be modified to accommodate any stand-alone testing the spacecraft requires. To accommodate spacecraft processing, additional days may be added to the vehicle processing schedule. One of the most important dates to note is the day the spacecraft is to be mated to the vehicle. Typically, "mate day" sets the timer to meet the contracted launch date. If you do not finish the spacecraft processing in time to make the mate day, a launch slip is likely.

Spacecraft Schedule

Prior to coming to the launch site, the spacecraft team will have developed a detailed schedule of day-to-day activities. One of the golden rules is to not cut the schedule short. Up front, plan to work an 8 hour, 5 day (40 hour) week. This allows the other 16 hours per day and weekends to stay as contingency hours that may be used for unexpected situations that occur. In my experience, you WILL use those contingency hours and wish you had many more!

Most launch sites (at least in the US) have a cap on the number of hours worked by any one person in a week: typically a maximum of 12 hours per day, or 60 hours per week. Additional personnel may be needed to perform certain activities in order to not exceed this limit. Careful planning will help to ensure this does not happen during the processing. One exception is if a specific test will take more than 8 hours. Again, this should be the exception…not the rule.

A Personnel Deployment Plan (PDP) is key for both the project and launch site management. The PDP correlates day-by-day with the LSOP and defines when the supporting personnel are required to be at the launch site to support a particular event during spacecraft processing. Proper planning and coordination of the PDP will help keep a project on time and on budget. After all, you do not want your blanket installation team at the launch site "standing by" if the Flight Operations Team (FOT) is scheduled to test for the next 3 days. The PDP is a living document that is continuously revised as the processing schedule is adjusted to complete the necessary operations or unexpected situations that may arise. Table 27-4 represents a partial snapshot of a Spacecraft PDP.

To determine the day the spacecraft should arrive at the launch site, add the number of days of spacecraft stand-alone processing to the L-day (Launch minus day) of the spacecraft/vehicle mate. For example, if the spacecraft/vehicle mate day is on L-45 days (L minus 45 days) and the spacecraft stand-alone testing will take 22 days, arrival at the launch site would be on day L-67 (L minus 67).

Integrated Schedule

The launch vehicle and spacecraft schedules are considered "stand-alone" until the spacecraft and launch vehicle either occupy the same facility or start the process of integration. These two independent schedules must then be integrated to ensure there are no conflicts between the two. For example, there may be times when the launch vehicle or spacecraft processing activities require non-essential personnel to clear the area or the entire facility. Fueling is a good example of a facility clear. If you are not considered essential for this activity, you will not be allowed to enter the facility. Most launch vehicle and spacecraft activities can be executed simultaneously. If this is not feasible (typically a safety issue), it may be possible to work "off" shifts to meet the objectives of the day. If neither is possible, full days will need to be added to the schedule to accomplish the required tasks.

Table 27-3◀, Fig. 27-5, Eq. 27-0

Table 27-4. A Spacecraft Personnel Deployment Plan.

DATE	PPF	PAD	# of Staff	L. Jones	D. Gates		A. Smith	D. Everett	H. Cooper	T. Jackson	T. Russell		G. Fraizer	H. Henley	S. Castro	B. Schena	DATE
14-Dec	S/C Arrival—Unpack/ Setup EGSE at PPF		7	1	1			1	1	1			1	1			14-Dec
15-Dec	S/C Aliveness Test		7	1	1			1	1	1			1	1			15-Dec
16-Dec	Thermal Closeouts		7	1	1			1	1	1			1	1			16-Dec
17-Dec	Fueling		7		1			1	1	1			1	1		1	17-Dec
18-Dec	RCS & Leak Test		7		1			1	1	1			1	1		1	18-Dec
19-Dec			0														19-Dec

Table 27-5. R-Day Schedule (based on a January 2nd launch date).

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
13-Dec –20	14-Dec –19	15-Dec –18	16-Dec –17	17-Dec –16	18-Dec –15	19-Dec –14
	S/C Arrival Unpack/ Setup EGSE at PPF	S/C Aliveness Test	Thermal Closeouts	S/C Fueling (Facility Clear ~ 4 hours)	RCS & Leak Test	
20-Dec –13	21-Dec –12	22-Dec –11	23-Dec –10	24-Dec –9	25-Dec –8	26-Dec –7
	S/C & Vehicle Elect. and Mech. Checks	S/C Mate to Vehicle Upper Stage	Fairing Installation	Transport to Pad Mate to Vehicle	Holiday	
27-Dec –6	28-Dec –5	29-Dec –4	30-Dec –3	31-Dec –2	1-Jan –1	2-Jan 0
	Comm. Checks	S/C Functional	Final Vehicle Closeouts	Launch Rehearsal	Management Review Crew Rest	Launch Window 1400–1430Z

Carried over from the discontinued Scout program is a schedule format I have used for over 20 years, called a R-Day (Ready-Day) schedule. The R-Day schedule can be used to easily show all of the daily activities that will be performed by the vehicle and the spacecraft during the launch site processing. Table 27-5 is an example of an Integrated R-Day Schedule. A simple ship and shoot mission (i.e., no on-site spacecraft testing) could begin as late as R-11. A more complex mission could start weeks or even months earlier. Note: The R- and L- (used above) can be used interchangeably.

Once you arrive at the launch site, daily meetings usually occur among all parties to discuss the previous day's results, activities for the current day, and planned activities for the upcoming day. Typically these meetings occur in the morning prior and, depending on the complexity of the processing, may occur again in the afternoon. Good communication between the spacecraft and vehicle personnel during processing is essential to staying on schedule, and ensuring mission success.

27.3 Readiness Reviews and Mission Dress Rehearsals

The Launch Vehicle Readiness Review (LVRR) is held to certify the readiness to proceed with spacecraft/launch vehicle integration activities and is typically held prior to the Spacecraft Readiness Review (SRR), which is also called the Mission Readiness Review (MRR) or the Spacecraft Flight Readiness Review (SC-FRR).

The Flight Readiness Review (FRR) is held to update the mission status, close out any action items from the earlier LVRR and SRR, and certify the readiness to proceed with initiation of the launch countdown.

The Launch Readiness Review (LRR) is held to update the mission status, close out any actions from the previously held FRR, authorize approval to proceed into launch countdown and sign the Certification of Flight Readiness (COFR). The COFR is signed at the conclusion of the LRR.

There may be many other reviews as appropriate and necessary in preparation for launch. These may include, but are not limited to, Mission Requirements Reviews, Critical Design Reviews, Design Certification Reviews, Pre-ship Reviews, Ground Operations Reviews, Project Manager's Reviews and other appropriate safety reviews.

A Mission Dress Rehearsal (MDR) is performed with both the vehicle and spacecraft support personnel at approximately launch minus 2 to 3 days, exercising each step up to and including launch. MDRs are an invaluable learning tool for the team, as this may be the first time many of the members have ever supported a launch.

During the MDR and actual launch countdown, support personnel (both managers and engineers) will monitor many displays and voice networks. Headsets are worn so each individual can monitor the proper networks for reasons discussed in Sec. 27.8.1.

One of the main goals of the MDR, and possibly the most challenging, is real-time problem resolution. Typically there will be 2 or 3 runs of the launch script. During at least one of the runs, anomalies or issues (and possibly an abort) will be interjected to test the reaction of the team. This is not meant to embarrass anyone, but give the team experience in dealing with potentially stressful situations that may occur during the launch, prior to the launch. The MDR test coordinators are looking for predefined reactions to the induced anomalies. After each run, the team's performance is reviewed and evaluated. The Test Conductor (TC) gives a debriefing, explaining what was expected during the run and both the positives and negatives on how the team performed.

Table 27-6 lists the reviews and rehearsals, and the time frame in which they typically occur.

Table 27-6. Readiness Reviews and Mission Dress Rehearsals Timeline.

Item	Timeline
Ground Operations Review	Arrival minus 30 days
Pre-Ship Review	Arrival minus 7 days
Flight Operations Readiness Review	Launch minus 90 days
Pre-Mate Review	Mate minus 1 day
Payload Mission Dress Rehearsal*	Launch minus 15–20 days
Flight Readiness Review	Launch minus 3 to 4 days
Launch Management Coordination Meeting	Launch minus 3 days
Mission Dress Rehearsal	Launch minus 2 to 3 days
Launch Readiness Review	Launch minus 1 day

* Not a requirement, but good practice is to do a Payload MDR for the spacecraft team prior to the full-up MDR. Members of the launch vehicle team participate reading the checklist. The spacecraft team gains invaluable experience on how to use the launch consoles, using voice protocol, anomaly resolutions and is a great preparation for the "real" MDR.

27.4 Launch Site Access

Access to a launch site can be obtained easily in most cases, as long as the proper procedures for that site are followed. Typically, a visit request (VR) is required, where you will provide your full name, company, date of birth (DOB), citizenship, purpose of visit, dates of visit and clearance (if any). This information is sent to the assigned launch site point-of-contact (POC) or sponsor for processing. Once the VR is approved, the Visitor Center will make available the appropriate credentials.

You must provide any requested information to your POC as early as possible. The number one cause of personnel being denied access to a launch site is procrastination.

If you are not a resident of the country in which you are launching, you will most likely be designated as a Foreign National (FN). You will provide the same information as the VR, but the VR may take longer to process (typically 30 days or more).

If you are launching in the US and require FN's at the launch site, make this known as early as possible. FNs will require an escort while on the site. Note: A foreigner possessing a green card (officially known as a United States Permanent Resident Card) is typically treated as a US Citizen.

A security clearance is typically not required to gain access to a launch site. If you require access to secure facilities at the launch site and do not have appropriate clearance, you will be escorted.

NASA missions require a minimum of a National Agency Check Inquiry (NAC-I). A NAC-I is the minimum level of investigation required of Federal employees as a condition to employment with the Federal government and now for contract employees as a condition to access to Federal facilities and information systems. This is essentially a check of law enforcement records, written inquiries to schools, police departments, contacting references and a fingerprint check to verify your status. While it takes about a year (or more) to get a formal security clearance processed, a NAC-I is usually issued in approximately 6 months.

27.5 Launch Site Training

The amount of training required for spacecraft personnel supporting at the launch site depends on the functions each individual will perform. At a minimum, personnel will go through some type of facility safety or familiarization training. In most cases, the written portion of the training can be completed at your facility prior to arrival. Ask the launch site POC to provide as much of the training materials as possible. The walk-through portion of the launch site training is completed upon arrival. Topics include emergency evacuation procedures, operational procedures, and facility safety, to name a few.

Specialized training may need to be completed at the launch site. This could include the use of an Emergency Life Support Apparatus (ELSA) unit or Self-Contained Atmospheric Protective Ensemble (SCAPE) suit, vehicle/spacecraft fueling, clean room procedures, magnetic procedures, or other site- and vehicle-specific content. In some cases, proof of a recent medical physical must be presented upon arrival for those personnel training to operate equipment, such as overhead cranes and forklifts. Make sure you have two trained operators (prime and a backup) for all time-sensitive operations.

Console training familiarizes personnel who are on console for the launch with the range processes, available displays and communication protocols.

Most likely, during Integration and Test (I&T), spacecraft personnel have not been using formal voice protocols when speaking to each other over voice lines. Rather than calling out "Bob" or "Helen," proper voice protocols are established for the launch and must be adhered to. Unique titles such as Mission Manager and Mission Director are used to signify the function being

performed by that individual. As the voice lines are recorded, open-air conversations are highly discouraged. If an anomalous situation occurs, the structured clarity of proper voice protocol can be very helpful in determining the cause of the problem.

The spacecraft safety person and the on-site launch site safety officer are responsible for ensuring personnel have received proper training. A training matrix (Fig. 27-6) is used to track the training required for each individual and the status of that training.

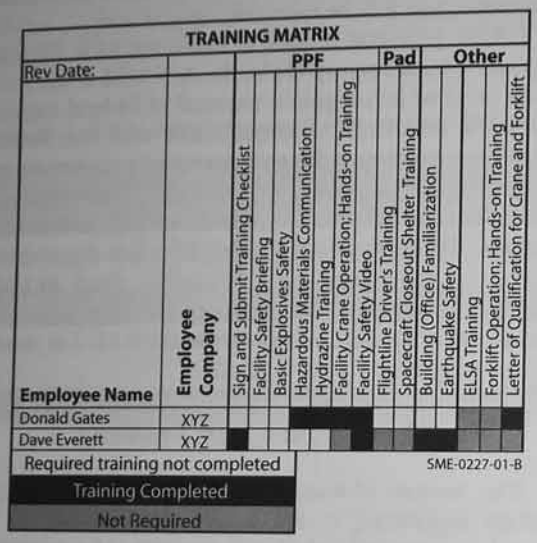


Fig. 27-6. Training Matrix.

27.6 Transporting the Spacecraft to the Launch Site

Depending on how far the launch site is from the spacecraft assembly facility, transporting the spacecraft to the launch site is accomplished either via air or ground. Government owned spacecraft (e.g., NASA, DoD) have the choice of using government (e.g., Air Force, Army) or commercial carriers, while commercial spacecraft typically use a commercial carrier. Coordination with the launch site personnel is crucial to ensure an uneventful transition once you have arrived.

Traveling by air has a few benefits over ground transport, such as taking less time to transport, especially if transporting across the country and having less physical impact on the spacecraft. Traveling by air does come with higher costs as additional resources are required once the plane arrives, such as offloading and transport equipment, personnel, airport fees, costs to procure the aircraft, and fuel.

Traveling by ground is typically much cheaper, but it may take longer and be much harder on the spacecraft due to the various road conditions encountered on the way. If you are purging the spacecraft (which you should be), have pre-planned stops along the way to check the levels and fittings. In my experience, spacecraft teams have followed the truck in some type of vehicle (usually a camper) from departure to arrival. Ensure you have enough personnel to support this type of operation, as trucking companies typically use two drivers, and only stop for fuel.

Whether you transport by ground or air, attach shock sensors to the spacecraft container. By monitoring and recording the vibrations from these sensors, you will be able to see if any shock like conditions were encountered along the way. After arrival review the sensor data to see if there should be any concerns.

Two points of caution that must be considered are:

1. Choose a carrier that has experience in moving spacecraft and the GSE (this is not the place to cut costs) and
2. Ensure you get enough insurance to replace the spacecraft and the effort to duplicate it (called an agreed upon value) in case a mishap should occur along the way.

When budgeting for transportation, it is important to do a trade study to weigh the costs versus the benefits of both air and ground travel. Though it may not always be apparent, saving time can be a way of saving costs.

Ensure you know how the spacecraft will be offloaded at the launch site. While you may use the lift gate of the transport truck to load the spacecraft at your facility, the launch site may have a NO lift gate policy and require the use of a crane or forklift. Depending on the launch site policy, this could be one of the determining factors of how you design the spacecraft-shipping container.

Another design driving consideration is the environment which the container may be exposed to. If offloading inside a facility, there may not be much of a concern. If offloading outside, determine if weather-proofing the container should be considered in the design. Consider the worst-case environmental conditions, as your launch date could slip, shifting your arrival date from the most optimal weather to the worst.

27.7 Launch Site Processing

Depending on the operations to be performed, once at the launch site, the spacecraft may proceed to either the Payload Processing Facility (PPF) or directly to the Vehicle Assembly Building (VAB). Figure 27-7 illustrates the Astrotech Space Operations (ASO) facility; one of the PPF's currently available at VAFB.

This facility, owned and operated by ASO, is located on a 60 acre leased site on Vandenberg Air Force Base, CA. They provide for non-hazardous and hazardous processing. An 8,000 ft^2 spacecraft processing bay that provides the capability to process 5-meter class payloads was completed in 2010.

27.7.1 Arrival

Upon arrival, the spacecraft is met by launch site personnel and escorted to the processing facility. At the processing facility, the spacecraft and GSE are unloaded

Table 27-6◀, Fig. 27-6, Eq. 27-0



Fig. 27-7. Astrotech Payload Processing Facility (VAFB).

and taken to the assigned area. Once this has been completed, all operations cease and any remaining safety training is completed.

In a matter of hours, the GSE can be connected to the spacecraft, configured and ready to begin testing. Typically the first test is a spacecraft "aliveness test" to ensure the spacecraft made the trip to the launch site from the processing facility successfully and establish a baseline for later testing.

27.7.2 Processing Flow

The processing flow or Integration and Test (I&T) activities, both spacecraft and launch vehicle, are well defined prior to the spacecraft arrival at the launch site. Unless you are a "ship and shoot" type mission (i.e., no spacecraft testing or processing is necessary), you will have allotted time on the schedule to do final "stand-alone" spacecraft and instrument testing, solar array installation, ordinance installation, and successful testing of each.

While you are at the launch site, work at the Mission Operations Center (MOC) is still in progress as well. A good, (although not necessary), rule of thumb is that anytime the spacecraft is powered on at the launch site, allow the MOC to receive the telemetry data (assuming the MOC is off-site). In many instances, the spacecraft team at the launch site will allow one last test (e.g., commanding the spacecraft) with the MOC prior to launch. This gives the MOC personnel an opportunity to continue to test their systems and further train personnel. When deciding if this is a good thing for your mission, keep in mind that this is NOT the time to try out anything new (this includes commands, automated scripts, or data base updates). The ONLY exception would be that if the change to the MOC system would delay the launch if it were not tested prior to the launch.

During the processing flow, a Payload Attach Fitting (PAF) is provided to the spacecraft team by the launch vehicle team to integrate the spacecraft with the launch vehicle. This is also the separation point between the two. The PAF consists of two halves held together by a clamp band and is modified on the spacecraft side for each mission to accommodate the attachment points of the spacecraft. A PAF fit check and separation test is done at the spacecraft facility long before arrival at the launch site. One half of the PAF will stay with the spacecraft after separation from the launch vehicle, so this must be included in your spacecraft design, weight measurements and orbit analysis.

A pre-mate review is held just prior to the spacecraft and vehicle integration, where each team presents what has been done (e.g., testing, assembly, repairs) since arrival at the launch site and the results to upper management. With a successful review and upper management approval, a GO is given for mate.

Depending on the launch vehicle, spacecraft mate is accomplished either in a horizontal or vertical position. Each position offers its own challenges and how this is successfully and efficiently accomplished is carefully thought out.

With vertical processing, the launch vehicle is assembled inside of a tower, which rolls out of the way on a railway for launch. Typically with vertical processing the spacecraft is mated with the upper stage motor at the PPF (now called an assembly) and then placed inside a special shipping container, provided by the launch vehicle team. The container is transported to the launch complex, where it is craned and set on a platform, located at a height just above the top of the launch vehicle. The crane is used to lift the assembly out of the container and lowers it slowly towards the launch vehicle. Once the PAF is aligned with the upper stage, the bolts are put in and torqued to specification. The container is removed and operations continue.

With horizontal processing, many of the above processes are the same; therefore I'll just discuss the differences. The launch vehicle is assembled inside a facility, instead of a tower. When mating the spacecraft to the launch vehicle, the spacecraft is lifted from the side, instead of the top. This type of mate is typically accomplished using a crane; however, some missions use a wheeled mechanical structure to have better control over the spacecraft movement. Determine early in the process if a crane lift can be performed inside a clean environment (such as the facility itself or a clean tent) as required by your contamination expert. If not, you may need to design (or modify) a rolling fixture to integrate with the launch vehicle. One could argue which process is better, but both processes will work, as long as either the spacecraft or launch vehicle can be rotated when it comes time to align the bolt holes. Finally, the launch vehicle is raised from the horizontal to vertical position for launch. Depending on the launch vehicle, once vertical, a tower may be rolled in to keep the launch vehicle safe from the weather and then rolled out of the way for launch. One notable exception to these final steps is the Pegasus launch vehicle, which features a horizontal integration and unique horizontal air launch.

Even though electrical and mechanical interface testing has been previously completed, this is the first time the spacecraft is integrated with the fully assembled launch vehicle. To ensure there are no interference issues, the launch vehicle team powers the launch vehicle, then requests the spacecraft to be powered, carefully monitoring each system for any abnormalities. After suc-

cessful testing, each half of the fairing is installed, allowing time for any spacecraft and launch vehicle Remove Before Flight (RBF) items and final closeouts. This completes the assembly and testing of the launch vehicle. You're just about ready to launch!

In Sec. 27.3, the MDR and launch reviews were explained in detail, therefore I won't say more than… it's that time. After successful completion of the MDR and launch reviews, a GO is given for launch. The supporting ground stations will go through a last pre-launch readiness test, flowing spacecraft data to the MOC. Depending on the mission requirements, these ground stations may flow data to the MOC during the launch count down—something to consider when you are planning how to ensure the MOC and your supporting ground stations are "green" for launch. Figure 27-8 is a snapshot of a processing flow from arrival through post-launch.

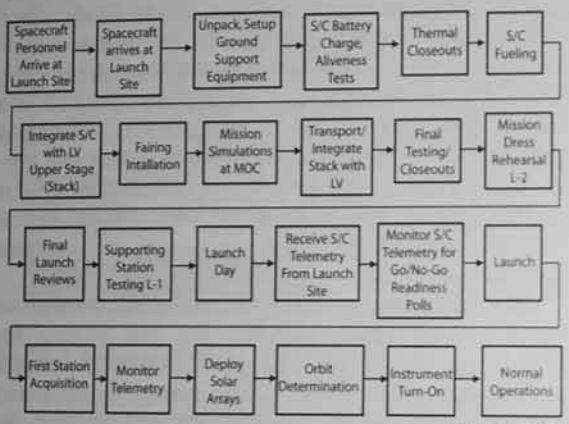


Fig. 27-8. Process Flow from Arrival Through Post-Launch.

27.8 Launch Day*

Depending on the vehicle and spacecraft configurations required prior to launch, you may be requested to arrive "on station" anywhere from 4 to 12 hours prior to launch (L-0). The Launch Conductor (LC) will "open" the Launch Checklist, perform voice checks with key positions and update the team with any last minute changes or corrections (black lines). Once the LC has completed the opening, NO changes to mandatory requirements are accepted.

Spacecraft and vehicle engineers arrive on station to perform final configurations and monitor their respective telemetry data. Once the direct spacecraft cables are disconnected, spacecraft telemetry is multiplexed, or mixed in, with the vehicle data and then decommutated, or separated out, on the ground. If any discrepancies are observed, they are reported to the appropriate personnel. After launch, spacecraft telemetry is received via the vehicle telemetry downlink until spacecraft separation or the first independent contact is established.

Typically, launch is targeted for the opening of the "launch window." *The launch window* is the period between the earliest and latest times during which the launch vehicle can be launched in order to achieve the orbit defined by the satellite. If an unexpected delay occurs during the count, it may be possible to recover from the anomaly and still launch, if the new time is still within the window. Depending on the required orbit, the launch window could be as small as a few seconds to a few hours.

27.8.1 Launch Checklist (Script)

The Launch Checklist (Fig. 27-9) is a step-by-step list (script) used on launch day by all supporting personnel on console during the countdown. The launch script specifies exactly what is to be said by the Launch Conductor (LC) for each call-out and the exact response expected from the respondent i.e., Ground Station Manager (GSM). Each item is checked off as it is completed. Certain pre-defined parameters are hand recorded in the checklist during the count, as they are not fixed, but are expected to be within a certain range. If the numbers read back do not fall within this range, there may be an issue that will be discussed. Another example is the exact launch time. This time is filled in at the opening of the checklist, as the time may have been shifted a few seconds to accommodate a Collision Avoidance (COLA) and ensure that the spacecraft (or launch vehicle) will not collide with another object in space. There may be times when items are "jumped" or skipped during the count. The LC will come back to that item to ensure it has been completed. ALL steps in the checklist must be completed (or waived) prior to giving the final GO for launch.

Embedded in the launch count, are periods called "built-in holds." These are blocks of time that allow the team to take a break or catch up if any of the previous steps being performed took a little longer than expected. The LC continues with each step in sequential order as defined in the Launch Script, but may occasionally jump steps if a task needs a little extra time to be completed.

At approximately L-2 minutes "terminal count" is entered (time actually depends on the vehicle). Once Terminal count is entered, all "required" requirements are now considered "mandatory" and cannot be waived. If any anomalies occur with the vehicle, spacecraft, or any of the supporting sites during this period, a launch scrub will most likely occur. A poor report from the Launch Weather Officer (LWO) would be a good example. If a scrub does occur, the LC will direct the team as to the next course of action. Otherwise, the LC continues to work to the L-0 time and the vehicle is launched.

If an anomaly occurs and can be fixed in time for all entities supporting the launch to complete their required work, prior to the end of the launch window, the LC will instruct the launch team to go to a point in the checklist and the count will begin again.

Get Stnd

*Applicable standards: ISO 10784 (Space Systems). See website for reference.

#	Action	Operation	Report	Net
1	LC	Call to Station/ Voice Checks ROC _____ RCO _____ MPCO _____ PM _____ LWO _____	Voice Checks Complete	Launch Ops Net
2	PM	Call to Station/ Voice Checks GSM _____ ACS _____ Thermal _____ LWO _____	Voice Checks Complete	S/C 1
3	RCO	Report status of Voice, Data, and Video Lines	Range Voice, Data, and Video Lines Operational	Launch Ops Net
4	ROC	Coordinate LWO Weather Brief on Ground Ops Net	Roger	Launch Ops Net
5	LWO	Confirm Weather Status	Weather is Green	Launch Ops Net

SME-0229-01-B

Fig. 27-8. Launch Checklist (Script).

27.8.2 Launch Commit Criteria

The Launch Commit Criteria (LCC) defines the Go/No-Go criteria for both the spacecraft and vehicle. Listed are the critical data points for each subsystem, along with the minimum and maximum limit specifications. Each subsystem being monitored must be in a nominal condition (i.e., within the specified limits) in order to proceed with the next task.

There are two levels of requirements defined in the LCC, "required" and "mandatory." Once the checklist has been opened, any "required" parameters that exceed the expected limits may be waived ONLY if the launch management team is in unanimous concurrence. Again, once Terminal count is entered, all "required" requirements are now considered "mandatory" and cannot be waived.

"Mandatory" parameters that exceed the expected limits cannot be waived for any reason. This process helps to keep a condition called "*Launch Fever*" from occurring. As discussed in Sec. 27.12, launch fever can occur when the excitement or pressure of getting the spacecraft launched on time overtakes making a common sense decision on an issue that arose, potentially delaying or scrubbing the launch. These parameters were defined as mandatory for a reason, with level heads, and cannot be changed for this launch attempt.

27.9 Post-Launch and Early Orbit Operations

Once the launch vehicle has lift-off, the Launch Vehicle Engineer will announce the key events (e.g., Stage 1 separation, Stage 2 ignition, and so on) over a specified network channel. The launch vehicle transmits both the launch vehicle and spacecraft telemetry data (multiplexed by the launch vehicle systems) in real-time during the flight. Once the ground equipment receives the telemetry, it is decommutated and transmitted to the respective systems specialists. The launch vehicle and spacecraft engineers monitor the telemetry data for any signs of off-nominal specifications and verification of nominal operations. Range safety personnel monitor the trajectory of the vehicle in-flight and are ready to send a destruct signal if the vehicle veers off course.

Ground stations are pre-scheduled to follow the launch vehicle using radar or other assets from lift-off through spacecraft separation. If the launch vehicle signal is not available at the time of separation (e.g., due to ground station support not being available), spacecraft separation is not verified until the next contact. Once spacecraft separation occurs, spacecraft telemetry is no longer received as part of the launch vehicle telemetry, but from a separate signal transmitted by the spacecraft to a supporting ground station.

When developing the automated on-board sequences for after spacecraft separation, ensure no spacecraft deployables are initiated until the launch vehicle is far enough away from the spacecraft to avoid interference or possible contamination (e.g., at least a 2 second delay).

At spacecraft separation or Loss of Signal (LOS) with the launch vehicle, spacecraft operations are handed over from the spacecraft team at the launch site to the Flight Operations Team (FOT) at the Mission Operations Center (MOC). It is now up to the FOT to begin the on-orbit checkout of the spacecraft.

In many cases, spacecraft teams have a limited number of support personnel. Key spacecraft personnel that were required to be at the launch site for launch may not physically make it to the MOC prior to the first acquisition. The solution to this problem is to transmit the spacecraft telemetry data received at the MOC to the ground support equipment still located at the launch site. This will allow those personnel to view the same data as the MOC and participate in the operations. Once it is determined that all is well with the spacecraft, those key personnel can then make their way to the MOC.

The FOT will continue to monitor the health and safety of the spacecraft and follow the agenda that is defined in the Launch and Early Orbit Plan (LEOP). This includes operations such as deploying the solar arrays, instrument covers and booms, and monitoring the status of the spacecraft (housekeeping data), as required. Eventually the spacecraft instruments are turned on, checked out, and the primary mission begins (e.g., science and military).

Launch Site Departure

What came to the launch site must leave the launch site. Coordinating with the launch site personnel, GSE shipping arrangements are made by the spacecraft team. Again, give the requested information for the truck drivers to the launch site personnel as early as possible, thus giving them time to process the badging. As most people are anxious to get home, it generally takes little time to get the equipment boxed and ready to go. Once the shipping truck arrives, is loaded and has departed, the launch site personnel will start getting the area ready to receive its next occupant.

Prepare yourself for an emotional experience, for some, both good and bad. What has been "home" for the past few weeks (or months) is coming to an end and could be a little sad. What has taken years to plan, build and test is now thousands of miles away in orbit, performing as designed. What an emotional high! Yes, it's a joyous occasion and time for celebration.

Through these past few years, co-workers have become friends and in many cases, as close as family. Nonetheless, it's time to go home. The space community is a small community. You never know where our paths may cross in the future, but one thing is for sure…if you stay in the business long enough, it will happen.

27.10 Modernizing Launch Operations

Budgets get smaller each year, so newer and smarter ways of doing business are quickly becoming a reality. The less time spent processing and testing at the launch time can equate to quite a savings. While the caption "Faster, Better, Cheaper" is echoing throughout the spacecraft community, doing more with less appears to be the motivating factor.

Cutting Costs

There is always the age-old question, how can we cut costs? Is there anything the mission (spacecraft) designer should know about how to make launch operations easier and cheaper?

In most cases the launch site can and will support the necessary requirements to process a spacecraft at their facility. There are, however, various operations that could be taken into consideration to shorten the process flow and cut costs (see Secs. 13.3 and 13.5).

Testing

- Limit testing to what is really necessary to verify the spacecraft is ready for launch.
- Consider whether full blown I&T simulation can be shortened at the launch site and achieve the same objectives.

Environmental Operations (Purge/Cleanliness Requirements)

- Typically the cleanliness requirements of the entire spacecraft are driven by the instruments.
- Devise ways to keep the instruments and spacecraft clean without specialized purging systems.
- Evaluate the spacecraft environment. Is a class 10K contamination environment essential or is class 100K sufficient? What could be done differently in the spacecraft design or operations to utilize a class 100K environment?

Mechanical Operations

- Keep the number of mechanical operations in the field to a minimum by installing as much hardware as possible at the manufacturing facility.

Spacecraft Battery

- Use a battery that does not require high maintenance. Avoid those with constant trickle charging, recycling, top-offs, and excessive cooling requirements (Sec. 21.2).

Support Personnel

- Limit the number of support personnel at the launch site. Only deploy those who are essential for the operation at hand, and do so in a timely manner.

Risks Versus Cost

With the constant need to cut costs comes higher risk. When designing the spacecraft, careful consideration should be given as to when the risks outweigh the costs. For example, many spacecraft today employ "single string" architecture. A failure using this design could mean a quick end to the mission. Risk vs. cost must also be considered while processing at the launch site. Questions to consider include:

- What spacecraft testing actually needs to occur vs. what is a nice thing to do?
- How much retesting is needed at the launch site when you have tested this system multiple times over the past few years?
- Will a quick aliveness test do, or will only a full 2 day comprehensive test fulfill the requirement?
- Do you need to do any additional testing at all? How confident are you the spacecraft will work on orbit without testing that one last time?

More time equals more money. Determine what risks you are willing to take **before** arriving at the launch site.

Remote Operations

Instead of bringing the entire spacecraft engineering support team to the launch site, send the data to them. With the technology and communications available today, this concept may not be too far from reach.

One could argue that key decision making personnel should be at the launch site. Those monitoring telemetry displays could be basically anywhere in the world if they can receive the data they need to review.

Alternatively, a smaller team could be deployed to the launch site. A trade study should be done to compare the communications costs versus what it would cost to deploy the larger team to the launch site.

Also, there is the risk factor (Sec. 24.2). The question is…would you be willing to launch the spacecraft if you were to lose communications with off-site personnel? Reliability of the communication lines is a huge factor to consider (Sec. 24.1). Consider the cost of redundant communications as this capability might be significantly less than the cost of sending an entire team to the launch site and reduce such a risk.

Table 27-6◀, Fig. 27-9◀, Eq. 27-0

Launching as a Secondary Payload

In many cases, launch vehicles can accommodate one or more smaller payloads, along with a primary mission. Often, the primary payload may not be using the full performance capability or "real estate" of that launch vehicle. Special adapters have been developed by launch vehicle operators to carry small payloads, allowing universities and low-cost missions to take advantage of an opportunity to be launched as a "secondary" payload.

As a secondary payload, you are subject to the same characteristics of the primary payload (e.g., shock, thermal and loads) and will have to meet the requirements defined for that launch vehicle and primary mission. If well managed, "hitching a ride" is a win-win for both the launch vehicle and spacecraft, as two satellites are launched for the price of one.

There are two very apparent advantages of flying as a secondary payload: time and cost savings. With the primary payload paying for the actual launch, you save a huge amount of money on launch costs. By using proven (qualified) systems developed by the launch vehicle team, time-to-flight will be significantly reduced by minimal testing with the launch vehicle.

While saving time and costs are a major advantage, launching as a secondary payload does come with risks:

- **Schedule.** You are reliant on the primary payload to hold to the advertised schedule (but not vice versa). If they slip, you slip. If you slip, you run the risk of them launching without you. In addition, you could be "called up" at the last minute to make a launch (e.g., 7 days).
- **Weight.** You are reliant on the primary payload to not exceed their predicted weight (total mass) during their processing. If they go over, you may be bumped from the flight if your mass is now too much for the launch vehicle to get both to orbit. Also, you must keep your spacecraft mass within the predicted amount as well.
- **Launch.** You are reliant on finding a launch that is in the "vicinity" of where you need to be placed and with a launch vehicle that can get you there AFTER the primary payload has been placed in orbit. Be ready to adjust your trajectory and schedule to accommodate the primary payload (as long as you can still meet your mission objectives).

These are only a few of the risks that could potentially jeopardize your mission when launching as a secondary payload. This, however, is a much better solution than never launching because of funding issues.

27.11 Common Mistakes to Avoid

After each launch, it is not uncommon to have a review to discuss what the team could have done differently to make the entire process faster, easier and less stressful. If your mission is similar to one that has already been launched, check for lessons learned from that mission. This information can be invaluable. Below are some common mistakes that are made at the launch site that can and should be avoided:

1. **Under Estimating the Schedule.** Too many times, projects have not allowed enough time in the schedule for contingencies, issues that may arise, and crew rest. The "going in" plan should be based on an 8 hour a day, 5 day a week schedule with nights and weekends kept for contingencies.
2. **Lack of Coordination for Base Access.** Since September 11, 2001, security has increased at most launch sites around the World. Getting the POC the information required in a timely manner is extremely important to get your personnel on site. Don't forget to include the truck drivers (or airplane crew) who are delivering the spacecraft!
3. **Lack of Planning and Coordination of Personnel Coming To/From the Launch Site.** The PDP is a "living" document and should be constantly reviewed and updated to ensure the correct personnel arrive at least 1 day PRIOR to when they will be needed. Careful thought and preparation will help make it easy for the necessary personnel to travel to the launch site to support the operations.
4. **Requirements "Creep."** It can manifest itself as higher, more stringent requirements at the launch site than at the manufacturing facility. (For example, Grade B N2 (Nitrogen) was used for purging during processing at your facility; however, Grade C N2 is now required at the launch site. A class 100K facility was used while you were at your facility; now a class 10K requirement has been levied on the launch site. Smocks were used during the build, but bunny suits are being required at the launch site.) Not only can such creep be unnecessary and expensive, but it also introduces last minute changes that can be a source of unexpected system errors.
5. **"Launch Fever."** Teams suffering from "launch fever" can fall into the trap of failing to make sound, common sense decisions, just to make a launch date. When it gets down to crunch time and things may not be going the right way, do not let schedule pressure be a reason to start cutting corners. Just about any anomaly can be fixed on the ground. Once the spacecraft is launched into space, well, not so easy. Procedures were put together with level heads; you want to keep it that way.
6. **Over Testing at the Launch Site.** Most likely you have been testing the reliability and performance of the spacecraft for many months. The spacecraft's trip to the launch site is generally far gentler than the launch itself and all of the environmental testing that was previously performed.

The logical extension of this belief is the minimization of testing at the launch site. You are not trying to "re-qualify" the spacecraft or the GSE, just ensuring it survived the transport to the launch site.

7. **Using Procedures for the first time at the Launch Site.** To stay on schedule, activities need to be well planned and thoroughly rehearsed ahead of time. Tests and countdown procedures that are used at the launch site should have been incorporated into the regular testing in the preceding months of qualification and acceptance testing at the manufacturer's facility. Do not perform a test or use a procedure for the first time at the launch site UNLESS it is mission critical!

8. **Expired GSE Certification.** Ensure the GSE that requires calibration arrives at the launch site calibrated and with a valid certification that well exceeds the planned launch date.

9. **Lack of Contingency Procedures.** Even though you cannot plan for every contingency, you can plan for the most common, including facility power outages, GN2 purge depletion, purge panel failure, personnel absences, contamination issues, lightning, temperature or humidity limits exceedance and many more. Have contingency plans in place and follow the procedures as defined in those plans. ALL contingency procedures should have been FULLY exercised at the manufacturer's facility prior to arrival at the launch site.

10. **Not Having a Pre-Determined Access List**

    There are times when operations will require a limitation on the number of personnel in the area. Having a pre-determined access list with the names of the individuals required to perform those operations will ensure the right personnel are there to support the launch and decrease the chance of distraction or confusion.

    This will become very apparent when transitioning operations to the launch pad. It seems that a lot of people believe that they have the right to go to the launch pad, but in reality a very small portion of the launch team actually needs to go to the launch pad. Even though you will still have a few personnel who will not understand (or agree), with a pre-determined access list there will be fewer problems to resolve. If possible, schedule a time where all personnel can have a short visit at the launch pad and an opportunity to take photos.

In addition, consider the following questions:

1. **Can the launch site environment affect the spacecraft?** Consider the temperature and humidity (both at the PPF and while transporting). What about helium exposure (possibly from those processing before or at the same time as you)? What about water exposure?

2. **When at the launch site, how is the spacecraft and instrument going to be purged?** Is there sufficient access to get to those areas (with and without the fairing installed) requiring purging? Will a high flow of GN2 be required (such as using a Vortex Cooler) increasing the amount used? Are covers required for certain areas of the spacecraft to avoid contamination?

3. **Can the spacecraft tolerate the levels of the range-radiated emissions that occur at the launch site?** Consider the conditions inside the shipping container, outside the shipping container, inside the processing facility and inside of the fairing. Also take into consideration self-exposure (by accidental radiating) inside of the fairing.

4. **If you are processing inside a clean tent at the PPF, is there enough room to test the deployable systems (such as a mag boom, instrument covers or solar arrays)?**

5. **Are there enough "inhibits" in place to meet the range safety requirements?** Typically any potentially hazardous conditions, such as a pyro firing or solar array deployment, must be at least 2-fault tolerant, including 1 via software AND 1 via hardware.

Any of the above could potentially be avoided or overcome with a modification in the design of the spacecraft or GSE. Something to consider!

Applicable Standards:
ISO 17400: Space Systems
See website for reference.

Applicable Standards:
ISO 10784: Space Systems
See website for reference.

References



For annotated bibliography of launch operations, see the book website.

Isakowitz, Steven J., Joshua B. Hopkins, and Joseph P. Hopkins, Jr. 2004. *International Reference Guide to Space Launch Systems*, 4th ed. Reston, VA: AIAA.

Strom, Steven R. 2006. *International Launch Site Guide*, 2nd ed. Reston, VA: AIAA.

28 Ground System Design

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28.1 Antenna Services
Ground Based Antenna Systems; Space Relay Systems
28.2 Data Accounting and Distribution Services
Data Capture/Transmission Verification; Standard Transmission Protocols; Bandwidth/Connectivity Between Ground Stations and User Facilities
28.3 Ground System Driving Requirements and Sizing
28.4 Mission Examples
28.5 Technology Trends
Shorter Wavelength Communications; Reduced Latency; New International Standards—Increasing Interoperability and Data Capture Percentage
28.6 Summary

Whether in LEO, GEO, or headed into deep space, spacecraft are only valuable if the data they collect can get to the end user. End users can include:

- University based researchers
- War fighters on the battlefield
- Satellite TV/radio subscribers
- Operations Engineers: responsible for the health of spacecraft systems
- Mission Planners: responsible for planning upcoming activities
- Flight Dynamicists/Navigators: responsible for ensuring the spacecraft maintains the desired orbit/trajectory and attitude

Collecting and processing data transmitted by the spacecraft through a RF stream as well as providing a method for sending instructions to the spacecraft via an RF stream is the job of the ground system. As shown in Fig. 28-1, spacecraft ground systems perform the following functions:

- **Antenna Services (covered in Sec. 28.1):** Enables transmission/reception and processing of RF signals with an embedded data stream.
  - Includes systems for encoding a data stream of instructions destined for the spacecraft and then modulating this data stream onto an RF carrier and amplifying the signal to ensure that it will have an acceptable signal-to-noise ratio when it arrives at the spacecraft.
  - Includes systems for recovering the spacecraft RF signal and then demodulating and decoding the embedded data stream before passing it off to the next function.
- **Data Accounting and Distribution Services (covered in Sec. 28.2):** Responsible for ensuring data streams headed to and from the spacecraft are complete and distributed to the proper users within the required latency period.
  - Includes systems that perform data accounting on data streams headed to and from the spacecraft to identify missing data that was not recovered from/received from the spacecraft, and to initiate actions to recover missing data as required.
  - Includes systems that perform storage, distribution, and data accounting of recovered spacecraft data streams that require transport to user locations while ensuring data integrity after recovery.
- **Command/Control and Data Processing Services (covered in Sec. 29.1):** Responsible for generating spacecraft instructions for uplink and for processing the data stream received from the spacecraft into information required to support end users.
  - Includes Mission Planning and Command Generation systems that are used to plan spacecraft activities, and then generates the specific command structures/sequences required to execute these activities.
  - Includes data processing systems used to develop specific engineering and scientific products that allow end users to utilize the data collected from the spacecraft.

Although almost all spacecraft efforts include the functions described above, the approach to performing these functions varies greatly based on specific mission requirements and mission complexity and cost. One common theme for many missions today is automation of ground system functions to minimize the cost of oper-

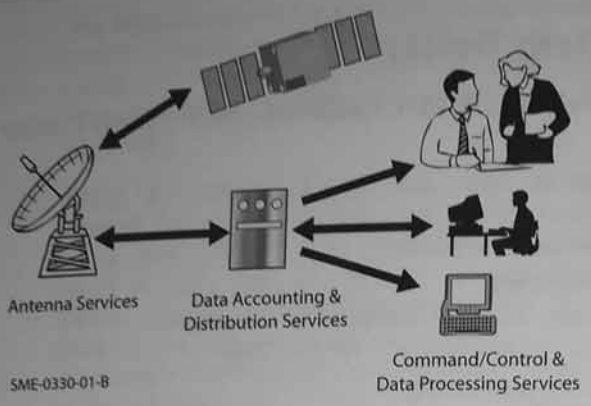


Fig. 28-1. Spacecraft Ground System Services.

ations. For many missions, everything from the pointing of the antenna to the data accounting and distribution functions are done without human involvement. These highly automated *lights-out* systems allow round-the-clock operations to execute with only periodic monitoring and intervention by operations personnel. In other cases, technology advances have opened the door to a new generation of small satellites, many of them developed by university student led teams, where the ground system functions have been greatly simplified to minimize both development and operations costs.

Looking beyond 2011, ground systems will continue to evolve as spacecraft technology advances enable higher data rates and as the synergy between ground system and spacecraft designs leads to increased interaction between the two with less and less human intervention required. These systems will rely increasingly on international standards to ensure interoperability between systems from different countries and commercial service providers and they will have an increasing reliance on Commercial Off-The-Shelf (COTS) products that are developed using these standards—reducing the need for developing expensive custom systems for each new mission.

The following sections provide information on the variety of choices available to ground system designers as they look for optimized solutions to meet their requirements.

28.1 Antenna Services

Every ground system architecture needs a service capable of embedding digital information onto an RF wave for uplink *(forward link)* to the spacecraft as well as performing the function of extracting the data from the downlink *(return link)* RF signal arriving from the spacecraft. These services fall into two primary categories:

1. Ground antennas that send/receive data directly to/from a user spacecraft, and
2. Space based relay antennas that send/receive data to/from user spacecraft and then take responsibility for getting this data to/from the ground

From a different perspective, these services can also be thought of as either:

- Provided by a government or commercial service provider who has a distributed network of antennas that are unified under an umbrella organization and where services are scheduled by a community of users, or
- Dedicated to an individual mission

A third way of thinking about antenna services are:

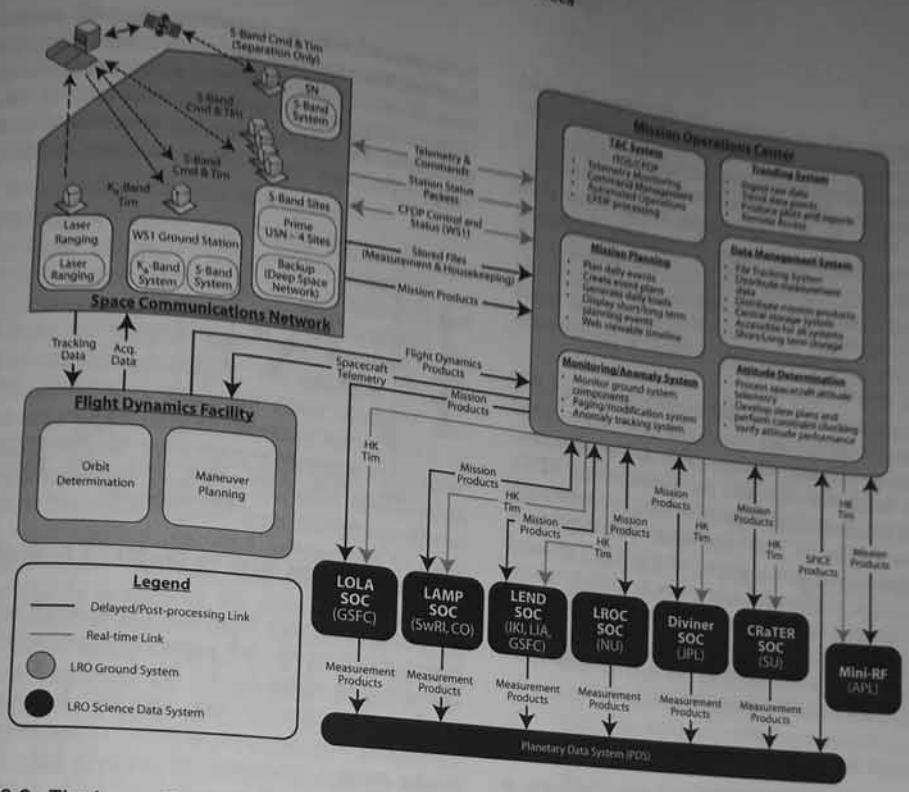
- Commercial fee for service
- Government multi-mission infrastructure, or
- Project unique funded

Each of these options should be evaluated in the mission design process. For example, the use of existing commercial or government multi-mission antenna network services has numerous advantages. These networks typically have numerous interoperable antennas that can provide a significant amount of coverage area on a predictable schedule with a well defined cost structure and interface. These geographically distributed networks can provide robustness through fault tolerance, establishing high levels of availability and reliability. These attributes may be extremely important for missions with high data capture percentage requirements or short latency requirements. Conversely, deploying a spacecraft dedicated antenna relieves the mission from the need to schedule time with competing, and possibly higher priority, missions. It also allows the antenna systems to be optimally customized for the specific spacecraft whereas existing networks will typically have standard services with which the mission must be compatible. Complying with these standards may require reducing a mission's capability by forcing compromises in areas such as data rates, modulation schemes, and encoding schemes to ensure network compatibility. Downsides of a dedicated antenna include the high development cost required and the lack of geographic distribution, especially if the desire is to have the dedicated antenna near a university or other spacecraft control center location. This may drive the spacecraft design to incorporate high levels of on board storage to support an operations concept that includes limited opportunities to transmit data to the ground.

For some missions, the optimal answer is a combination of approaches. The ground system design for NASA's Lunar Reconnaissance Orbiter (LRO), for example, is shown in Fig. 28-2 [NASA, 2011b].

The LRO ground system utilized a combination of mission dedicated, government owned, and commercially provided antenna assets to fulfill its mission.

First, LRO deployed a single, mission dedicated, 18 m K_a-/S-band antenna at White Sands, NM to support high rate science data retrieval from on-board recorders. With the Moon above the horizon for approximately half of every day, the on-board recorders were sized to accommodate using this single dedicated antenna architecture to download 473 Gbits of data each day (see Sec. 14.5.5). This LRO dedicated K_a-band antenna was originally go-



SME-0381-01-B

Fig. 28-2. The Lunar Reconnaissance Orbiter Ground System.

ing to be a single point failure since it had no back-up. Coincidentally, during the same period that the LRO mission was being developed, work was underway on a new NASA solar observatory that would communicate in K_a-band from geosynchronous orbit, the Solar Dynamics Observatory (SDO). SDO's ground system was designed with two new 12 m mission dedicated K_a-/S-Band antennas that would also be located in the desert environment of White Sands where K_a-band attenuation effects would be minimized. To address the need for a back-up to the single LRO K_a-band antenna, an agreement was established to allow LRO to use one of the SDO K_a-Band antennas in situations where the single LRO antenna becomes inoperable. In addition, the SDO antennas were increased to 18 m to be able to provide this back-up LRO support without negatively impacting their ability to support their primary mission.

Another consideration that impacted the design of the LRO ground system architecture was that mission designers had to establish a method for collecting radiometric data during each LRO orbit of the moon (~2 hour orbital period) to meet orbit determination requirements driven by lunar mapping accuracy requirements. To support this requirement, the LRO Operations Team scheduled numerous daily contacts on a commercial ground network that used 9 m to 15 m S-band antennas distributed throughout the world to support LRO's radiometric data collection requirements. These contacts provided tracking data collection from both the northern and southern hemisphere. LRO orbit determination accuracy was greatly improved by having this frequent, two-hemisphere tracking. A secondary benefit of these frequent passes was that LRO mission designers could reduce the amount of spacecraft on-board autonomy since spacecraft housekeeping data (downlinked in S-band which could close a link budget at lunar distances on a 9 m ground antenna in combination with spacecraft HGA) could be collected and commanding performed during these once/orbit contacts at the commercial stations. As it turned out, LRO's orbit determination accuracy requirement, driven by a scientific goal of generating a highly accurate lunar gravity model, was too stringent to be met using only RF ranging techniques. To meet this requirement, NASA's Satellite Laser Ranging (SLR) [NASA, 2011e] network was used to augment the RF ranging described above. SLR used laser pulse time-of-flight measurements to provide extremely precise measurements of the distance to LRO.

In addition to the nominal links described above, LRO also received support during specific mission phases from the NASA's Deep Space Network (DSN) and Space Network (SN). Immediately after separation from the launch vehicle, LRO utilized the Tracking and Data Relay Satellites (TDRS) of the SN to provide constant contact during this crucial early phase (the SN is described in Sec. 28.1.2.1). In addition, LRO established an agreement with the DSN (the DSN is described, in Sec. 28.1.1.1) to provide emergency support using 34 m

antennas distributed around the globe. This support would be provided if the LRO spacecraft entered a safe-mode that transitioned the spacecraft RF-string from the high gain antenna (HGA) to an omni antenna (where closing the Moon-Earth link budget at even low data rates in S-band, would not be possible with the dedicated White Sands 18 m antenna). DSN support was also required during lunar orbit maintenance maneuvers when the LRO HGA could not be accurately pointed at Earth and omni antenna operations were used to maintain contact with the spacecraft.

The challenge of using this approach is that the LRO Control Center interface and spacecraft interface needed to be compatible not only with the LRO dedicated antenna at White Sands but also with the commercial, SDO, SN and DSN antennas used to provide support. Although some standardization of interfaces made this multi-network compatibility a minor issue for the LRO ground system designers, continuing advances in the development and deployment of international interface standards by the world's antenna service networks will continue to drive down interface development costs for future missions requiring multi-network support.

28.1.1 Ground Based Antenna Systems

The biggest challenge of any ground based antenna is the limited amount of coverage it can provide, especially for the lowest of LEOs. Specific ground station contacts times are driven not only by orbit altitude but by the geometry between the ground station and the spacecraft during each specific pass and the *antenna mask*. The antenna mask is defined as the boundary between a clear transmission path and an obstacle for all directions around a ground station. When plotted along with satellite ground tracks, antenna masks (example in Fig. 28-3) provide an indication of when or if satellite contacts will be possible at that specific ground-station during each orbit. Masks are satellite altitude dependent, with higher orbits providing extended view periods of a specific ground station.



Fig. 28-3. Antenna Mask for the NASA Wallops Island, VA Ground Station shown for 500 km (inner circle), 750 km (middle circle), and 1,000 km (outer circle) Satellite Orbit Altitudes.

A second way to represent a ground station mask is based on the elevation angle that a satellite must reach in a specific direction to clear obstructions and establish a communications path. Figure 28-4 shows the elevation mask for one of the 18 m antennas at White Sands, NM used to support the Solar Dynamics Observatory (SDO) mission.



Fig. 28-4. Elevation Mask for the SDO-2 Antenna.

The approximate time that a spacecraft will be able to communicate with each supporting ground station can be calculated by using Eq. (8-77) found in Sec. 8.4.

To accommodate the spacing between contacts, spacecraft utilizing ground based antennas need to incorporate enough on-board memory to hold data between contacts. Spacecraft memory should be sized to:

- Store both payload and scientific instrument data as well as spacecraft housekeeping data generated during non-contact periods
- Continue storing payload and housekeeping data generated during ground station contact periods since memory will typically not be freed for overwriting until memory recovery during a contact has been completed and verified (possibly late in the contact)
- Accommodate at least one missed contact without reaching capacity, since problems on the spacecraft and/or at the ground may result in periodic missed contacts

In addition, care should be taken to determine the actions that will happen if spacecraft memory reaches capacity in cases where numerous contacts are missed or others issues arise. One option is to allow the spacecraft to begin overwriting the oldest on-board data. This option will ensure that the most recent data will be available when a playback capability is restored and memory can be recovered. But, by overwriting the oldest on-board memory, clues as to what might have happened during the earliest periods when issues arose that prevented the downlinking of data may be lost or overwritten. The other option is to have spacecraft memory stop recording when it reaches capacity. In this option, the oldest data will be available for examination once recorder playback capability is restored, but, more recent data generated by the spacecraft will be lost.

In addition, the mission design needs to incorporate high enough data rates to ensure that enough time is available during contact periods so that on-board stored data can be transferred to the ground, and possibly partially or totally retransmitted if the initial transmission is not completely recovered.

28.1.1.1 Government Network Examples

As the sponsor for many of the world's scientific and national defense satellites, individual country Space Agencies, Defense Departments, Land Use Agencies and Weather Agencies, develop and operate distributed antenna networks to support their own satellites, and, through international agreements, the satellites of other countries. Building and operating these complex networks is not only technically challenging but includes an aspect of diplomacy as well, since providing significant downlink opportunities typically requires establishing ground stations beyond a sponsor country's geographical boundaries, requiring international agreements to have other countries "host" antenna sites on their territory. Whether attempting to provide, for example, deep-space links, through stations spread evenly around the globe, or polar orbit links where high latitude ground stations could provide coverage of a high percentage of orbits/day, international agreements become the valuable tool for establishing optimized networks. Almost all countries involved in flying spacecraft have some sort of antenna network. The following few examples are of country sponsored networks that are highly leveraged by various space agencies around the world due to their broad reach or unique capabilities:

NASA's Deep Space Network (DSN)

The California Institute of Technology's Jet Propulsion Laboratory (JPL) is responsible for managing NASA's Deep Space Network (DSN) of large aperture ground based antennas used to support spacecraft traveling beyond 30,000 km from Earth. Developed in the 1960s to support robotic lunar, planetary and other deep space missions, the DSN also provided support to the Apollo human missions to the Moon, where its powerful antennas provided complimentary and emergency support for NASA's smaller Manned Space Flight Network (MSFN) antennas.

DSN antennas are located at 3 Deep Space Communications Complexes (DSCC), separated by ~120 deg of longitude. Two are located in the northern hemisphere: Madrid, Spain (40° 25' 53" North/4° 14' 53" West) and Goldstone, California (35° 25' 36" North/116° 53' 24" West)—while the third, Canberra, Australia (35° 24' 05" South/148° 58' 54" East), is located in the southern hemisphere (this two hemisphere configuration is optimal for collecting tracking data for precision orbit/trajectory determination). Overall coordination and operational management of the DSN is accomplished through the Deep Space Operations Center (DSOC) located at JPL in Pasadena, California.

The DSN uses four different types of antennas—they are:

- **34 m HEF**: The design of the High Efficiency (HEF) antennas incorporate recent design features in electronics and mechanical systems with the antenna dish itself being precision shaped for maximum signal gathering capability.
- **34 m BWG**: The design of the Beam Waveguide (BWG) antennas is based on that of the 34 m HEF antennas—providing similar performance. Five radio frequency mirrors are used to reflect signals along a beam waveguide tube from the vertex of the antenna to a below-ground equipment room. This allows for relocation of sensitive equipment from the feed cone in the center of the main reflector to the pedestal equipment room—making this equipment more accessible, easier to temperature control, and easier to replace and upgrade as requirements change.
- **34 m HSB**: The single High Speed Beam Waveguide (HSB) antenna is located at the Goldstone complex. These antennas can move at 3 deg/sec in azimuth and 2 deg/sec in elevation compared with the 0.8 deg/sec in azimuth and elevation of the other DSN 34 m antennas. These faster slew rates enable this antenna to track spacecraft closer to Earth versus the other 34 m antennas.
- **70 m**: NASA's largest and therefore most sensitive ground based antennas—capable of tracking spacecraft 16B km from Earth.
- **26 m (decommissioned)**: These antennas were deployed at the DSN sites to support Apollo lunar missions. They were decommissioned in the 2008 timeframe and are no longer available for mission support.

The capabilities of the DSN antennas are listed in Table 28-1:

A few additional attributes/constraints of the DSN antennas include:

- **Maximum Data Rates:** The maximum telemetry capture rates for DSN antennas are:
  - 8 Mbps (deep space) and 125 Mbps (near Earth) for Convolutional encoded data ($r = ½$, $k = 7$) concatenated with Reed-Solomon encoding
  - 1.6 Mbps for Turbo coded data
  - 10 bps—minimum data rate supported
  - 13.5 Mbps—maximum data rate out from each DSCC (can constrain the data rate for each individual mission being supported concurrently at a DSCC)

  Note: specific mission maximum data rates are driven by coding scheme, modulation method, and other link parameters (see Sec. 21.1.4)
- **Signal Strength**: The maximum received signal strength for X-band transmissions is –90 dBm. Spacecraft must limit their Equivalent Isotropically

Table 28-1. DSN Antenna Capabilities.

Antenna Type	Complex	DSN ID	Transmit Frequency (MHz)	EIRP (dBW)	Receive Frequency (MHz)	Gain (dBi)
34 m BWG	Goldstone	DSS24	S: 2,025–2,120	78.7–98.7	S: 2,200–2,300	
	Canberra	DSS34	S: 2,025–2,120	78.7–98.7	S: 2,200–2,300	
	Madrid	DSS54	S: 2,025–2,120	78.7–98.7	S: 2,200–2,300	
	Goldstone	DSS25	X: 7,145–7,235	89.5–109.5	X: 8,400–8,500	
	Goldstone	DSS26	X: 7,145–7,235	89.5–109.5	X: 8,400–8,500	
	Canberra	DSS34	X: 7,145–7,235	89.5–109.5	X: 8,400–8,500	
	Madrid	DSS54	X: 7,145–7,235	89.5–109.5	X: 8,400–8500	
	Madrid	DSS55	X: 7,145–7,235	89.5–109.5	X: 8,400–8,500	
	Goldstone	DSS24	—	—	K_a: 25,500–27,000	
	Goldstone	DSS25	K_a: 34,315–34,415	98.2–108.2	K_a: 31,800–32,300	
	Goldstone	DSS26	—	—	K_a: 31,800–32,300	
	Canberra	DSS34	—	—	K_a: 25,500–27,000 K_a: 31,800–32,300	
	Madrid	DSS54	—	—	K_a: 25,500–27,000 K_a: 31,800–32,300	
	Madrid	DSS55	—	—	K_a: 31,800–32,300	
34 m HEF	Goldstone	DSS15	—	—	S: 2,200–2,300	
	Canberra	DSS45	S: 2,025–2,110	71.8–78.8	S: 2,200–2,300	
	Madrid	DSS65	S: 2,025–2,110	71.8–78.8	S: 2,200–2,300	
	Goldstone	DSS15	X: 7,145–7,190	89.8–109.8	X: 8,400–8,500	
	Canberra	DSS45	X: 7,145–7,190	89.8–109.8	X: 8,400–8,500	
	Madrid	DSS65	X: 7,145–7,190	89.8–109.8	X: 8,400–8,500	
34 m HSB	Goldstone	DSS27	S: 2,025–2,120	70.7–76.7	S: 2,200–2,300	
70 m	Goldstone	DSS14	S: 2,110–2,118 S: 2,090–2,091	85.6–105.6 85.6–97.4	S: 2,200–2,300	
	Canberra	DSS43	S: 2,110–2,118 S: 2,110–2,118 S: 2,090–2,091	85.6–105.6 106.7–118.7 85.6–97.4	S: 2,270–2,300	
	Madrid	DSS63	S: 2,110–2,118 S: 2,090–2,091	85.6–105.6 85.6–97.4	S: 2,270–2,300	
	Goldstone	DSS14	X: 7,145–7,190	95.8–115.8	X: 8,400–8,500	
	Canberra	DSS43	X: 7,145–7,190	95.8–115.8	X: 8,400–8,500	
	Madrid	DSS63	X: 7,145–7,190	95.8–115.8	X: 8,400–8,500	

* referenced to 45° elevation with 90% weather condition (CD=0.90) and diplexed configuration

Radiated Power (EIRP) so as to not exceed this level at the DSN receiver. For K_a-band transmissions (25.5–27 GHz), the maximum received signal strength can be up to ~ –45 dBm at the cost of lower G/T performance of the receiving antenna.

- **Multi-Spacecraft Per Antenna (MSPA)**: Enables, with some constraints, DSN antennas to receive telemetry from multiple spacecraft when they are all within the antenna's beamwidth (although only one spacecraft can be commanded at a time).
- **Antenna Arraying**: Combining signals from various DSN antennas [NASA/JPL, 2011] at the Intermediate Frequency (IF) level is used to create a "virtual" antenna with performance greater than the individual contributing antennas. In the 8 GHz band, for example, five 34 m antennas can be used to provide the same performance as a DSN 70 m antenna. Arraying can be used in S, X and K_a-Band although S-band arraying capability will be phased out with upgrades planned for the ~2014 timeframe.

Although downlink telemetry processing using antenna arraying is most typical, uplink arraying research is ongoing as well. Moving beyond 2011, DSN plans to slowly phase-out their expensive-to-maintain 70 m antennas with the addition of more 34 m antennas at each site that can be arrayed or used individually as required. The general equation for establishing the maximum increase in gain that can be achieved through arraying a number of antennas (N) (assuming no combining loss) is:

$$(G/T)_{array} = \sum_{i=1}^{N} (G/T)_i \qquad (28\text{-}1)$$

- Simply the sum of the sensitivity for each arrayed antenna. In the case of a homogeneous array—the equation becomes (shown in Fig. 28-5):

$$(G/T)_{array} = (G/T)_0 \sum_{i=1}^{N} 1 = N(G/T)_0 \qquad (28\text{-}2)$$

Table 28-1, Fig. 28-4◀, Eq. 28-2

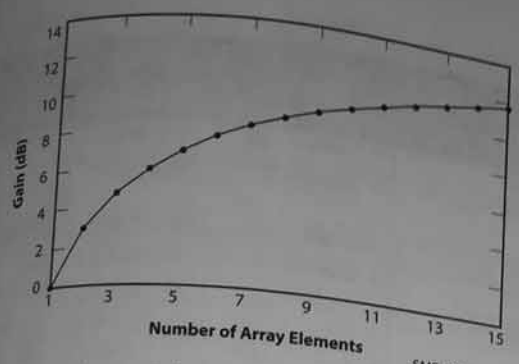


Fig. 28-5. Antenna Gain for Addition of Arrayed Antenna Elements.

NASA's Near Earth Network (NEN)

The Near Earth Network (NEN) [NASA, 2011c] is a Goddard Space Flight Center managed NASA network consisting of a range of antenna apertures between 4.3 m and 18 m in diameter and capable of receiving VHF, UHF, S, X and K_a-band transmissions and transmitting in the S, VHF and UHF-bands. The NEN has evolved to where ~50% of its current capability is provided by NASA owned antennas while the other ~50% is procured through commercial contracts with the Norwegian Kongsberg Satellite Services (KSAT) AS and the Universal Space Network (USN) subsidiary of the Swedish Space Corporation (SSC). Since KSAT and SSC are covered later as individual networks, this section will focus on the NASA-owned NEN antennas. NEN ground stations are located at:

- Wallops Island Virginia (Wallops Ground Station (WGS)) (38°N/75°W),
- The University of Alaska, Fairbanks (Alaska Satellite Facility (ASF)) (64°N/147°W),
- McMurdo Station Antarctica (McMurdo Ground Station (MGS)) (78°S/193°W),
- Merritt Island Florida (Merritt Island Launch Annex (MILA)) (29°N/81°W)
- New Smyrna Beach Florida (Ponce De Leon (PDL)) (29°N/81°W)
- White Sands New Mexico (White Sands (WS)) (32°N/106°W)
- Svalbard Norway (Svalbard Ground Station (SGS)) (78°N/15°E)
  - Note: The Svalbard Ground Station is actually owned and operated by the KSAT commercial organization but NASA still owns the first antenna that was deployed on the Svalbard archipelago.

The NEN has its heritage in the Spaceflight Tracking and Data Network (STDN), established in 1975, and earlier NASA ground networks developed in the 1960s to support early human spaceflight (the Manned Space Flight Network (MSFN)) as well as robotic missions in Earth orbit (the Spacecraft Tracking and Data Acquisition Network (STADAN)). In evolving from the STDN to today's NEN, the number of antennas was significantly reduced as NASA phased in a space-based relay network, the Tracking and Data Relay Satellite System (TDRSS), in the 1980s. The TDRSS based Space Network (discussed in Sec. 28.1.2.1) could provide constant coverage for Earth-orbiting missions, which was highly desired for human spaceflight applications as well as for some robotic missions. In addition, the 1990s saw a further reduction in NASA-owned NEN stations as commercial ground station utilization became a more cost effective option for many applications.

In general, the remaining NASA-owned NEN ground stations are focused on supporting missions between LEO and the Moon with a specific focus on supporting LEO, polar-orbiting missions and missions wherein NASA has unique requirements. For example, NASA's NEN ground stations in Florida are primarily used to support Space Shuttle launch/ascent data capture. These assets will be retired at the conclusion of the Space Shuttle Program. Much of the history of the NEN has been in supporting S/X-band operations of LEO spacecraft. With the recent addition of an 18 m S/K_a-band antenna in White Sands, New Mexico, the NEN now has an antenna capable of supporting K_a-band operations out to lunar distances. This antenna was initially developed as a dedicated asset to support NASA's LRO mission, but will be available to future NEN K_a-band missions. In addition, NEN stations support 1-way and 2-way Doppler as well as Tone Ranging (see Sec. 14.4.5).

Looking beyond 2011, the NEN will be focused on expanding K_a-band operations to additional sites to increase the capability to support future higher data rate missions. The NEN is also standardizing capabilities across all of their antennas and increasing interoperability with other networks by implementing recently developed international communication standards and protocols.

The capabilities of the NASA owned NEN antennas are listed in Table 28-2.

ESA's ESTRACK

ESTRACK consists of 10 ESA operated ground stations that provide communication support for launch, Earth-orbital and deep space missions [ESA, 2011]. Each ground station hosts one or more antennas that are either: 13 m, 13.5 m or 15 m. These antennas are used for launch and Earth orbit communication. Additionally, 35 m antennas are being added to ESTRACK (as of 2011) to be used for deep space communication. Data rates supported vary by antenna but range from 256 Kbps to 105 Mbps. All antennas operate at the following frequencies:

- S-band transmit/receive frequencies: 2,025–2,120 MHz / 2,200–2,300 MHz
- X-band transmit/receive frequencies: 7,145–7,235 MHz / 8,400–8,500 MHz

Table 28-2. NASA NEN Antenna Capability Summary.

Antenna Type	Complex	NEN ID	Transmit Frequency (MHz)	EIRP (dBW)	Receive Frequency (MHz)	Data Rate[1]	G/T (dB/K)
4.3 m	Merritt Island[2]	MILA	S: 2,025–2,120	58	S: 2,200–2,300	S: 2 Mbps	11
	New Smyrna Beach[2]	PDL	S: 2,025–2,120	58	S: 2,200–2,300	S: 2 Mbps	17
4.7 m	Wallops	LEO-T	S: 2,025–2,120	59	S: 2,200–2,300	S: 8 Mbps	16.99
9 m	Merritt Island[2]	MILA (2)	S: 2,025–2,120	63	S: 2,200–2,300	S: 10 Mbps	24
10 m	McMurdo	MG1	S: 2,025–2,120	63	S: 2,200–2,400 X: 8,025–8,400	S: 15 Mbps X: 110 Mbps	21.11 32.48
	University of Alaska	ASF10	No Uplink Available	No Uplink Available	S: 2,200–2,400 X: 8,025–8,350	S: 8 Mbps X: 150 Mbps[3]	22.59 36.36
11.3 m	Svalbard	SG1	S: 2,025–2,120	66	S: 2,200–2,400 X: 8,025–8,500	S: 8 Mbps X: 150 Mbps[3]	23.63 35.84
	Wallops	WGS11m	S: 2,025–2,120	66	S: 2,200–2,400 X: 8,025–8,400	S: 36 Mbps X: 150 Mbps[3]	23.63 34.50
	University of Alaska	ASF11	S: 2,025–2,120	66	S: 2,200–2,400 X: 8,025–8,400	S: 8 Mbps X: 150 Mbps[3]	23.01 35
18 m	White Sands	WS1	S: 2,025–2,120	80	S: 2,200–2,300 K_a: 25,500–27,000	S: 20 Mbps K_a: 470 Mbps	29 45
Yagi (VHF)	Wallops	WFF VHF-1	VHF: 139.208	45.4	VHF: 143.625	NA	N/A
	Wallops	WFF VHF-2	VHF: 130.167	45.4	VHF: 121.750	NA	N/A
	White Sands	WSC VHF-1	VHF: 139.208	43.4	VHF: 143.625	NA	N/A
	White Sands	WSC VHF-2	VHF: 130.167	45.4	VHF: 121.750	NA	N/A
Quad Helix (UHF)	Merritt Island[2]	MILA	UHF: 225–300	28	UHF: 225-300	NA	11

Notes:
1. Specific mission data rates need to take into consideration modulation and coding schemes as well as other link parameters (see Sec. 21.1.4). These rates only consider the downlink rate received at the antenna; the rate at which the site delivers data to customer control centers is driven by transmission network bandwidth available.
2. Antennas at Merritt Island and New Smyrna Beach will be decommissioned at the end of the Space Shuttle Program
3. As of 2011, a NEN network-wide upgrade of X-band capabilities was underway to enable data rates of 300+ Mbps using a single polarization and 600+ Mbps using dual polarization.

ESTRACK ground antennas are all remotely operated from the ESTRACK Control Center (ECC) at the European Space Operations Center (ESOC) in Darmstadt, Germany. ESTRACK ground stations are located at:

- Perth Australia (31.8°S/115.9°E): 15 m S/X-band antenna
- Redu Belgium (50.0°N/5.15°E): 15 m S-band, 13.5 m K_a-band, 2.4 m S-band antennas
- Kourou French Guiana (5.25°N/52.81°W): 15 m S/X-band antenna
- Cebreros Spain (40.53°N/4.37°W): 35 m X/K_a-band antenna
- New Norcia Australia (31.05°S/116.19°E): 35 m S/X-band antenna
- Maspalomas Canary Islands (27.76°N/15.63°W): 15 m S/X-band antenna
- Kiruna Sweden (67.86°N/20.96°E): 15 m S/X-band, 13 m S/X-band antennas
- Villafranca Spain (40.44°N/3.95°W): Two 15 m S-band antennas
- Santa Maria Portugal (36.997°N/25.14°W): 5.5 m S-band antenna
- Malargue Argentina (5.25°N/52.81°W): 35 m X/K_a-band antenna (under construction as of 2011)

Table 28-2, Fig. 28-5◀, Eq. 28-2◀

28.1.1.2 Commercial Networks

Beginning in the late 1990s, a number of commercial organizations entered the ground station marketplace, offering network services on a price-per-pass basis. By offering standard telemetry, tracking, and command services that utilize international standards as well as mission customized interfaces, these providers have extended their services around the world, in support of commercial and government agency space missions. Although this commercial sector is still fairly new and constantly evolving, the two companies that have the most extensive networks of commercial ground stations are:

Swedish Space Corporation (SSC)

SSC and their US subsidiary, Universal Space Network (USN) Inc. have established a worldwide fee for service network known as PrioraNet. SSC focuses on supporting spacecraft launches as well as polar and

equatorial LEO missions. Scheduling and control of PrioraNet are performed at three Network Management Centers located in: Newport Beach, California; Kiruna, Sweden; and Horsham, Pennsylvania. SSC ground stations are located at:

- Dongara Australia (29°S/115°E): 13 m S/X/K_u-band, 7.3 m S/X-band antennas
- Poker Flat Alaska (65°N/147°W): 7.3 m S/X-band, 11 m S/X-band antennas
- North Pole Alaska (64.8°N/147.5°W): 13 m S/X-band, 5 m S/X-band antennas
- Inuvik Canada (68.4°N/133.5°w): Two 13 m S/X-band antennas
- Santiago Chile (33°S/70°W): Six antennas—VHF to S/L/X-band (up to 12 m)
- South Point Hawaii (19°N/155.6°W): Two 13 m S/X/K_u-band antennas
- Esrange Sweden (67.9°N/21°E): Seven antennas—UHF to S/L/X-band (up to 13 m)

As of 2011, SSC continues to add antennas to PrioraNet in support of both equatorial and polar orbiting missions [SSC, 2011].

Kongsberg Satellite Services (KSAT) AS

KSAT is a commercial Norwegian enterprise headquartered in Tromsø. KSAT is a fee for service organization that supports both commercial and government space programs around the world. The KSAT focus has historically been on supporting polar orbiting LEO satellites with ground stations within the Arctic Circle and in Antarctica where they can view almost all passes for these satellites. The Tromsø Network Operations Center (TNOC) is responsible for scheduling, operating and controlling KSAT antennas. KSAT Ground Stations are located at:

- Tromsø Norway (69.5°N/18.8°E): Numerous L/X/S antennas up to 13 m diameter
- Svalbard Norway (78.2°N/15.4°E): Numerous L/X/S antennas up to 13 m diameter
- Troll Research Station Antarctica (72.01°S/2.5°E): Two 7.3 m S/X-band antennas

Looking beyond 2011, KSAT continues plans to continue adding antennas at Svalbard and Troll. In addition, they are looking to expand their equatorial support through the development of a number of new tracking stations [KSAT, 2011].

28.1.1.3 Project Dedicated Antennas

There are times when sharing antenna assets with other missions is impractical due to an ops concept that may include security requirements, cost restrictions, scheduling requirements, or latency requirements, that cannot be met through the use of a multi-user antenna network. For these missions, the development of project dedicated antenna services is an option. These project dedicated systems can be highly tuned to support their specific satellite's requirements. For example, for missions with significant on-board storage capability, and the ability to quickly downlink stored information to the ground, a single dedicated ground station may be adequate to support the mission needs (keeping in mind that this single station may only view a small number of orbits per day depending on the orbital inclination and the ground station latitude). In addition, dedicated antennas need to be located at sites where the bandwidth available from the site back to the end user of the data is adequate to support the latency requirements of the mission. For example, placing a mission-dedicated antenna at the North Pole to support a polar orbiting mission may sound like a perfect solution to view every orbit; however, getting that data back to the mission's Control Center in Europe, Asia or the United States in a reasonable amount of time can be a difficult or impossible challenge. Below are examples of GEO, LEO and SmallSat missions that use dedicated antenna services architectures.

GEO: NOAA Geostationary Operations Environmental Satellites (GOES)

NOAA's Geostationary Operations Environmental Satellite (GOES) constellation consists of two operational geosynchronous satellites located at: ~75° W (GOES-East) and ~135° W (GOES-West) that provide data on weather across the United States. Each of these satellites transmits ~100 images to Earth each day. In addition, after processing on the ground, some GOES weather imagery is retransmitted, on a frequent basis, from the ground to the GOES spacecraft where it is then broadcast to users throughout the country who have small Earth terminals able to receive and process this data.

To support these almost constant uplink/downlink requirements, NOAA established a single ground terminal at Wallops Island, Virginia where 13+ m antennas are dedicated to each operational GOES. These antennas ensure constant contact is maintained with each spacecraft so that time critical weather imagery can flow unimpeded to the National Weather Service and many other GOES data customers. In addition, direct communication links between this NOAA Wallops Command & Data Acquisition Station (WCDAS) and the GOES Satellite Operations Center located at NOAA's Satellite Operations Facility (NSOF) in Suitland, MD, provide satellite operations personnel at the NSOF with a real-time view of the GOES spacecraft state-of-health and an available command link for sending up instructions as required.

One major challenge of deploying mission dedicated antennas is building in fault tolerance while controlling spending on back-up capability that may seldom be needed. Since the WCDAS is located near the Atlantic Ocean, concerns have been raised about the possibility of hurricanes coming up the East Coast, bringing wind speeds that could damage or at least prohibit accurate pointing of GOES ground antennas, disabling this critical weather

Table 28-2◀, Fig. 28-5◀, Eq. 28-2◀